An Introduction to

Relativistic Celestial Mechanics

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The source for these notes are available online (via `git`):
`https://github.com/ashcat2005/RCM`

Extra material for these notes are available online (via `git`):
`https://github.com/ashcat2005/RCM_Extra`

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Part I

Celestial Mechanics

Chapter 1

Newtonian Celestial Mechanics

The starting point to study the relativistic celestial mechanics is to review the formulation of the Keplerian systems using the well known Newtonian gravitational interaction. This is based on the *Poisson equation*,

$$\nabla^2 \phi(t, \mathbf{x}) = 4\pi G \rho(t, \mathbf{x}), \tag{1.1}$$

where $\phi(t, \mathbf{x})$ represents the *gravitational potential* depending on time and space coordinates $\mathbf{x} = (x, y, z)$, $G = 6.67 \times 10^{-11}$ N m^2/kg^2 is the Newtonian gravitational constant and $\rho(t, \mathbf{x})$ is the matter density. As is shown in Appendix B, a general solution of this equation can be written in terms of the corresponding Green function as

$$\phi(t, \mathbf{x}) = -G \int \frac{\rho(t, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x'. \tag{1.2}$$

Considering the case of a point particle with mass m_1 as the source in equation (1.1) and due to its spherical symmetry, the general solution is simplified to the potential

$$\phi(r) = -\frac{Gm_1}{r} \tag{1.3}$$

where $r = |\mathbf{x}|$ represents the distance from the particle to the point in space where the potential is evaluated. From this expression, it is also possible to define the *potential energy* stored in a system of two particles with masses m_1 and m_2 separated by a distance r,

$$U(r) = m_2 \phi(r) = -\frac{Gm_1 m_2}{r}. \tag{1.4}$$

Now, we will present the description of a two-body system using Newtonian gravity and the Lagrangian formulation.

1.1 The Two-Body Problem

Consider two point particles with masses m_1 and m_2, located at positions $\mathbf{x}_1$ and $\mathbf{x}_2$ with respect to some coordinate system, as shown in Figure 1.1.

We define the relative position vector $\mathbf{x}$ through the relation

$$\mathbf{x} = \mathbf{x}_2 - \mathbf{x}_1, \tag{1.5}$$

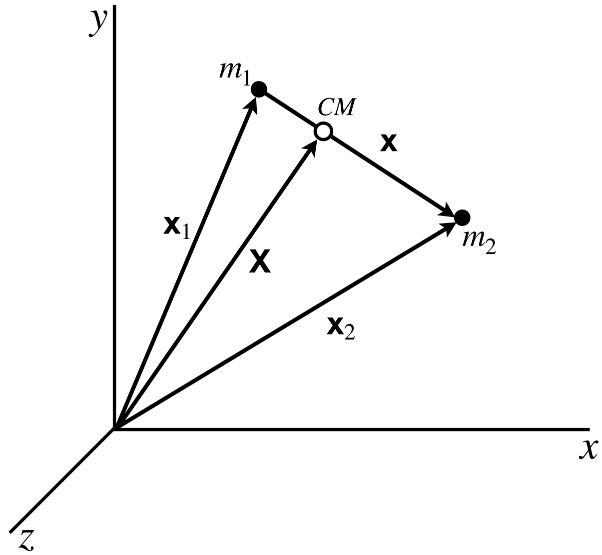

Figure 1.1: Two point particles moving under the mutual gravitational influence. Using the inertial frame shown in the figure, the position vectors of the particles are $\mathbf{x}_1$ and $\mathbf{x}_2$.

and thus, the gravitational force between the two point particles can be derived from the potential energy $U(r) = -\frac{Gm_1m_2}{r}$. We write the Lagrangian of the two body system as

$$L = \frac{1}{2}m_1\dot{\mathbf{x}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{x}}_2^2 - U(r) \tag{1.6}$$

where overdot indicates differentiation with respect to t. Using the center of mass definition,

$$\mathbf{X} = \frac{m_1\mathbf{x}_1 + m_2\mathbf{x}_2}{M}, \tag{1.7}$$

where $M = m_1 + m_2$, the Lagrangian becomes

$$L = \frac{1}{2}M\dot{\mathbf{X}}^2 + \frac{1}{2}\left(\frac{m_1m_2}{M}\right)\dot{\mathbf{x}}^2 - U(r). \tag{1.8}$$

This function can be split into two components $L = L_{CM} + L_{rel}$, where $L_{CM} = \frac{1}{2}M\dot{\mathbf{X}}^2$ is the Lagrangian describing the center of mass and L_{rel} is the Lagrangian describing the relative motion of the two particles. It is easily checked that the center of mass will have a uniform rectilinear motion and therefore, we will restrict our attention to the description of the relative motion*. Defining the reduced mass of the system as $\mu = \frac{m_1m_2}{M}$, we write the relevant Lagrangian to describe the relative motion as

$$L = \frac{1}{2}\mu\dot{\mathbf{x}}^2 - \mu\phi(r), \tag{1.9}$$

which can be interpreted as the Lagrangian describing a particle with mass μ moving under the interaction described by the gravitational potential

$$\phi(r) = \frac{U(r)}{\mu} = -\frac{GM}{r}. \tag{1.10}$$

*The uniform rectilinear motion of the center of mass provides the definition of an inertial reference system, called *barycentric*, in which the center of mass is located at rest at the origin of the system. We will adopt this system to derive the equations of motion.

Exercise 1.1

Equation of Motion for the Two Body System
Show that in the barycentric frame and using cartesian coordinates such that $\boldsymbol{x} = (x^1, x^2, x^3)$ *the Euler-Lagrange equations*

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}^j}\right) = \frac{\partial L}{\partial x^j} \tag{1.11}$$

with $j = 1, 2, 3$ *yields the equations of motion for the two-body system as*

$$\ddot{\boldsymbol{x}} = -\boldsymbol{\nabla}\phi(r) = -\frac{GM}{r^3}\boldsymbol{x}. \tag{1.12}$$

1.1.1 Conserved Quantities in the Two-body Problem

In order to obtain the conserved quantities in the two-body system description, we will consider the usual spherical coordinates $\mathbf{x} = (r, \theta, \varphi)$. Hence, the Lagrangian (1.9) becomes

$$L = \frac{1}{2}\mu\left[\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\varphi}^2\right] + \frac{GM\mu}{r}. \tag{1.13}$$

The canonical conjugated momenta are defined as $p_j = \frac{\partial L}{\partial \dot{x}^j}$. From the above Lagrangian it is clear that the φ component of the momentum is conserved,

$$p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = \mu r^2 \sin^2\theta\dot{\varphi}, \tag{1.14}$$

and it is identified with the azimuthal component of the total angular momentum. It is also easy to see that the Hamiltonian

$$H = \sum_{j=1}^{3} p_j\dot{x}^j - L = \frac{1}{2}\mu\dot{\mathbf{x}}^2 - \frac{GM\mu}{r} \tag{1.15}$$

is conserved and that it corresponds to the total energy of the system.

The total angular momentum of the two-body system, assuming that the particles have no spin, is given by $\mathbf{L} = m_1\mathbf{x}_1 \times \dot{\mathbf{x}}_1 + m_2\mathbf{x}_2 \times \dot{\mathbf{x}}_2 = \mu\mathbf{x} \times \dot{\mathbf{x}}$ and it is also a constant of motion, i.e. $\frac{d\mathbf{L}}{dt} = 0$, due to the fact that the acceleration lies in the radial direction (1.12).

Since $\mathbf{L}$ is constant and orthogonal to both position and velocity vectors, $\mathbf{L}\cdot\mathbf{x} = \mathbf{L}\cdot\dot{\mathbf{x}} = 0$, the motion of the system lies in a fixed plane that is always perpendicular to the total angular momentum. Introducing a coordinate system with unit vectors $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ in which the orbit lies in the XY plane, we take $\mathbf{L}$ along the Z-axis to write

$$\mathbf{L} = \mu\boldsymbol{\ell} \tag{1.16}$$

where

$$\boldsymbol{\ell} = \mathbf{x} \times \dot{\mathbf{x}} = \ell\mathbf{k}, \tag{1.17}$$

$\ell = |\boldsymbol{\ell}| = |\mathbf{x} \times \dot{\mathbf{x}}|$ and $\mathbf{k}$ is a unit vector in the Z-direction. The motion in this plane can be described by taking $\theta = \frac{\pi}{2}$ and using only the polar coordinates r and φ. Hence, the position vector has components $\mathbf{x} = (r\cos\varphi, r\sin\varphi, 0)$, where we are measuring the angle φ with respect to the X-axis, as usual.

However, it is usual to introduce a time dependent basis to describe the motion in which the first vector points from particle 1 to particle 2, the second vector lies in the orbital plane and is orthogonal to the first one and the third vector points in the Z-direction,

$$\begin{cases} \mathbf{n} = \frac{\mathbf{x}}{r} = \cos\varphi \mathbf{i} + \sin\varphi \mathbf{j} \\ \boldsymbol{\lambda} = -\sin\varphi \mathbf{i} + \cos\varphi \mathbf{j} \\ \mathbf{k} \end{cases} \tag{1.18}$$

It is easy to show that these vectors satisfy the relations

$$\frac{d\mathbf{n}}{d\varphi} = \boldsymbol{\lambda} \tag{1.19a}$$

$$\frac{d\boldsymbol{\lambda}}{d\varphi} = -\mathbf{n}. \tag{1.19b}$$

Exercise 1.2

Position,Velocity and Acceleration Vectors.
Show that the position, velocity and acceleration vectors in the time-dependent basis are

$$\boldsymbol{x} = r\boldsymbol{n} \tag{1.20a}$$

$$\dot{\boldsymbol{x}} = \dot{r}\boldsymbol{n} + r\dot{\varphi}\boldsymbol{\lambda} \tag{1.20b}$$

$$\ddot{\boldsymbol{x}} = \left(\ddot{r} - r\dot{\varphi}^2\right)\boldsymbol{n} + \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\varphi}\right)\boldsymbol{\lambda}. \tag{1.20c}$$

Comparison of the acceleration in equation (1.20c) with the equation (1.12) yields to the conclusion that the factor $r^2\dot{\varphi}$ is a constant of motion due to the radial behavior of the gravitational interaction. This conserved quantity corresponds to the angular momentum magnitude,

$$\ell = r^2\dot{\varphi}. \tag{1.21}$$

Another conserved quantity is the *Laplace-Runge-Lenz vector* (also called the *eccentricity vector* for reasons that we will show below), defined as

$$\mathbf{e} = \frac{\dot{\mathbf{x}} \times \boldsymbol{\ell}}{GM} - \mathbf{n}. \tag{1.22}$$

Due to its construction, it is clear that

$$\boldsymbol{\ell} \cdot \mathbf{e} = 0, \tag{1.23}$$

i.e. the eccentricity vector lies in the orbital plane.

Exercise 1.3

Conservation of the Laplace-Runge-Lenz vector
Show explicitly that

$$\frac{d\boldsymbol{e}}{dt} = \boldsymbol{0}. \tag{1.24}$$

1.1.2 Equations of Motion

The radial equation of motion is easily obtained from the Hamiltonian (1.15) and (1.20b) together with (1.21), giving the first order differential equation

$$\frac{1}{2}\dot{r}^2 + \frac{\ell^2}{2r^2} - \frac{GM}{r} = \mathcal{E}, \tag{1.25}$$

where we've defined the total energy per unit mass $\mathcal{E} = \frac{H}{\mu}$. Introducing an effective potential energy,

$$V_{eff}(r) = \frac{\ell^2}{2r^2} - \frac{GM}{r} \tag{1.26}$$

the radial equation can be written as

$$\frac{1}{2}\dot{r}^2 = \mathcal{E} - V_{eff}(r). \tag{1.27}$$

Plenty of information about the possible trajectories can be obtained by studying the function $V_{eff}(r)$. For example, from the plot in Figure 1.2 it is possible to note that there exist circular orbits, determined by the condition

$$\left.\frac{dV_{eff}}{dr}\right|_{r=r_c} = 0, \tag{1.28}$$

which gives the value of the orbit radius as $r_c = \frac{\ell^2}{GM}$. By replacing this value in equation (1.25), we obtain the energy of the particle at the circular orbit,

$$\mathcal{E}_c = -\frac{GM}{2r_c} = -\frac{G^2M^2}{2\ell^2}. \tag{1.29}$$

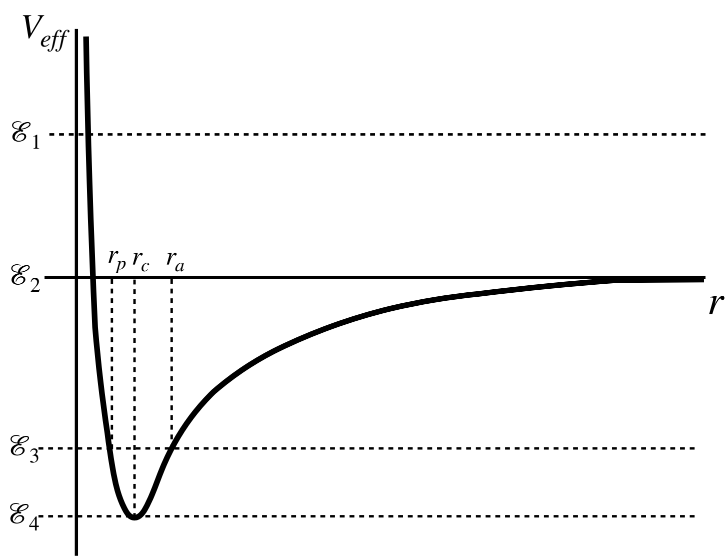

Figure 1.2: Typical behavior of the effective potential for the 2-Body system as function of the separation distance r with $\ell \neq 0$. The horizontal doted lines indicate constant values of the total energy per unit mass $\mathcal{E}$. The motion is allowed only in regions in which $\mathcal{E} \geq V_{eff}$ and thus, intersection of the dotted lines with the effective potential curve correspond to return points in the trajectory. The values $\mathcal{E}_1$ and $\mathcal{E}_2$ correspond to open trajectories (hyperbolic and parabolic motion). The value $\mathcal{E}_3 < 0$ corresponds to elliptic motion and shows two return points: the pericenter r_p (minimum distance) and the apocenter r_a (maximum distance). The particular value of energy $\mathcal{E}_4 = V_{eff}^{min}$ corresponding to the minimum of the effective potential energy, describes a circular orbit with radius r_c.

The analytic form of the trajectories can be found by using φ as the independent variable instead of the time, i.e. considering $r = r(\varphi)$. It is also usual to introduce a new variable, $u(\varphi) = \frac{1}{r(\varphi)}$.

Exercise 1.4

Differential Equation of the Trajectory
Show that the second order differential equation of the trajectory obtained from equation (1.27) and the new variable $u(\varphi)$ is

$$u'' + u = \frac{GM}{\ell^2}, \tag{1.30}$$

where prime denotes differentiation with respect to φ.

The solution of the differential equation (1.30) are the conic sections

$$r(\varphi) = \frac{p}{1 + e\cos(\varphi - \omega)} \tag{1.31}$$

where

$$p = \frac{\ell^2}{GM} \tag{1.32}$$

is known as the *semi-latus rectum* of the orbit† and e and ω are constants of integration.The constant e defines to the eccentricity of the trajectory and, since we are interested only in elliptic orbits, we will assume that $0 \leq e < 1$. On the other hand, the constant ω defines the angle φ at which r achieves a minimum, and therefore it is called the *longitude of the pericenter*. In fact, when $\varphi = \omega$ we obtain the pericenter radius

$$r_p = \frac{p}{1+e} \tag{1.33}$$

while $\varphi = \omega + \pi$ gives the maximum value of the radius or *apocenter*,

$$r_a = \frac{p}{1-e}. \tag{1.34}$$

From these two values we obtain the *semi-major axis* as

$$a = \frac{1}{2}\left(r_p + r_a\right) = \frac{p}{1-e^2}. \tag{1.35}$$

Using equations (1.31) and (1.21) we can write the radial velocity as

$$\dot{r} = \sqrt{\frac{GM}{p}}e\sin(\varphi - \omega) \tag{1.36}$$

and replacing this expression in equation (1.25) gives the energy per unit mass for the elliptic motion as

$$\mathcal{E} = -\frac{GM(1-e^2)}{2p} = -\frac{GM}{2a}. \tag{1.37}$$

Similarly, the angular momentum vector per unit mass in equation (1.21) can be written as

$$\boldsymbol{\ell} = \sqrt{GMp}\mathbf{k}. \tag{1.38}$$

†The name *semi-latus rectum* comes from the latin words *semi* (half), *latus* (side) and *rectum* (straight). Geometrically, the *latus rectum* is defined as the length of a chord passing through one focus and perpendicular to the major axis. Therefore, the *semi-latus rectum* is one half of this length.

Exercise 1.5

Magnitude of the Laplace-Runge-Lenz vector
Show that the Laplace-Runge-Lenz vector for elliptic trajectories can be written as

$$\boldsymbol{e} = e\left[\cos(\varphi - \omega)\boldsymbol{n} - \sin(\varphi - \omega)\boldsymbol{\lambda}\right] \tag{1.39}$$

and therefore, this vector always points in the direction of the pericenter of the orbit and its magnitude is equal to the eccentricity of the orbit, $|\boldsymbol{e}| = e$.

1.1.3 Time Dependence

A formal integration of the energy equation (1.27) solves the radial problem as a function of time,

$$\int_{r_p}^{r} \frac{dr}{\sqrt{2[\mathcal{E} - V_{eff}(r)]}} = t - t_p, \tag{1.40}$$

where t_p corresponds to the *time of pericenter passage*. Similarly, using equation (1.21), it is possible to write the angular velocity as

$$\dot{\varphi} = \sqrt{\frac{GM}{p^3}}\left[1 + e\cos(\varphi - \omega)\right]^2 \tag{1.41}$$

and a formal integration of this expression gives

$$\sqrt{\frac{p^3}{GM}} \int_{\omega}^{\varphi} \frac{d\varphi}{\left[1 + e\cos(\varphi - \omega)\right]^2} = t - t_p. \tag{1.42}$$

Note that the lower limit in the integral corresponds again to the passage through the pericenter, located at $\varphi = \omega$. Although these expressions formally solve the motion problem with time dependence, it is usual to use other approach, based on the so-called *Kepler's equation*, to obtain the position of the particle as function of time.

Kepler's Equation

We begin this approach by defining two new variables, the *true anomaly* $f = \varphi - \omega$ and the *eccentric anomaly* E. Figure 1.3 illustrates these angular variables and from the geometry of the plot, it is possible to show that these variables are related through the relations

$$\cos E = \frac{\cos f + e}{1 + e\cos f} \tag{1.43a}$$

$$\sin E = \frac{\sqrt{1 - e^2}\sin f}{1 + e\cos f}. \tag{1.43b}$$

Equivalently, these expressions can be written as

$$\tan\frac{f}{2} = \sqrt{\frac{1 + e}{1 - e}}\tan\frac{E}{2} \tag{1.44}$$

which clearly states that at the pericenter, $f = 0$, the eccentric anomaly is $E = 0$. Also note that $e = 0$ implies that $f = E$ for circular motion.

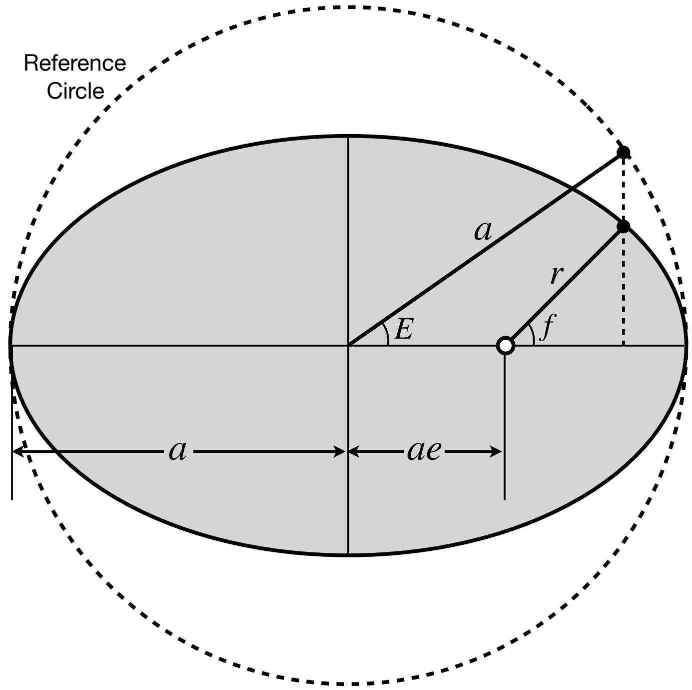

Figure 1.3: Elliptic orbit in the 2-body problem. The plot indicates some of the geometrical quantities such as the semi-major axis a, the relative distance vector $\mathbf{r}$ and the true anomaly f. It is also shown the *reference circle*, which coincides with the elliptic orbit both at the pericenter and at the apocenter, and the *Eccentric Anomaly*, E.

Exercise 1.6

Eccentric and True Anomalies

Show that equations (1.43) can be inverted to obtain

$$\cos f = \frac{\cos E - e}{1 - e\cos E} \tag{1.45a}$$

$$\sin f = \frac{\sqrt{1-e^2}\sin E}{1 - e\cos E}, \tag{1.45b}$$

and that

$$\frac{df}{dE} = \frac{\sqrt{1-e^2}}{1 - e\cos E} \tag{1.46a}$$

$$\frac{dE}{df} = \frac{\sqrt{1-e^2}}{1 + e\cos f}. \tag{1.46b}$$

In terms of the eccentric anomaly, we can write an alternatives set of equations describing the orbit.In particular, equations (1.31), 1.30) and (1.21) become

$$r = a(1 - e\cos E) \tag{1.47a}$$

$$\dot{r} = \sqrt{\frac{GM}{a}}\frac{e\sin E}{1 - e\cos E} \tag{1.47b}$$

$$\dot{E} = \sqrt{\frac{GM}{a^3}}\frac{1}{1 - e\cos E}. \tag{1.47c}$$

Equation (1.47c) is integrated to obtain the transcendental *Kepler's Equation*,

$$l = \sqrt{\frac{GM}{a^3}} \left(t - t_p\right) = \left(E - e \sin E\right), \tag{1.48}$$

where we defined the *mean anomaly*, l. Using this equation we can obtain the position at each time as follows:

1. Given a time t, we invert Kepler's equation to obtain the eccentric anomaly $E(t)$.
2. Then, we use equation (1.44) to obtain the true anomaly f which immediately gives the angular position $\varphi(t)$.
3. The radial position $r(t)$ is obtained via equation (1.31).
4. Additionally, we can get the components of the velocity $\dot{r}(t)$ and $\dot{\varphi}(t)$ using equations (1.36) and ((1.41)).

Kepler's equation also permit us to calculate the period of the orbit because when E goes from 0 to 2π, time incresases from t_p to the period T. This gives

$$T = 2\pi \sqrt{\frac{a^3}{GM}}, \tag{1.49}$$

which is exactly Kepler's third law. Using this expression together with Kepler's equation (1.48) let us write the mean anomaly as

$$l = n\left(t - t_p\right) \tag{1.50}$$

where

$$n = \frac{2\pi}{T} = \sqrt{\frac{GM}{a^3}} \tag{1.51}$$

is called the *mean motion*. It is also usual to use a reference time or *epoch*, $t_0 \neq t_p$ such that the mean anomaly is written as the linear function of time

$$l = l_0 + n(t - t_0) \tag{1.52}$$

where l_0 is the mean anomaly at the epoch. Note that these expressions imply that for circular motion, $e = 0$, all the anomalies coincide, $f = E = l$[‡].

1.2 Orbital Elements

Until now, we've been working in the orbit plane and we've defined two basis. The first one corresponds to a barycentric inertial frame with basis vectors $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ while the second uses a time-dependent basis given by the vectors $\{\mathbf{n}, \boldsymbol{\lambda}, \mathbf{k}\}$. Now we will consider a more general coordinate system (x, y, z) in which the orbit is no longer in the xy plane (which will be called now the *fundamental plane*) and we will denote the basis for this new system as $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$. It is clear that, we will need additional information to completely define the orbit in three-dimensional space. As shown in Figure 1.4, we will introduce the *inclination angle*, ι, as the angle between the orbital plane and the fundamental plane[§]. The line of intersection between the orbital plane and the fundamental plane is called the *line of nodes* and the point along this line in which the trajectory cuts the fundamental plane from below is called the *ascending node*. Also in Figure 1.4, it can be

[‡]In fact, the name *anomaly* comes from the early belief that circular motion is perfect. Therefore, deviations from perfection, such as elliptic motion, were considered *anomalies*

[§]The inclination can also be defined as the angle between the z axis and the $\mathbf{k}$ vector defined before.

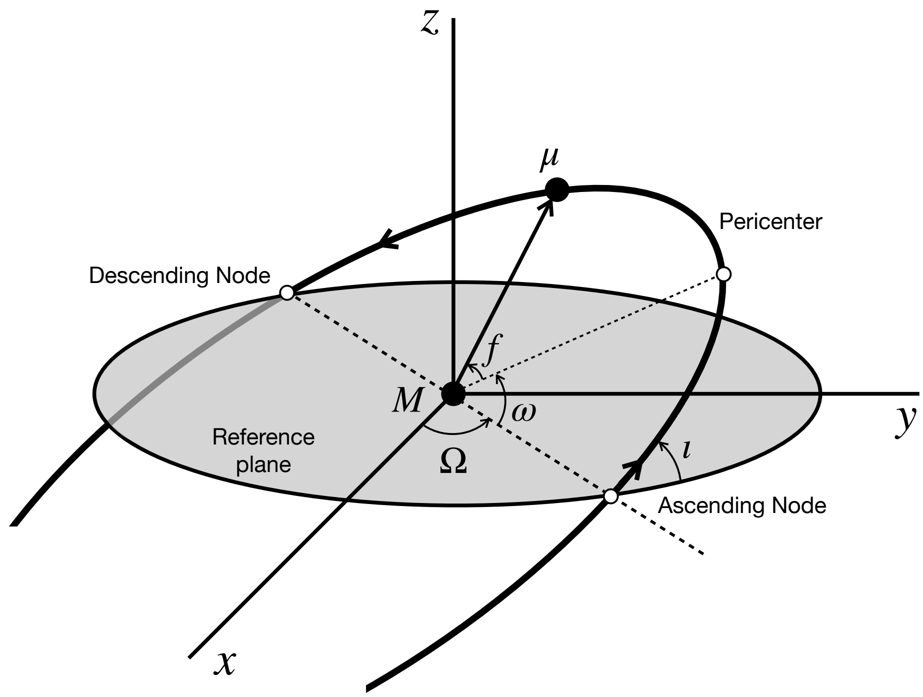

Figure 1.4: Elliptic Orbit in the 2-body problem shown in 3-dimensional space. The orbital elements ι, Ω and ω are shown.

seen the angle between the x axis and the line of nodes, which is called the *longitude of the ascending node*, Ω.

Thus, there are six *orbital elements* or parameters that define completely the orbit,

- Principal Elements:

  1. p : *Semi-latus rectum* or a : *Semi-major axis*
  2. e : *Eccentricity*

- Positional Elements:

  3. ω : *Argument of the pericenter*
  4. ι : *Inclination*
  5. Ω : *Longitude of the ascending node*

- Time element:

  6. T : *Orbital period* or l_0 : *Mean anomaly at the epoch.*

The transformation of a vector $\mathbf{A}$, with components $\mathbf{A} = A^1\mathbf{i} + A^2\mathbf{j} + A^3\mathbf{k}$ in the barycentric frame, to the fundamental coordinate system, in which its components will be denoted $\mathbf{A} = A^x\mathbf{e}_x + A^y\mathbf{e}_y + A^z\mathbf{e}_z$, can be represented as

$$\begin{pmatrix} A^x \\ A^y \\ A^z \end{pmatrix} = R_3(\Omega)R_1(\iota)R_3(\omega)\begin{pmatrix} A^1 \\ A^2 \\ A^3 \end{pmatrix}, \tag{1.53}$$

and involves three rotations matrices,

$$R_3(\omega) = \begin{pmatrix} \cos\omega & -\sin\omega & 0 \\ \sin\omega & \cos\omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.54a}$$

$$R_1(\iota) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\iota & -\sin\iota \\ 0 & \sin\iota & \cos\iota \end{pmatrix} \tag{1.54b}$$

$$R_3(\Omega) = \begin{pmatrix} \cos\Omega & -\sin\Omega & 0 \\ \sin\Omega & \cos\Omega & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{1.54c}$$

For example, the vectors of the time dependent basis (1.18) written now in terms of the true anomaly,

$$\begin{cases} \mathbf{n} = \cos f \mathbf{i} + \sin f \mathbf{j} \\ \boldsymbol{\lambda} = -\sin f \mathbf{i} + \cos f \mathbf{j} \\ \mathbf{k} \end{cases} \tag{1.55}$$

can be transformed using (1.53) to obtain their components in the fundamental coordinate system,

$$\begin{aligned} \mathbf{n} = {} & [\cos\Omega\cos(f+\omega) - \cos\iota\sin\Omega\sin(f+\omega)]\,\mathbf{e}_x \\ & + [\sin\Omega\cos(f+\omega) + \cos\iota\cos\Omega\sin(f+\omega)]\,\mathbf{e}_y \\ & + [\sin\iota\sin(f+\omega)]\,\mathbf{e}_z \end{aligned} \tag{1.56a}$$

$$\begin{aligned} \boldsymbol{\lambda} = {} & [-\cos\Omega\sin(f+\omega) - \cos\iota\sin\Omega\cos(f+\omega)]\,\mathbf{e}_x \\ & + [-\sin\Omega\sin(f+\omega) + \cos\iota\cos\Omega\cos(f+\omega)]\,\mathbf{e}_y \\ & + [\sin\iota\cos(f+\omega)]\,\mathbf{e}_z \end{aligned} \tag{1.56b}$$

$$\mathbf{k} = \sin\iota\sin\Omega\mathbf{e}_x - \sin\iota\cos\Omega\mathbf{e}_y + \cos\iota\mathbf{e}_z. \tag{1.56c}$$

The first of these relations gives immediately the components of the separation vector, because $\mathbf{x} = r\mathbf{n}$ with $r = \frac{p}{1+e\cos f}$. Denoting the components of this vector in the fundamental system as $\mathbf{x} = (x, y, z)$, we have

$$\begin{cases} x &= r\,[\cos\Omega\cos(f+\omega) - \cos\iota\sin\Omega\sin(f+\omega)] \\ y &= r\,[\sin\Omega\cos(f+\omega) + \cos\iota\cos\Omega\sin(f+\omega)] \\ z &= r\sin\iota\sin(f+\omega), \end{cases} \tag{1.57}$$

and differentiating with respect to time we obtain the components of the relative velocity, $\dot{\mathbf{x}}$, in the fundamental frame,

$$\begin{cases} \dot{x} &= -\sqrt{\frac{GM}{p}}\,[\cos\Omega\,(\sin(f+\omega) + e\sin\omega) + \cos\iota\sin\Omega\,(\cos(f+\omega) + e\cos\omega)] \\ \dot{y} &= -\sqrt{\frac{GM}{p}}\,[\sin\Omega\,(\sin(f+\omega) + e\sin\omega) - \cos\iota\cos\Omega\,(\cos(f+\omega) + e\cos\omega)] \\ \dot{z} &= \sqrt{\frac{GM}{p}}\sin\iota\,[\cos(f+\omega) + e\cos\omega]\,. \end{cases} \tag{1.58}$$

Another interesting transformation is to obtain the components of the eccentricity vector in the fundamental coordinate system. Replacing the basis vectors (1.56) in equation (1.39) gives

$$\begin{aligned} \mathbf{e} = e\,[\, & (\cos\Omega\cos\omega - \cos\iota\sin\Omega\sin\omega)\,\mathbf{e}_x \\ & + (\sin\Omega\cos\omega + \cos\iota\cos\Omega\sin\omega)\,\mathbf{e}_y \\ & + (\sin\iota\sin\omega)\,\mathbf{e}_z\,]\,. \end{aligned} \tag{1.59}$$

All these equations let us write the set of orbital elements through the relations

$$p = \frac{\ell^2}{GM} \tag{1.60a}$$

$$a = \frac{p}{1 - e^2} \tag{1.60b}$$

$$e = |\mathbf{e}| \tag{1.60c}$$

$$\cos\iota = \mathbf{k} \cdot \mathbf{e}_z \tag{1.60d}$$

$$\sin\iota \sin\Omega = \mathbf{k} \cdot \mathbf{e}_x \tag{1.60e}$$

$$\sin\iota \sin\omega = \frac{\mathbf{e} \cdot \mathbf{e}_z}{e}. \tag{1.60f}$$

Chapter 2

Perturbation Theory

2.1 Perturbation of the Orbital Elements

In the previous chapter we've solved the 2-body problem in Newtonian mechanics and the solution involves six initial conditions, e.g. components of the initial relative position and initial relative velocity, that we have replaced by the six orbital elements, which will be denoted as $\vartheta^a = \{a, e, \iota, \Omega, \omega, l_0\}$. Hence, the solution of the Newtonian 2-body problem is formally expressed by the functions

$$\mathbf{x}(t) = \mathbf{x}_N(t, \vartheta^a) \tag{2.1a}$$

$$\dot{\mathbf{x}}(t) = \dot{\mathbf{x}}_N(t, \vartheta^a) \tag{2.1b}$$

where the Newtonian vectors satisfy the equations of motion

$$\dot{\mathbf{x}}_N = \frac{d\mathbf{x}_N}{dt} \tag{2.2a}$$

$$\ddot{\mathbf{x}}_N = \frac{d\dot{\mathbf{x}}_N}{dt} = -\frac{GM}{r_N^3}\mathbf{x}_N\,, \tag{2.2b}$$

where $M = m_1 + m_2$ is the total mass of the system.

In this chapter, we will introduce a perturbing force per unit mass $\delta\mathbf{A} = \frac{\mathbf{F}}{\mu}$ which will account for small external influences on the 2-body system. Using the time dependent basis $(\mathbf{n}, \boldsymbol{\lambda}, \mathbf{k})$, introduced in (1.18), we write the equation of motion, including the perturbing force, as

$$\ddot{\mathbf{x}} = -\frac{GM}{r^2}\mathbf{n} + \delta\mathbf{A}, \tag{2.3}$$

where $\mathbf{n} = \frac{\mathbf{x}}{r}$ is the unit vector in the relative radial direction and the components of $\delta\mathbf{A}$ are written as

$$\delta\mathbf{A} = \mathcal{R}\mathbf{n} + \mathcal{T}\boldsymbol{\lambda} + \mathcal{W}\mathbf{k} \tag{2.4}$$

or

$$\begin{cases} \mathcal{R} &= \delta\mathbf{A}\cdot\mathbf{n} \\ \mathcal{T} &= \delta\mathbf{A}\cdot\boldsymbol{\lambda} \\ \mathcal{W} &= \delta\mathbf{A}\cdot\mathbf{k}. \end{cases} \tag{2.5}$$

We will assume that $\delta\mathbf{A}$ is small compared with the Newtonian gravitational force and therefore, the description in the previous chapter is approximately correct. The effect of the perturbing force will be that the orbit is no longer a fixed conic in space. Hence, the approach that we will use to obtain the perturbed solution is to use the Newtonian solution (2.1) considering that the orbital elements are now functions of time, $\vartheta^a = \vartheta^a(t)$. The perturbed solution will have the form

$$\mathbf{x}(t) = \mathbf{x}_N(t, \vartheta^a(t)) \tag{2.6a}$$
$$\dot{\mathbf{x}}(t) = \dot{\mathbf{x}}_N(t, \vartheta^a(t)). \tag{2.6b}$$

In order to obtain the differential equations defining the orbital elements, we note that the angular momentum per unit mass, $\boldsymbol{\ell} = \ell\mathbf{k} = \mathbf{x} \times \dot{\mathbf{x}}$, is now a vector that changes with time,

$$\frac{d\boldsymbol{\ell}}{dt} = \mathbf{x} \times \delta\mathbf{A}. \tag{2.7}$$

Using the perturbation in equation (2.4), we get

$$\frac{d\boldsymbol{\ell}}{dt} = -r\mathcal{W}\boldsymbol{\lambda} + r\mathcal{T}\mathbf{k}. \tag{2.8}$$

from which we conclude that

$$\frac{d\ell}{dt} = r\mathcal{T} \tag{2.9a}$$
$$\ell\frac{d\mathbf{k}}{dt} = -r\mathcal{W}\boldsymbol{\lambda}. \tag{2.9b}$$

Exercise 2.1

Laplace-Runge-Lenz variation in time
Consider the eccentricity vector given in equations (1.22) and (1.39),

$$\boldsymbol{e} = \frac{\dot{\boldsymbol{x}} \times \boldsymbol{\ell}}{GM} - \boldsymbol{n} = e\left[\cos f\boldsymbol{n} - \sin f\boldsymbol{\lambda}\right]. \tag{2.10}$$

Show that its time derivative is

$$\begin{aligned}\frac{d\boldsymbol{e}}{dt} &= \frac{1}{GM}\left[\delta\boldsymbol{A} \times \boldsymbol{\ell} + \dot{\boldsymbol{x}} \times (\boldsymbol{x} \times \delta\boldsymbol{A})\right] \\ &= \frac{1}{GM}\left[2\ell\mathcal{T}\boldsymbol{n} - (\ell\mathcal{R} + r\dot{r}\mathcal{T})\boldsymbol{\lambda} - r\dot{r}\mathcal{W}\boldsymbol{k}\right].\end{aligned} \tag{2.11}$$

Also show that this equation implies the relations

$$\frac{de}{dt} = \frac{1}{GM}\left[\ell\sin f\mathcal{R} + (2\ell\cos f + r\dot{r}\sin f)\,\mathcal{T}\right] \tag{2.12}$$

and

$$\begin{aligned}\frac{d}{dt}\left(\frac{\boldsymbol{e}}{e}\right) &= \frac{1}{GMe}\left[-\ell\cos f\mathcal{R} + (2\ell\sin f - r\dot{r}\cos f)\,\mathcal{T}\right]\left[\cos f\boldsymbol{n} + \sin f\boldsymbol{\lambda}\right] \\ &\quad - \frac{r\dot{r}\mathcal{W}}{GMe}\boldsymbol{k}.\end{aligned} \tag{2.13}$$

2.1.1 Osculating Elements

Now we are ready to obtain the time dependence of the orbital elements. Differentiating equations (1.60) and using the above results, we obtain the differential equations known as *osculating elements* or as *Lagrange's* *planetary equations*,

$$\frac{da}{dt} = \frac{2}{n\sqrt{1-e^2}}\left[e\sin f\mathcal{R} + (1+e\cos f)\mathcal{T}\right] \tag{2.14a}$$

$$\frac{de}{dt} = \frac{\sqrt{1-e^2}}{na}\left[\sin f\mathcal{R} + \frac{2\cos f + e\left(1+\cos^2 f\right)}{1+e\cos f}\mathcal{T}\right] \tag{2.14b}$$

$$\frac{d\iota}{dt} = \frac{\sqrt{1-e^2}}{na}\frac{\cos(f+\omega)}{1+e\cos f}\mathcal{W} \tag{2.14c}$$

$$\frac{d\Omega}{dt} = \frac{\sqrt{1-e^2}}{na}\frac{\sin(f+\omega)}{(1+e\cos f)\sin\iota}\mathcal{W} \tag{2.14d}$$

$$\frac{d\omega}{dt} = \frac{\sqrt{1-e^2}}{nae}\left[-\cos f\mathcal{R} + \frac{2+e\cos f}{1+e\cos f}\sin f\mathcal{T} - e\frac{\sin(f+\omega)}{1+e\cos f}\cot\iota\mathcal{W}\right] \tag{2.14e}$$

$$\frac{dl_0}{dt} = -\sqrt{1-e^2}\left[\frac{d\omega}{dt} + \cos\iota\frac{d\Omega}{dt} - \frac{2p}{na^2(1+e\cos f)}\mathcal{T}\right], \tag{2.14f}$$

where $n = \sqrt{\frac{GM}{a^3}}$ is the mean motion. We also include into this set the differential equation for the semi-latus rectum,

$$\frac{dp}{dt} = \frac{2}{n}\frac{(1-e^2)^{3/2}}{1+e\cos f}\mathcal{T}. \tag{2.15}$$

From the relation

$$\cos f = \frac{\mathbf{n}\cdot\mathbf{e}}{e} \tag{2.16}$$

it is possible to obtain, after differentiation, the time evolution equation for the true anomaly,

$$\frac{df}{dt} = \frac{n\,(1+e\cos f)^2}{\sqrt{(1-e^2)^3}}\left[1 + \frac{(1-e^2)^2}{n^2 ae}\left(\frac{\cos f}{(1+e\cos f)^2}\mathcal{R} - \frac{2+e\cos f}{(1+e\cos f)^3}\sin f\mathcal{T}\right)\right]. \tag{2.17}$$

Similarly, the time variation of the eccentric anomaly is obtained from Kepler's equation (1.48),

$$\frac{dE}{dt} = \frac{na}{r} - \frac{r}{a\sqrt{1-e^2}}\left[\frac{d\omega}{dt} + \cos\iota\frac{d\Omega}{dt} + \frac{\sin f}{1-e^2}\frac{de}{dt}\right]. \tag{2.18}$$

2.1.2 Secular Changes of the Orbital Elements

The orbital elements may present periodic and non-periodic changes. The periodic contributions are not interesting because they average to zero after some revolutions. On the other hand, the changes in the orbital elements that doesn't average but accumulate over time are called *secular changes* and are important because, after some time interval, they can produce a large departure from the initial unperturbed orbit. In order to obtain the secular changes in the orbital elements ϑ^a, we integrate over one period of the orbit,

$$\Delta\vartheta^a = \int_0^T \frac{d\vartheta^a}{dt}dt. \tag{2.19}$$

The time integration over one period can be replaced by an integration with the true anomaly using (2.17) to change variables,

$$\Delta\vartheta^a = \int_0^{2\pi} \frac{d\vartheta^a}{df}df. \tag{2.20}$$

Assuming that both the perturbing force and the changes in the orbital elements are small, the first order approximation of the osculating elements as function of the true anomaly gives the equation

$$\frac{da}{df} = \frac{2(1-e^2)}{n}\left[\frac{e\sin f}{(1+e\cos f)^2}\mathcal{R} + \frac{1}{1+e\cos f}\mathcal{T}\right] \tag{2.21a}$$

$$\frac{de}{df} = \frac{(1-e^2)^2}{n^2 a}\left[\frac{\sin f}{(1+e\cos f)^2}\mathcal{R} + \frac{2\cos f + e\left(1+\cos^2 f\right)}{(1+e\cos f)^3}\mathcal{T}\right] \tag{2.21b}$$

$$\frac{d\iota}{df} = \frac{(1-e^2)^2}{n^2 a}\frac{\cos(f+\omega)}{(1+e\cos f)^3}\mathcal{W} \tag{2.21c}$$

$$\frac{d\Omega}{df} = \frac{(1-e^2)^2}{n^2 a}\frac{\sin(f+\omega)}{(1+e\cos f)^3\sin\iota}\mathcal{W} \tag{2.21d}$$

$$\frac{d\omega}{df} = \frac{(1-e^2)^2}{n^2 ae}\left[-\frac{\cos f}{(1+e\cos f)^2}\mathcal{R} + \frac{2+e\cos f}{(1+e\cos f)^3}\sin f\mathcal{T} - e\frac{\sin(f+\omega)}{(1+e\cos f)^3}\cot\iota\mathcal{W}\right] \tag{2.21e}$$

$$\frac{dl_0}{df} = -\sqrt{1-e^2}\left(\frac{d\omega}{df} + \cos\iota\frac{d\Omega}{df}\right) + \frac{2(1-e^2)^3}{n^2 a(1+e\cos f)^3}\mathcal{T}, \tag{2.21f}$$

together with a similar equation for the semi-latus rectum,

$$\frac{dp}{df} = \frac{2}{n^2}\frac{(1-e^2)^3}{(1+e\cos f)^3}\mathcal{T}. \tag{2.22}$$

2.2 Examples

In order to illustrate the perturbation theory treatment, we will present two of the most important examples: the precession of the orbit of Mercury under the influence of another planet and the Kozai-Lidov effect on the orbit of an asteroid. Both examples consider the gravitational field of a third body as the perturbing force.

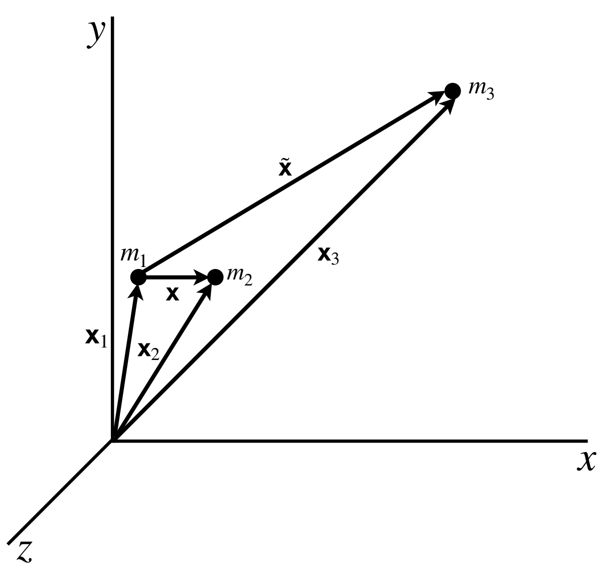

Figure 2.1: Two point particles moving under the mutual gravitational influence and perturbed by a third body. Using the inertial frame shown in the figure, the position vectors of the particles are $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$.

The initial two-body problem is formed by the particles with masses m_1 and m_2, while m_3 represents the mass of a perturbing third body. As before, $\mathbf{x} = \mathbf{x}_2 - \mathbf{x}_1$ represents the relative position of m_2 with respect to $m_!$ and we introduce $\tilde{\mathbf{x}} = \mathbf{x}_3 - \mathbf{x}_1$ as the relative position of the third body with respect to m_1, see Figure 2.1. The equations of motion for particles m_1 and m_2 are,

$$\ddot{\mathbf{x}}_1 = Gm_2 \frac{\mathbf{x}}{r^3} + Gm_3 \frac{\tilde{\mathbf{x}}}{\tilde{r}^3} \tag{2.23a}$$

$$\ddot{\mathbf{x}}_2 = -Gm_1 \frac{\mathbf{x}}{r^3} + Gm_3 \frac{\tilde{\mathbf{x}} - \mathbf{x}}{|\tilde{\mathbf{x}} - \mathbf{x}|^3}. \tag{2.23b}$$

The relative acceleration of the 2-Body system is

$$\ddot{\mathbf{x}} = -GM \frac{\mathbf{x}}{r^3} + Gm_3 \left[\frac{\tilde{\mathbf{x}} - \mathbf{x}}{|\tilde{\mathbf{x}} - \mathbf{x}|^3} - \frac{\tilde{\mathbf{x}}}{\tilde{r}^3} \right], \tag{2.24}$$

from which we identify the perturbing force (per unit mass) $\delta\mathbf{A}$ produced by the third body as

$$\delta\mathbf{A} = Gm_3 \left[\frac{\tilde{\mathbf{x}} - \mathbf{x}}{|\tilde{\mathbf{x}} - \mathbf{x}|^3} - \frac{\tilde{\mathbf{x}}}{\tilde{r}^3} \right]. \tag{2.25}$$

In order to incorporate this force as a perturbation to the 2-Body system, we introduce the normalized vectors $\mathbf{n} = \frac{\mathbf{x}}{r}$ and $\tilde{\mathbf{n}} = \frac{\tilde{\mathbf{x}}}{\tilde{r}}$ and we will assume that $\tilde{r} \gg r$. Hence, we expand the perturbing force in powers of $\frac{r}{\tilde{r}}$ to obtain

$$\delta\mathbf{A} = -\frac{Gm_3 r}{\tilde{r}^3} \left[\mathbf{n} - 3(\mathbf{n} \cdot \tilde{\mathbf{n}})\tilde{\mathbf{n}} + \mathcal{O}\left(\frac{r}{\tilde{r}}\right) \right]. \tag{2.26}$$

Using the time dependent basis $(\mathbf{n}, \boldsymbol{\lambda}, \mathbf{k})$ in the plane of the 2-Body system's orbit, we obtain the components of the perturbing force as given in equations (2.5),

$$\mathcal{R} = -\frac{Gm_3 r}{\tilde{r}^3} \left[1 - 3(\mathbf{n} \cdot \tilde{\mathbf{n}})^2 + \mathcal{O}\left(\frac{r}{\tilde{r}}\right) \right] \tag{2.27a}$$

$$\mathcal{T} = 3\frac{Gm_3 r}{\tilde{r}^3} (\mathbf{n} \cdot \tilde{\mathbf{n}})(\tilde{\mathbf{n}} \cdot \boldsymbol{\lambda}) + \mathcal{O}\left(\frac{r}{\tilde{r}}\right) \tag{2.27b}$$

$$\mathcal{W} = 3\frac{Gm_3 r}{\tilde{r}^3} (\mathbf{n} \cdot \tilde{\mathbf{n}})(\tilde{\mathbf{n}} \cdot \mathbf{k}) + \mathcal{O}\left(\frac{r}{\tilde{r}}\right). \tag{2.27c}$$

2.2.1 Perturbation by a Third Body in the Solar System.

One of the more important examples of application of the perturbation theory in celestial mechanics involves the motion the planets of the Solar System and is known as the *advance of the perihelion*. This effect was first recognized for Mercury by Urbain Le Verrier, back in 1859, by reanalyzing the observations of transits of Mercury dated from 1697 to 1848. Using Newtonian mechanics, together with perturbation theory, it was shown that some circumstances in the Solar System, such as the presence of other planets and the effect of solar oblateness, may cause the advance of the perihelion. It is clear that all of the planets experience the same effect but, due to the involved masses, distances and periods (see Table A.1), the advance of the perihelion is most notorious for the orbit of Mercury [4].

Now, we will calculate the advance of the perihelion by considering a planet (say Mercury) moving in the Solar System under the influence of the Sun and including a third body (for example Jupiter) affecting its orbital motion. The equations of motion are given by (2.24) with the perturbing force (2.27). Its important to note that the assumption $\tilde{r} \gg r$ is justified in this example because of the data given in Table A.1. The values of the involved masses show that the second major gravitational influence on Mercury is produced by Jupiter and thus, the corresponding ratio of distances is $\frac{r}{\tilde{r}} \sim 10^{-2}$.

In order to simplify the components of the force in equations (2.27), we will consider two more assumption for the system,

- First, the orbit of the third body is approximated by a circumference with radius equal to the value of the semi-major axis, $\tilde{r} = \tilde{a}$. As a result of this assumption, the orbital angular frequency of the perturbing body has the Keplerian value

$$\frac{2\pi}{\tilde{T}} = \sqrt{\frac{G(m_1 + m_3)}{\tilde{r}^3}}, \tag{2.28}$$

and therefore its true anomaly is equal to its mean anomaly,

$$\tilde{f} = \sqrt{\frac{G(m_1 + m_3)}{\tilde{r}^3}} t. \tag{2.29}$$

Since the Keplerian period of the planet is $T = 2\pi \left[\frac{G(m_1+m_2)}{r^3}\right]^{-1/2}$, after one period of the planet, the change in the true anomaly of the perturbing body is

$$\Delta\tilde{f} = \sqrt{\frac{G(m_1 + m_3)}{\tilde{r}^3}} T = 2\pi \sqrt{\frac{m_1 + m_3}{m_1 + m_2}} \left(\frac{r}{\tilde{r}}\right)^{3/2} \ll 1, \tag{2.30}$$

where we have applied the assumption $\tilde{r} \gg r$. This result means that the third body doesn't changes appreciably its position after one orbit of Mercury.

- Second, the orbit of the planet and the orbit of the third body lie in the same plane. This assumption is understood from the data in Table A.1 where, for example, the inclination of Jupiter's orbit with respect to Mercury's orbit is very small (about 5.7°). We will choose the orientation of the fundamental frame $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$ such that $\mathbf{e}_x$ and $\mathbf{e}_y$ both lie in the orbital plane and $\mathbf{e}_x$ points in the direction of the pericenter. Then, the time dependent basis is

$$\mathbf{n} = \cos(f + \omega)\mathbf{e}_x + \sin(f + \omega)\mathbf{e}_y \tag{2.31a}$$
$$\boldsymbol{\lambda} = -\sin(f + \omega)\mathbf{e}_x + \cos(f + \omega)\mathbf{e}_y \tag{2.31b}$$
$$\mathbf{k} = \mathbf{e}_z. \tag{2.31c}$$

Similarly, the unitary vector $\tilde{n}$, indicating the direction of the third body, is given by

$$\tilde{n} = \cos\tilde{f}\mathbf{e}_x + \sin\tilde{f}\mathbf{e}_y. \tag{2.32}$$

As a result of these assumptions, the components of the perturbing force become

$$\mathcal{R} = -\frac{Gm_3 r}{\tilde{r}^3}\left[1 - 3\cos^2(f + \omega - \tilde{f})\right] \tag{2.33a}$$
$$\mathcal{T} = -3\frac{Gm_3 r}{\tilde{r}^3}\sin(f + \omega - \tilde{f})\cos(f + \omega - \tilde{f}) \tag{2.33b}$$
$$\mathcal{W} = 0. \tag{2.33c}$$

Replacing these components in equations (2.21) and integrating with respect to f over one period of the motion of the planet, assuming that $\tilde{f}$ is a constant, we obtain the variation of the orbital elements

$$\Delta a = 0 \tag{2.34a}$$
$$\Delta e = \frac{15\pi}{2}\frac{Gm_3}{n^2\tilde{r}^3} e\sqrt{1-e^2}\sin 2(\omega - \tilde{f}) \tag{2.34b}$$
$$\Delta \iota = 0 \tag{2.34c}$$
$$\Delta \Omega = 0 \tag{2.34d}$$
$$\Delta \omega = \frac{3\pi}{2}\frac{Gm_3}{n^2\tilde{r}^3}\sqrt{1-e^2}\left[1 + 5\cos 2(\omega - \tilde{f})\right] \tag{2.34e}$$
$$\Delta p = -15\pi\frac{Gm_3 a}{n^2\tilde{r}^3} e^2\sqrt{1-e^2}\sin 2(\omega - \tilde{f}). \tag{2.34f}$$

Precession of the Perihelia.

Equations (2.34) give the change of the orbital elements per obit of the planet. It is easy to see that the longitude of the perihelion is the only one that presents a secular contribution*. It corresponds to a shift in ω of

$$(\Delta\omega)_{sec} = \frac{3\pi}{2}\frac{Gm_3}{n^2\tilde{r}^3}\sqrt{1-e^2} = \frac{3\pi}{2}\frac{m_3}{m}\left(\frac{a}{\tilde{r}}\right)^3\sqrt{1-e^2} \tag{2.35}$$

per orbit. As was discussed in the beginning of this section, the greater effect of advance of the perihelion for planets in the Solar System is obtained for Mercury. Using the data from Table A.1, the secular contribution for the advance of the perihelion of Mercury produced by all of the planets of the Solar System, within this approximation, give the values shown in Table 2.1. We also include a column with the calculated values obtained by Le Verrier [9, 10] for comparison. The evident difference between our calculations and Le Verrier results (specially for Earth and Venus' contributions), are due to the approximation $\tilde{r} \gg r$ in equations (2.27). To obtain a better result for the perihelion advance we need to include higher terms in $\frac{r}{\tilde{r}}$.

Planet	Eccentricity	Inclination (w.r.t. Mercury's orbit)	Perihelion shift (radians/orbit)	Perihelion shift (arcsec/century)	Perihelion shift Le Verrier (arcsec/century)
Venus	0.006772	3.6104°	1.7996×10^{-6}	148.0730	280.6
Earth	0.0167086	7.005°	8.0323×10^{-7}	68.7656	83.9
Mars	0.0934	5.155°	2.4297×10^{-8}	2.0801	2.6
Jupiter	0.0489	5.702°	1.8109×10^{-6}	155.0343	152.6
Saturn	0.0565	4.52°	8.6853×10^{-8}	7.4356	7.2
Uranus	0.046381	6.232°	1.6450×10^{-9}	0.1408	0.1
Neptune	0.009456	5.2370°	5.0456×10^{-10}	0.0432	–
Total				381.5727	526.7

Table 2.1: Perihelion Advance of Mercury due to the planets in the Solar System. We include a column with the information about the eccentricity of the orbit and the inclination with respect to the plane of Mercury's orbit for each planet in order to justify the approximations discussed in the text. The last column in this Table presents the values of the perihelion advance calculated by Le Verrier (1859) [9, 10] for comparison

The estimates of Le Verrier show that the combined effect of Newtonian perturbations doesn't account for the complete observational estimated for Mercury's orbit of 574.10 ± 0.65 arcsec per century† and the complete explanation of the effect did have to wait to 1916 with the general theory of relativity [4].

*The terms depending on $\sin 2(\omega - \tilde{f})$ or $\cos 2(\omega - \tilde{f})$ are periodic contributions that average to zero when calculating over many revolutions. Hence the only secular contribution corresponds to the first term in the $\Delta\omega$ equation.

†According to Le Verrier (1859) the discrepance between the observed and the calculated advance was estimated in 38 arcsec per century but the improved result of Simon Newcomb (1882) gives a discrepance of 43 arcsec per century that was explained later by general relativity.

The Kozai-Lidov Effect

Now, we will consider the case in which and object such an asteroid moves under the influence of the Sun and is perturbed by a third body that moves in an orbit that is not coplanar with the asteroid's orbit. The resulting evolution of the orbital elements will show a periodic exchange between eccentricity and inclination, known as the Kozai-Lidov Effect‡.

In Figure 2.2 we show the orientation of the fundamental frame, now having the orbit of the perturbing body in the xy-plane. The orbit of the 2-body system will have an inclination ι with respect to the reference plane and we have chosen the direction of the x-axis such that the longitude of the ascending node is $\Omega = 0$. This election of the fundamental frame makes that the basis vectors $\{\mathbf{n}, \boldsymbol{\lambda}, \mathbf{k}\}$ given in equations (1.56) reduce to

$$\mathbf{n} = \cos(f+\omega)\mathbf{e}_x + \cos\iota\sin(f+\omega)\mathbf{e}_y + \sin\iota\sin(f+\omega)\mathbf{e}_z \tag{2.36a}$$
$$\boldsymbol{\lambda} = -\sin(f+\omega)\mathbf{e}_x + \cos\iota\cos(f+\omega)\mathbf{e}_y + \sin\iota\cos(f+\omega)\mathbf{e}_z \tag{2.36b}$$
$$\mathbf{k} = -\sin\iota\mathbf{e}_y + \cos\iota\mathbf{e}_z. \tag{2.36c}$$

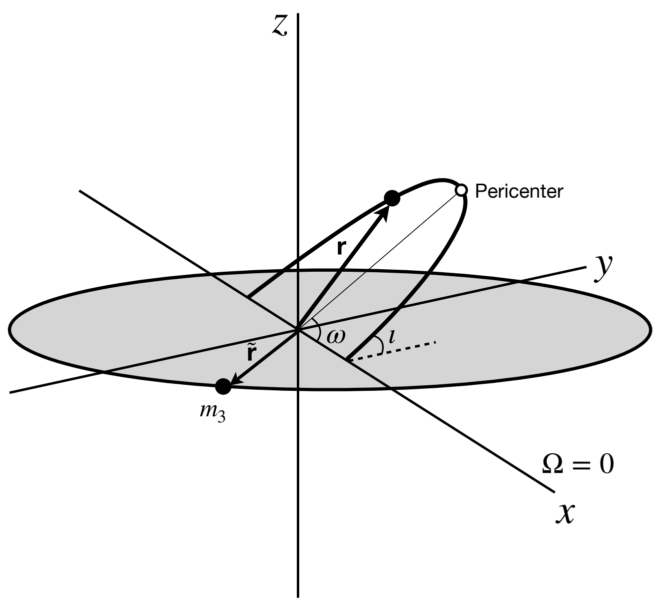

Figure 2.2: The three-body system to analyze the Kozai-Lidov effect. We set the fundamental frame such that the orbit of the perturbing body lies in the xy-plane and such that the longitude of the ascending node of the 2-body system is $\Omega = 0$.

We will assume that the perturbing body moves in a circular orbit, defining the normal vector

$$\tilde{n} = \cos\tilde{f}\mathbf{e}_x + \sin\tilde{f}\mathbf{e}_y. \tag{2.37}$$

‡This effect was discovered independently by the japanese astronomer Yoshihide Kozai and the soviet scientist Michael Lidov in 1962.

Using these vectors, we obtain the components of the perturbing force (2.27) as

$$\begin{aligned}\mathcal{R} =& -\frac{Gm_3 r}{\tilde{r}^3}\left[1-3\cos^2\tilde{f}\cos^2(f+\omega)-3\cos^2\iota\sin^2\tilde{f}\sin^2(f+\omega)\right.\\ &\left.-6\cos\iota\sin\tilde{f}\cos\tilde{f}\sin(f+\omega)\cos(f+\omega)\right] &(2.38a)\\ \mathcal{T} =& \frac{Gm_3 r}{\tilde{r}^3}\left[\left(\cos^2\iota\sin^2\tilde{f}-\cos^2\tilde{f}\right)\sin(f+\omega)\cos(f+\omega)\right.\\ &\left.-\cos\iota\sin\tilde{f}\cos\tilde{f}\left(\sin^2(f+\omega)-\cos^2(f+\omega)\right)\right] &(2.38b)\\ \mathcal{W} =& -\frac{Gm_3 r}{\tilde{r}^3}\sin\iota\sin\tilde{f}\left[\cos\tilde{f}\cos(f+\omega)+\cos\iota\sin\tilde{f}\sin(f+\omega)\right]. &(2.38c)\end{aligned}$$

By replacing these relations in 2.21), we obtain a set of differential equations to obtain the changes in the orbital elements. Averaging over the position of the perturbing body,

$$\langle\vartheta(\tilde{f})\rangle = \frac{1}{2\pi}\int_0^{2\pi}\vartheta(\tilde{f})d\tilde{f}, \tag{2.39}$$

and integrating with respect to f over one period, we obtain the variation of the orbital elements as

$$\begin{aligned}\langle\Delta a\rangle &= 0 &(2.40a)\\ \langle\Delta e\rangle &= \frac{15\pi}{2}\frac{m_3}{M}\left(\frac{a}{\tilde{r}}\right)^3 e\sqrt{1-e^2}\sin^2\iota\sin\omega\cos\omega &(2.40b)\\ \langle\Delta\iota\rangle &= -\frac{15\pi}{2}\frac{m_3}{M}\left(\frac{a}{\tilde{r}}\right)^3\frac{e^2}{\sqrt{1-e^2}}\sin\iota\cos\iota\sin\omega\cos\omega &(2.40c)\\ \langle\Delta\Omega\rangle &= \frac{3\pi}{2}\frac{m_3}{M}\left(\frac{a}{\tilde{r}}\right)^3\frac{\cos\iota}{\sqrt{1-e^2}}\left(5e^2\cos^2\omega-4e^2-1\right) &(2.40d)\\ \langle\Delta\omega\rangle &= \frac{3\pi}{2}\frac{m_3}{M}\left(\frac{a}{\tilde{r}}\right)^3\frac{1}{\sqrt{1-e^2}}\left[5\cos^2\omega(1-e^2-\cos^2\iota)-(3(1-e^2)-5\cos^2\iota)\right] &(2.40e)\\ \langle\Delta p\rangle &= -15\pi\frac{m_3}{M}\left(\frac{a}{\tilde{r}}\right)^3 ae^2\sqrt{1-e^2}\sin^2\iota\sin\omega\cos\omega. &(2.40f)\end{aligned}$$

Angular Momentum Conservation

From the Figure 2.2 , it can be seen that the z-component of the total angular momentum of the system can be written as

$$\boldsymbol{\ell}^z_{total} = \boldsymbol{\ell}^z + \tilde{\boldsymbol{\ell}}^z \tag{2.41}$$

where $\boldsymbol{\ell}$ is the angular momentum of the binary system and $\tilde{\boldsymbol{\ell}}$ is the angular momentum associated with the orbital motion of the third body. Assuming that the objects m_1 and m_2 don't affect the motion of the third body, we consider that $\tilde{\boldsymbol{\ell}}$ is conserved. Hence, the question arising is about the conservation of the component $\boldsymbol{\ell}^z$ of the binary system, which is

$$\ell^z = \boldsymbol{\ell}\cdot\mathbf{e}_z = \sqrt{GMa(1-e^2)}\cos\iota. \tag{2.42}$$

From this relation it is possible to obtain the change in this quantity, averaged over one orbit of the third body, as

$$\frac{\langle\Delta\ell^z\rangle}{\sqrt{GM}} = \frac{(1-e^2)\langle\Delta a\rangle - 2ae\langle\Delta e\rangle}{2\sqrt{a(1-e^2)}}\cos\iota - \sqrt{a(1-e^2)}\sin\iota\langle\Delta\iota\rangle. \tag{2.43}$$

Using the results (2.40), we confirm immediately the conservation law $\langle \Delta \ell^z \rangle = 0$. Since $\langle \Delta a \rangle = 0$, the conservation statement is equivalent to the relation

$$\frac{e}{1-e^2} \langle \Delta e \rangle + \tan \iota \langle \Delta \iota \rangle = 0. \tag{2.44}$$

This result is known as the *Kozai-Lidov effect* and implies an exchange between eccentricity and inclination of the orbit. In order to analyze this effect, consider the factor $\left[5\cos^2\omega(1-e^2-\cos^2\iota) - (3(1-e^2) - 5\cos^2\iota)\right]$ in the equation for $\langle \Delta\omega \rangle$. From it, we find a critical value ω_c for which $\langle \Delta\omega \rangle = 0$,

$$\cos^2 \omega_c = \frac{3(1-e^2) - 5\cos^2\iota}{5(1-e^2-\cos^2\iota)}. \tag{2.45}$$

If $\cos^2\iota > \frac{3}{5}(1-e^2)$, i.e. for small initial inclination angles, the critical value ω_c doesn't exists and therefore, the argument of the pericenter ω increases monotonically while eccentricity and inclination present a periodic exchange, as seen in Figure 2.3. In this case, the longitude of the ascending node Ω also increases monotonically.

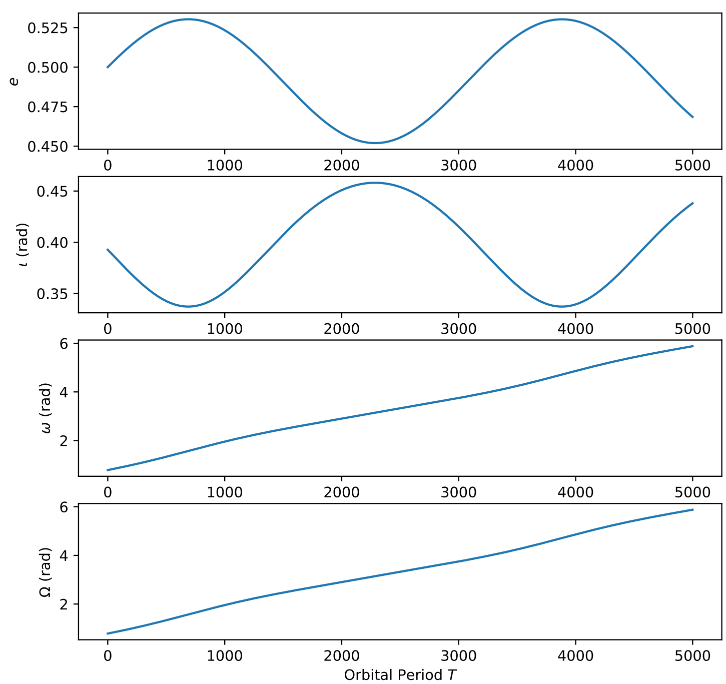

Figure 2.3: This plot shows the Kozai-Lidov effect as a periodic exchange between eccentricity and inclination of the orbit. This oscillatory behavior is obtained from equations (2.40) and occurs when the inclination angle of the orbit is small (see text for details). We also include the advance of the pericenter and the shift in the longitude of he ascending node. We use the values $m_3 = 0.1M$ and $a = 0.1\tilde{r}$, together with the initial values $e_0 = 0.5$, $\omega_0 = \frac{\pi}{4}$, $\iota_0 = \frac{\pi}{8}$ and $\Omega_0 = 0$.

On the other hand, a high initial inclination angle, such that $\cos^2\iota < \frac{3}{5}(1-e^2)$ gives a real value for the critical value ω_c. Figure 2.4 shows that in this case the argument of the pericenter increases until it reaches the critical value and, since this gives $\langle \Delta\omega \rangle = 0$, the evolution of ω stops. This also stops the evolution of e, ι and Ω so that there is no periodical behavior in the orbital parameters but they reach a stable equilibrium state.

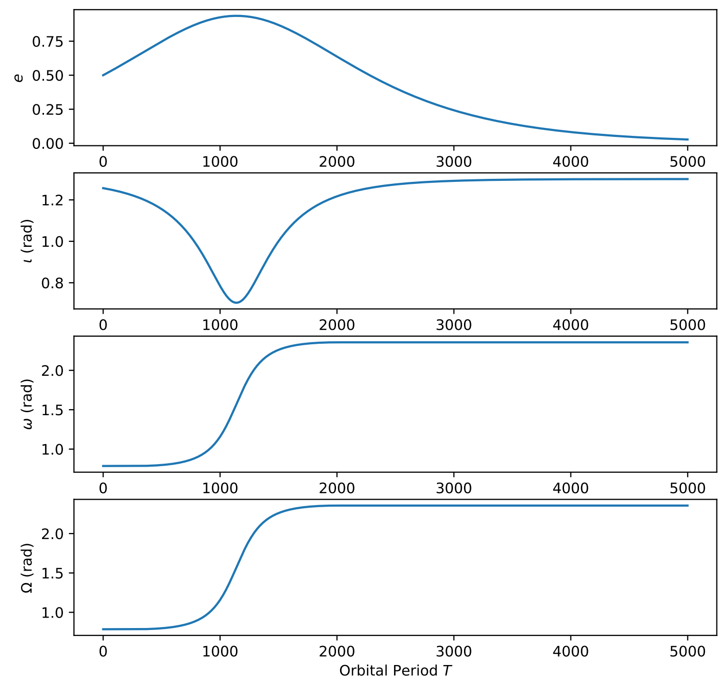

Figure 2.4: When the orbit has a high initial inclination angle, the Kozai-Lidov effect doesn't have an oscillatory exchange. In this case, the argument of the pericenter approaches a critical value for which $\langle \Delta\omega \rangle = 0$ and therefore the perihlion shift stops as well as the changes in eccentricity and inclination(see text for details). We use the values $m_3 = 0.1M$ and $a = 0.1\tilde{r}$, together with the initial values $e_0 = 0.5$, $\omega_0 = \frac{\pi}{4}$, $\iota_0 = \frac{2\pi}{5}$ and $\Omega_0 = 0$.

Part II

General Relativity

Chapter 3

General Relativity Fundamentals

In this chapter we will present a brief review of the General Theory of Relativity formulation. It is not intended to be a deep introduction to the field and therefore, we recommend to the un-experienced reader to consult other references for a detailed approach [1].

3.1 Basic Principles

General relativity is a 4-dimensional geometrical model of gravity based on two basic principles. The first one is the *general principle of relativity*, which states that

> *The laws of physics are the same in all reference frames.*

Introducing the idea of a *general covariant transformation* as a transformation between two arbitrary reference frames*, the general principle o relativity can be stated in an alternative form, known as the *principle of general covariance*,

> *The form of the laws of physics is invariant under arbitrary differentiable coordinate transformations.*

Mathematically, an equation is said to be covariant if it is written using only *tensors* and therefore we will write all physical quantities using these objects. For example, under a general coordinate transformation $x^\mu \to x'^\mu = x'^\mu(x^\alpha)$, an arbitrary third-order tensor, $T^{\alpha\beta}_\gamma$, obeys the well known transformation law,

$$T^{\mu'\nu'}_{\rho'} = \frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta}\frac{\partial x^\gamma}{\partial x'^\rho}T^{\alpha\beta}_\gamma \tag{3.1}$$

Exercise 3.1

Vanishing of a Tensor
Probe that if some arbitrary fourth-order tensor tensor vanishes in a certain coordinate system, $R_{\alpha\mu\beta\nu} = 0$, it will vanish in any coordinate system.

The second postulate of general relativity is the *Einstein Equivalence Principle* (EEP), which states that

1. Weak Equivalence Principle (WEP): *The trajectory of a freely-falling test-particle is independent of its internal structure and composition.*

*In the general theory of relativity, inertial and non-inertial frames are allowed.

2. Local Lorentz Invariance: *The outcome of any local non-gravitational experiment is independent of the velocity of the freely-falling reference frame in which it is performed.*

3. Local Position Invariance: *The outcome of any local non-gravitational experiment is independent of where and when it is performed.*

Based on these principles, general relativity is one of the so-called *metric theories of gravity*, for which the space-time will be equipped with a symmetric metric from which we will derive all the geometric quantities. For example, the 4-dimensional line element is written in terms of the metric as

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu, \tag{3.2}$$

where x^μ are the space-time coordinates and $g_{\mu\nu}$ are the general metric tensor components†. Due to the WEP, free-falling test-particles will follow trajectories known as *geodesics*, that are also defined by the metric tensor. Finally, it is important to note that a consequence of the Einstein equivalence principle is that the non-gravitational laws of physics in local freely-falling reference frames will be the same as those in special relativity, which are described in terms of a Minkowskian metric.

3.1.1 Covariant Derivative

In general relativity, the partial derivative operator ∂_μ is not the adequate to write the equations of physics because it is not covariant under general coordinate transformations. Therefore, it must be generalized to the *covariant derivative* operator ∇_μ which acts, for example, over a (3,2)-tensor as

$$\nabla_\mu T^{\alpha\beta}_\nu = \partial_\mu T^{\alpha\beta}_\nu + \Gamma^\alpha_{\mu\sigma} T^{\sigma\beta}_\nu + \Gamma^\beta_{\mu\sigma} T^{\alpha\sigma}_\nu - \Gamma^\sigma_{\mu\nu} T^{\alpha\beta}_\sigma \tag{3.3}$$

where the *connections* or *Christoffel symbols the second kind* are defined by the metric,

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2} g^{\alpha\sigma} \left[\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu} \right], \tag{3.4}$$

and therefore $\nabla_\mu g_{\alpha\beta} = 0$.

Exercise 3.2

Covariant Divergence
The divergence of a vector field A^μ is defined as the contracted covariant derivative $\nabla_\mu A^\mu$. Show that this can be written as

$$\nabla_\mu A^\mu = \frac{1}{\sqrt{-g}} \partial_\mu \left(\sqrt{-g} A^\mu \right), \tag{3.5}$$

where g represents the determinant of the metric $g_{\mu\nu}$.

3.1.2 Curvature Tensors

The complete information about the curvature of a space-time manifold is stored in the *Riemann tensor*, which is defined in terms of the Christoffel symbols as

$$R^\alpha_{\mu\beta\nu} = \partial_\beta \Gamma^\alpha_{\mu\nu} - \partial_\nu \Gamma^\alpha_{\mu\beta} + \Gamma^\alpha_{\sigma\beta} \Gamma^\sigma_{\mu\nu} - \Gamma^\alpha_{\sigma\nu} \Gamma^\sigma_{\mu\beta}. \tag{3.6}$$

Using the metric, we obtain the components $R_{\alpha\mu\beta\nu}$,

$$R_{\alpha\mu\beta\nu} = g_{\alpha\sigma} R^\sigma_{\mu\beta\nu}, \tag{3.7}$$

†In this work we will use the metric signature (-+++)

which satisfy the following symmetry properties

$$R_{\alpha\mu\beta\nu} = -R_{\mu\alpha\beta\nu} = -R_{\alpha\mu\nu\beta} = R_{\beta\nu\alpha\mu}. \tag{3.8}$$

Exercise 3.3

Bianchi identities. *Show explicitly that the Riemann tensor also satisfies the first Bianchi identities,*

$$R^{\alpha}_{\mu\beta\nu} + R^{\alpha}_{\nu\mu\beta} + R^{\alpha}_{\beta\nu\mu} = 0, \tag{3.9}$$

and the second Bianchi identities,

$$\nabla_\rho R^{\alpha}_{\mu\beta\nu} + \nabla_\beta R^{\alpha}_{\mu\nu\rho} + \nabla_\nu R^{\alpha}_{\mu\rho\beta} = 0. \tag{3.10}$$

From the Riemann tensor, we can define other curvature tensors such as the Ricci tensor,

$$R_{\mu\nu} = R^{\sigma}_{\mu\sigma\nu}, \tag{3.11}$$

which is symmetric $R_{\mu\nu} = R_{\nu\mu}$. The contraction of this tensor is known as the *Ricci curvature scalar*,

$$R = g^{\mu\nu} R_{\mu\nu}. \tag{3.12}$$

The Einstein tensor is a second order symmetric tensor defined as

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R. \tag{3.13}$$

Exercise 3.4

Bianchi identities for the Einstein tensor
Probe that the Einstein tensor satisfy the Bianchi identity

$$\nabla_\mu G^{\mu\nu} = 0. \tag{3.14}$$

3.2 Energy-Momentum Tensor

Two very important concepts in the construction of general relativity are those of *energy* and *momentum*. The simplest construction concerning these quantities corresponds to the momentum vector, which provides a complete description of the energy and momentum for a single relativistic particle. The components of this vector are given by

$$p^\mu = (mc, m\mathbf{u}) = \left(\frac{E}{c}, \mathbf{p}\right). \tag{3.15}$$

In the co-moving frame (i.e. the proper frame of the particle), the components of this vector are simply

$$p^\mu = (m_0 c, \mathbf{0}) = \left(\frac{E_0}{c}, \mathbf{0}\right) \tag{3.16}$$

However, the momentum vector is not enough to give the relevant information in the case of extended or continuous systems. In these cases, it is necessary to introduce the *energy-momentum* tensor‡, which will provide a complete description of the energy and momentum. The components $T^{\mu\nu}$ of this tensor are defined as the *"flux of 4-momentum P^μ across a surface of constant x^ν"*. Hence, it results in a symmetric second-order tensor and its components include the energy density, the pressure and the stresses.

‡It is also common to call it the *stress-energy* tensor.

Single Isolated Particle

The simplest example of an energy-momentum tensor corresponds to that associated to a single, isolated, relativistic particle. Denoting the proper time of this particle by τ, its position will be represented by the function

$$x_p^\mu(\tau) = \left(t(\tau), \mathbf{x}_p(\tau)\right), \tag{3.17}$$

while its velocity will be

$$u^\mu(\tau) = \frac{dx_p^\mu}{d\tau} \tag{3.18}$$

The corresponding energy-momentum tensor is defined as

$$T^{\alpha\beta} = m_0 u^\alpha(\tau) u^\beta(\tau) \delta\left(\mathbf{x} - \mathbf{x}_p(\tau)\right) = \frac{E_0}{c^2} u^\alpha(\tau) u^\beta(\tau) \delta\left(\mathbf{x} - \mathbf{x}_p(\tau)\right), \tag{3.19}$$

where $E_0 = m_0 c^2$ is the rest energy/mass of the particle and $\delta\left(\mathbf{x} - \mathbf{x}_p(\tau)\right)$ is the Dirac's delta function . In the co-moving frame, the components of this tensor take the simple form

$$T^{\mu\nu} = \begin{pmatrix} m_0 c^2 \delta\left(\mathbf{x} - \mathbf{x}_p(\tau)\right) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{3.20}$$

where the non-vanishing component corresponds to the rest energy density of the particle,

$$\varepsilon_0 = m_0 c^2 \delta\left(\mathbf{x} - \mathbf{x}_p(\tau)\right). \tag{3.21}$$

In other frame, the components of the energy-momentum tensor are

$$T^{00} = m_0 c^2 \gamma^2(u) \delta\left(\mathbf{x} - \mathbf{x}_p(\tau)\right) \tag{3.22a}$$

$$T^{0j} = m_0 c^2 \gamma^2(u) \delta\left(\mathbf{x} - \mathbf{x}_p(\tau)\right) \frac{u^j}{c} \tag{3.22b}$$

$$T^{ij} = m_0 c^2 \gamma^2(u) \delta\left(\mathbf{x} - \mathbf{x}_p(\tau)\right) \frac{u^i u^j}{c^2}, \tag{3.22c}$$

from which it is clear that the T^{00} component corresponds to the relativistic energy density, the T^{0j} components form the momentum density and the T^{ij} components represent the stresses, i.e. flux of i-momentum in the j-direction.

Incoherent Matter. Dust

The terms *incoherent matter* or *dust* are used to describe a collection of non-interacting particles at rest with respect to each other, i.e. a perfect fluid with zero pressure. Therefore, dust is characterized only by its density and by a velocity field. The corresponding energy-momentum tensor is defined as

$$\mathbf{T}_{Dust} = p \otimes N \tag{3.23}$$

where

$$p^\mu = m_0 u^\mu \tag{3.24}$$

is the momentum vector for each particle in the system and

$$N^\mu = n u^\mu \tag{3.25}$$

is the *number-flux* vector, with n the number density of particle and u^μ the velocity field. The components of the energy-momentum tensor are defined as

$$T^{\mu\nu}_{Dust} = \rho_0 u^\mu u^\nu = \frac{\varepsilon_0}{c^2} u^\mu u^\nu \tag{3.26}$$

where $\rho_0 = m_0 n$ is the rest mass density of the system and $\varepsilon_0 = \rho_0 c^2$ is the corresponding rest energy density. In the co-moving frame, the components of this tensor become

$$T^{\mu\nu}_{Dust} = \begin{pmatrix} \varepsilon_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{3.27}$$

In other frame of reference, the components of the energy-momentum tensor are

$$T^{00}_{Dust} = \rho_0 c^2 \gamma^2(u) = \varepsilon_0 \gamma^2(u) = \varepsilon \tag{3.28a}$$

$$T^{0j}_{Dust} = \rho_0 c^2 \gamma^2(u) \frac{u^j}{c} = \varepsilon \frac{u^j}{c} \tag{3.28b}$$

$$T^{ij}_{Dust} = \rho_0 c^2 \gamma^2(u) \frac{u^i u^j}{c^2} = \varepsilon \frac{u^i u^j}{c^2}, \tag{3.28c}$$

where $\varepsilon = \varepsilon_0 \gamma^2(u)$ is the relativistic energy density.

Exercise 3.5

Conservation of Energy
Show explicitly that the energy-momentum tensor for incoherent matter in absence of gravitational fields, i.e. in a Minkowskian background, satisfies the conservation equation

$$\partial_\nu T^{\mu\nu}_{Dust} = 0. \tag{3.29}$$

Perfect Fluid

A perfect fluid consist of a collection of particles with no heat conduction and no viscosity, so it will be completely characterized by its pressure, p, and its energy density, ε_0. This system looks isotropic in its rest frame, which implies that the energy-momentum tensor will be diagonal in that frame and the non-zero spatial components must be equal, $T^{11} = T^{22} = T^{33}$. Therefore, the two independent components of the energy-momentum tensor will be chosen as $T^{00} = \varepsilon_0$ and $T^{ii} = p$, which gives the following form in the co-moving frame

$$T^{\mu\nu}_{PF} = \begin{pmatrix} \varepsilon_0 & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}. \tag{3.30}$$

Generalizing the definition of the energy-momentum tensor for dust, we propose the general form for the perfect fluid,

$$T^{\mu\nu}_{PF} = \frac{\varepsilon_0}{c^2} u^\mu u^\nu + p s^{\mu\nu} \tag{3.31}$$

where $s^{\mu\nu}$ is a symmetric tensor. Considering a Minkowskian background, the only second order symmetric tensor built with the metric and the 4-velocity is

$$s^{\mu\nu} = \alpha u^\mu u^\nu + \beta \eta^{\mu\nu}. \tag{3.32}$$

In order to fix the constants α and β we impose the form (3.30) in the co-moving frame and the conservation equation $\partial_\nu T^{\mu\nu}_{PF} = 0$. This gives the components of the energy-momentum tensor as

$$T^{\mu\nu}_{PF} = \frac{\varepsilon_0 + P}{c^2} u^\mu u^\nu + P\eta^{\mu\nu}. \tag{3.33}$$

In a general curved space-time, the energy-momentum tensor for the perfect fluid is defined as

$$T^{\mu\nu}_{PF} = \frac{\varepsilon_0 + P}{c^2} u^\mu u^\nu + P g^{\mu\nu}. \tag{3.34}$$

3.3 Einstein's Field Equations

The field equations of Einstein's general relativity are constructed following some physical assumptions concerning gravity and its mathematical structure:

1. The gravitational field should be completely described by the metric tensor.
2. The field equations must be a set of covariant second order differential equations and they must lead to a correct Newtonian limit, recovering Poisson equation for the gravitational field.
3. The source term of the field equations must include the mass/energy density. Because of the covariant nature of the field equations, the obvious candidate to use is the energy-momentum tensor.
4. In the absence of mass/energy sources, we should obtain Minkowski space-time.

The resulting field equations, according to Einstein's proposal, are[§]

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \tag{3.35}$$

This is a system of ten second-order differential equations which determine the ten components of the metric tensor if the components of the energy-momentum tensor are given.

Exercise 3.6

Another form of the Field Equations
Use index contraction to show that

$$R = -\frac{8\pi G}{c^4} T \tag{3.36}$$

where $T = g_{\mu\nu}T^{\mu\nu}$ is the trace of the energy-momentum tensor. Using this result, show that Einstein's field equations can be written in the form

$$R_{\mu\nu} = \frac{8\pi G}{c^4} S_{\mu\nu} \tag{3.37}$$

where

$$S_{\mu\nu} = T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T. \tag{3.38}$$

[§]It is possible to include an additional term, $G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$, where Λ is called the *cosmological constant*. However, we will assume that this constant is zero because its effects in the typical problems of celestial mechanics are negligible.

Einstein's field equations can be derived from a variational principle. The total action is written as

$$S = S_{EH} + S_M, \tag{3.39}$$

where S_M is the matter action, which defines the energy-momentum tensor through

$$T^{\mu\nu} = \frac{1}{\sqrt{-g}} \frac{\delta S_M}{\delta g_{\mu\nu}} \tag{3.40}$$

and S_{EH} is the Einstein-Hilbert action

$$S_{EH} = \frac{c^4}{16\pi G} \int R\sqrt{-g} d^4 x, \tag{3.41}$$

which gives the left hand side of the field equations.

3.4 Geodesics

The WEP implies that free-falling test-particles will follow the trajectories known as *geodesics*. Geometrically, these are defined by the metric tensor and the covariant derivative through the concept of parallel transport. However, they can be obtained applying a variational principle to the action of the test particle,

$$S = -m_0 \int_1^2 ds = -m_0 c^2 \int_{\tau_1}^{\tau_2} d\lambda = -m_0 c \int_{\tau_1}^{\tau_2} \sqrt{g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda, \tag{3.42}$$

where λ represents the proper time associated to a massive particle or an affine parameter in the case of mass-less particles.

Exercise 3.7

Lagrangian for a single particle
Use the lagrangian arising from equation (3.42),

$$L = \sqrt{g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}}, \tag{3.43}$$

and the corresponding Euler-Lagrange equations to obtain the geodesic equations

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\alpha}{d\lambda} = 0. \tag{3.44}$$

Show that the alternative lagrangian

$$L = -\frac{1}{2} g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \tag{3.45}$$

leads to the same geodesic equations.

Chapter 4

Schwarzschild Spacetime

Given a certain energy-momentum distribution defined by the source tensor $T_{\mu\nu}$, Einstein's field equations (3.35) can be solved to obtain the metric tensor which gives all the geometric properties of space-time, and therefore the associated gravitational field. This procedure is not trivial, because the field equations are a system of non-linear second order differential equations. However, when space-time exhibits some symmetries the task is less difficult.

Schwarzschild's space-time is one of the most important solution of Einstein's field equations in the astrophysical context because it represents the gravitational field produced by static or slowly-rotating objects such as planets, stars or black holes. The corresponding metric defines the geometric properties of the space-time manifold in the exterior of a spherical mass distribution, and it is the only spherically symmetric vacuum solution of the field equations.

4.1 The Spherically Symmetric Solution

Assuming spherical symmetry, it is convenient to use spherical polar coordinates (t, r, θ, φ) and to propose a line element ansatz with the form

$$ds^2 = -\mathcal{A}(r)c^2dt^2 + \mathcal{B}(r)dr^2 + \mathcal{R}^2(r)\left(d\theta^2 + \sin^2\theta d\varphi^2\right), \tag{4.1}$$

where the functions $\mathcal{A}(r)$, $\mathcal{B}(r)$ and $\mathcal{R}(r)$ will be obtained by replacing this metric into Einstein's field equations (3.35). In vacuum, $T_{\mu\nu} = 0$, the field equations are integrated to obtain the well-known Schwarzschild's solution

$$ds^2 = -\left(1 - \frac{2m}{r}\right)c^2dt^2 + \left(1 - \frac{2m}{r}\right)^{-1}dr^2 + r^2\left(d\theta^2 + \sin^2\theta d\varphi^2\right), \tag{4.2}$$

where $m = \frac{GM}{c^2}$ and M is the total mass. If R denotes the radius of the spherical distribution of mass, this solution is valid in the exterior of the distribution, i.e. for radii $r > R$.

4.1.1 The Schwarzschild Radius and the Event Horizon

From equation (4.2) it is clear that the radius

$$r_S = 2m = \frac{2GM}{c^2} \tag{4.3}$$

produces an ill behavior of the line element. However, this singular point in the description always lies inside the mass distribution for astrophysical sources, with the only exception of black holes. In fact, the

Schwarzschild's radius r_S defines the boundary of a black hole, known as the *Event Horizon*, and gives the black hole their special properties. Since we will work in the weak field limit of the gravitational field, we won't cover this subject in our description of the relativistic celestial mechanics.

4.1.2 Isotropic Coordinates

Scharzschild's coordinates presented in the line element (4.2) give the simplest expression for the metric. When transforming to quasi-cartesian coordinates, the final metric has many off-diagonal components, making a complicated line element. Therefore, we may introduce another set of coordinates which help with the description. The Schwarzschild metric in *isotropic coordinates* $(t, r_{iso}, \theta, \varphi)$ is obtained by the transformation

$$r = r_{iso}\left(1 + \frac{m}{2r_{iso}}\right)^2, \tag{4.4}$$

giving the line element

$$ds^2 = -\left(\frac{1 - \frac{m}{2r_{iso}}}{1 + \frac{m}{2r_{iso}}}\right)^2 c^2dt^2 + \left(1 + \frac{m}{2r_{iso}}\right)^4 \left[dr_{iso}^2 + r_{iso}^2 d\theta^2 + r_{iso}^2 \sin^2\theta d\varphi^2\right]. \tag{4.5}$$

Note that in these coordinates, the spatial part of the metric is conformally flat. This fact implies that transforming to Cartesian coordinates defined by

$$x = r_{iso} \sin\theta \cos\varphi \tag{4.6a}$$
$$y = r_{iso} \sin\theta \sin\varphi \tag{4.6b}$$
$$z = r_{iso} \cos\theta, \tag{4.6c}$$

gives a line element with the form

$$ds^2 = -\left(\frac{1 - \frac{m}{2r_{iso}}}{1 + \frac{m}{2r_{iso}}}\right)^2 c^2dt^2 + \left(1 + \frac{m}{2r_{iso}}\right)^4 \delta_{ij}dx^i dx^j. \tag{4.7}$$

4.1.3 Harmonic Coordinates

The *harmonic coordinates* are introduced by the simple transformation

$$r = r_h + m, \tag{4.8}$$

from which we obtain the line element

$$ds^2 = -\left(\frac{1 - \frac{m}{r_h}}{1 + \frac{m}{r_h}}\right) c^2dt^2 + \left(\frac{1 + \frac{m}{r_h}}{1 - \frac{m}{r_h}}\right) dr_h^2 + r_h^2\left(1 + \frac{m}{r_h}\right)^2 \left[d\theta^2 + \sin^2\theta d\varphi^2\right]. \tag{4.9}$$

Using Cartesian coordinates defined as

$$x = r_h \sin\theta \cos\varphi \tag{4.10a}$$
$$y = r_h \sin\theta \sin\varphi \tag{4.10b}$$
$$z = r_h \cos\theta, \tag{4.10c}$$

the line element becomes

$$ds^2 = -\left(\frac{1 - \frac{m}{r_h}}{1 + \frac{m}{r_h}}\right) c^2dt^2 + \left(1 + \frac{m}{r_h}\right)^2 \left[\delta_{ij} + \frac{\left(\frac{m}{r_h}\right)^2}{1 - \left(\frac{m}{r_h}\right)^2} \frac{x_i x_j}{r_h^2}\right] dx^i dx^j. \tag{4.11}$$

Exercise 4.1

Determinant of the Schwarzschild's Metric Tensor in Harmonic Coordinates
Show that the determinant of the metric tensor in harmonic coordinates is

$$\sqrt{-g} = \left(1 + \frac{m}{r_h}\right)^2 r_h^2 \sin\theta. \tag{4.12}$$

Properties of the Harmonic Coordinates

The notion of "harmonic" for these coordinates is understood by defining a set of four scalar fields

$$\chi^{(0)} = ct \tag{4.13a}$$
$$\chi^{(1)} = (r-m)\sin\theta\cos\varphi \tag{4.13b}$$
$$\chi^{(2)} = (r-m)\sin\theta\sin\varphi \tag{4.13c}$$
$$\chi^{(3)} = (r-m)\cos\theta \tag{4.13d}$$

which we will denote together as $\chi^{(\alpha)}$. Clearly, the definitions of these fields coincide with the cartesian harmonic coordinates in equations (4.10) and the adjective "harmonic" is given because each of these scalar fields satisfy the relation (6.12),

$$\Box\chi^{(\alpha)} = 0. \tag{4.14}$$

Exercise 4.2

Harmonic Condition in Schwarzschild's Metric
Using equation (3.5), show that the harmonic condition (4.14) can be written as

$$\frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}g^{\mu\nu}\partial_\nu\chi^{(\alpha)}\right) = 0 \tag{4.15}$$

and probe explicitly that each one of the four scalar fields $\chi^{(\alpha)}$ satisfies this equation.

We will return to Schwarzschild's metric in harmonic coordinates in a subsequent chapter to define the parameterized post-Newtonian metric for the one body problem.

4.2 Particle Motion in the Schwarzschild's Background

Using the line element (4.2), the general lagrangian (3.45) describing the motion of a test particle in Schwarzschild's background becomes

$$L = \frac{1}{2}\left[\left(1-\frac{2m}{r}\right)c^2\left(\frac{dt}{d\lambda}\right)^2 - \left(1-\frac{2m}{r}\right)^{-1}\left(\frac{dr}{d\lambda}\right)^2 - r^2\left(\frac{d\theta}{d\lambda}\right)^2 - r^2\sin^2\theta\left(\frac{d\varphi}{d\lambda}\right)^2\right], \tag{4.16}$$

where λ is an adequate parameter for the description (for massive particle we chose the particle's proper time $\lambda = \tau$ while for massless particles λ is a so-called *affine parameter*).

Exercise 4.3

Lagrangian for a Particle in Schwarzschild's Background
Use the angular momentum conservation to show that the motion of a particle in the Schwarzschild's background is restricted to a plane.
Choose the plane $\theta = \frac{\pi}{2}$, to write the above Lagrangian for a particle moving in the Schwarzschild spacetime as

$$L = \frac{1}{2}\left[\left(1 - \frac{2m}{r}\right)c^2\left(\frac{dt}{d\lambda}\right)^2 - \left(1 - \frac{2m}{r}\right)^{-1}\left(\frac{dr}{d\lambda}\right)^2 - r^2\left(\frac{d\varphi}{d\lambda}\right)^2\right]. \tag{4.17}$$

4.2.1 Conserved Quantities and the Effective Potential

From the above Lagrangian, we identify the conservation of three quantities: the energy per unit mass,

$$\mathcal{E} = \frac{\partial L}{\partial\left(\frac{dt}{d\lambda}\right)} = \left(1 - \frac{2m}{r}\right)c^2\left(\frac{dt}{d\lambda}\right), \tag{4.18}$$

the angular momentum per unit mass

$$\ell = -\frac{\partial L}{\partial\left(\frac{d\varphi}{d\lambda}\right)} = r^2\left(\frac{d\varphi}{d\lambda}\right) \tag{4.19}$$

and the Hamiltonian*,

$$H = L = \delta \tag{4.20}$$

where we use $\delta = \frac{c^2}{2}$ for massive particles and $\delta = 0$ for massless particles.

Replacing the conserved quantities in the Lagrangian (4.17) gives the radial equation of motion

$$\left(\frac{dr}{d\lambda}\right)^2 = \frac{\mathcal{E}^2}{c^2} - V_{eff}^2(r) \tag{4.21}$$

where we have defined the effective potential

$$V_{eff}^2(r) = \left(1 - \frac{2m}{r}\right)\left(2\delta + \frac{\ell^2}{r^2}\right), \tag{4.22}$$

from which we can extract many of the information concerning the motion of particles.

4.2.2 Circular Motion

Circular Motion of Massive Particles

In Figure 4.1, we plot the typical behavior of the effective potential for a massive particle, $\delta = \frac{c^2}{2}$ and $\lambda = \tau$, as function of r and for some values of the angular momentum. As in the Newtonian case, motion is possible as long as $\mathcal{E}^2 \geq V_{eff}^2$. Figure 4.1 shows that particles can move following both bound and unbound orbits. Particularly, there are circular orbits determined by the condition

$$\left.\frac{dV_{eff}}{dr}\right|_{r=r_c} = 0 \tag{4.23}$$

*The Hamiltonian is mathematically equal to the Lagrangian and its conservation implies that the proper mass of the moving particle is a constant.

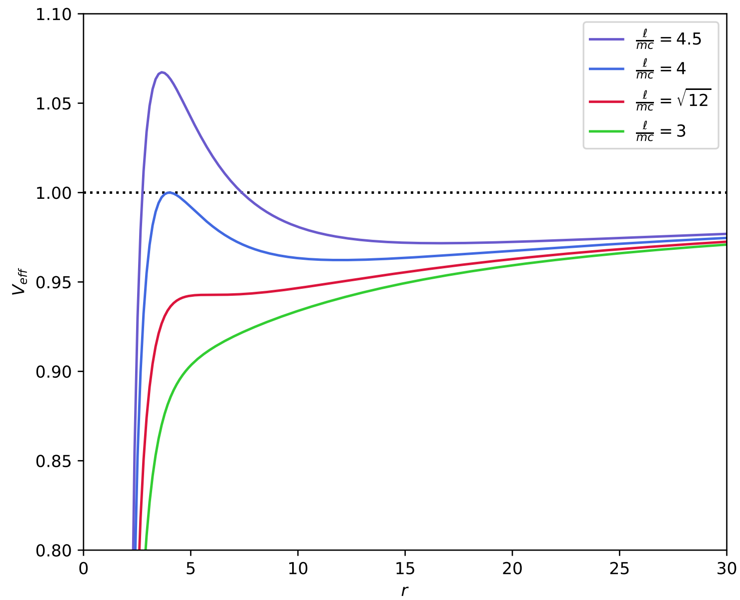

Figure 4.1: Effective potential for a massive particle moving in the Schwarzschild spacetime as function of the radial coordinate r with $m = c = 1$ and $\ell \neq 0$. The maximum and minimum in the curves correspond to possible circular orbits (unstable and stable) around the central mass for a given value of ℓ. The red curve corresponds to the extreme case $\ell^2 = 12m^2c^2$ for which it is possible to see the inflection point defining the Innermost Stable Circular Orbit (ISCO). Smaller angular momenta produce curves as the green one, with no maxima, minima or inflection points and therefore, no circular motion.

which gives the radius of the circular orbits as

$$r_c = \frac{\ell^2}{2mc^2}\left(1 \pm \sqrt{1 - 12\frac{m^2c^2}{\ell^2}}\right). \tag{4.24}$$

This result can be compared directly with the Newtonian result, $r_c^N = \frac{\ell^2}{GM}$, showing that, unlike the Newtonian case, for a given value of the angular momentum there are two possible circular orbits. One of the differences between these trajectories is their stability, which is determined by the second derivative $\left.\frac{d^2V_{eff}}{dr^2}\right|_{r=r_c}$.

Exercise 4.4

Circular Orbits in Schwarzschild Spacetime

Show that for large angular momentum, $\ell \gg \frac{GM}{c}$, the radii of the circular orbits and their stability are

$$r_c = \begin{cases} \frac{\ell^2}{mc^2} & \rightarrow \textit{Stable Orbit} \\ 3m & \rightarrow \textit{Unstable Orbit.} \end{cases} \tag{4.25}$$

Due to the argument under the square root, there is an extreme value for the angular momentum, $\ell^2 = 12m^2c^2$, which produces just one degenerate radius

$$r_{ISCO} = 6m. \tag{4.26}$$

The corresponding trajectory is called the *Innermost Stable Circular Orbit* (ISCO) and, as its name suggest, this corresponds to the smallest radius of the permitted stable circular orbits for massive particles moving in Schwarzschild's spacetime. In Figure 4.1, it is possible to visualize that this extreme value of the angular momentum corresponds to the curve for which the maximum and minimum values of the effective potential have degenerate into an inflection point.

From equations (4.25) and (4.26) we conclude that free particles in the Schwarzschild background may have both stable and unstable circular orbits with radii $r \geq 6m$ but there are only unstable circular for $3m < r < 6m$.

Circular Motion of Massless Particles

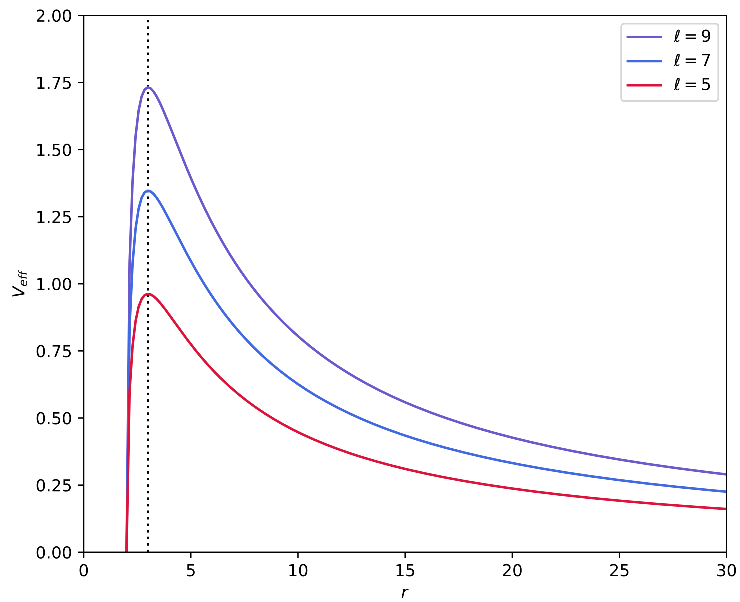

Figure 4.2: Effective potential for a massless particle moving in the Schwarzschild spacetime as function of the radial coordinate r with $m = 1$ and $\ell \neq 0$. The maximum in the curves correspond to the photon sphere. Note that the radius of this trajectory, represented by the dotted line, is always $r = 6m$, independent of the value of the angular momentum.

The effective potential for massless particles, $\delta = 0$, takes the simpler form

$$V_{eff}^2(r) = \left(1 - \frac{2m}{r}\right)\frac{\ell^2}{r^2}. \tag{4.27}$$

The plot of this function for some particular values of the angular momentum is shown in Figure 4.2. Note that the behavior of these curves is different from that of massive particles. In particular, there is just one possible circular orbit determined by the condition (4.23). The circular motion occurs at the radius

$$r_{ps} = 3m, \tag{4.28}$$

known as the *photon radius* and it is independent of the angular momentum of the particle. The surface at which photons (or in general particles with zero rest mass) move following circular trajectories is called the *photon sphere*. The value of the second derivative $\left.\frac{d^2V_{eff}}{dr^2}\right|_{r=r_{sp}}$ shows that this circular orbit is unstable.

4.3 Differential Equation of the Orbit

The differential equation defining the trajectories for particles moving in the Schwarzschild spacetime is found by using φ as the independent variable, instead of the proper time, and defining the new variable $u(\varphi) = \frac{1}{r(\varphi)}$. This change of variable in equation (4.21), following the same procedure used in the Newtonian case, gives the second order differential equation of the orbit,

$$u'' + u = \frac{2m}{\ell^2}\delta + 3mu^2, \tag{4.29}$$

where prime denotes differentiation with respect to φ. Comparing with equation (1.30) we note that the relativistic contribution corresponds to the last term in the right hand side. This equation is valid for massive and massless particles and in the following subsections, we will study the relativistic effect on the known Newtonian trajectories.

4.3.1 Bound Orbits. The Motion of the Planets in the Solar System

For massive particles, $\delta = c^2$, the differential equation of the orbit becomes

$$u'' + u = \frac{mc^2}{\ell^2} + 3mu^2. \tag{4.30}$$

In order to solve this equation we may note that the relativistic term $3mu^2$ is very small compared with the Newtonian term $\frac{mc^2}{\ell^2} = \frac{GM}{\ell^2}$. Indeed, a simple estimation for the planets in the Solar System gives

$$\frac{3mu^2}{\left(\frac{mc^2}{\ell^2}\right)} \sim \frac{v_\perp^2}{c^2} \sim 10^{-8} \tag{4.31}$$

where the values of the tangential velocity at the perihelion $v_\perp$ of the planets are taken from Table A.2. This small value implies that, at least in the Solar System, we may consider the relativistic term as a perturbation to the Newtonian orbits.

Solution of the Orbital Equation

Since the relativistic term will be treated as a perturbation, we consider the zeroth-order approximation differential equation (Newtonian contribution only) as

$$^{(0)}u'' + {}^{(0)}u = \frac{mc^2}{\ell^2}. \tag{4.32}$$

The solution of this equation are the conic functions (1.31),

$$^{(0)}u(\varphi) = \frac{1}{p_0}\left[1 + e\cos(\varphi - \omega)\right] \tag{4.33}$$

where the semi-latus rectum is $p_0 = \frac{\ell^2}{mc^2}$ and the integration constant are the eccentricity e and the longitude of the pericenter ω.

Without loosing generality, we will chose $\omega = 0$, so that the perihelion will be located at $\varphi = 0$ and we will plug this solution into the relativistic term of equation (4.30) to obtain the first order approximation differential equation

$$^{(1)}u'' + {}^{(1)}u = \frac{mc^2}{\ell^2} + \frac{3m^3c^4}{\ell^4} + \frac{6m^3c^4}{\ell^4}e\cos\varphi + \frac{3m^3c^4}{\ell^4}e^2\cos^2\varphi. \tag{4.34}$$

Exercise 4.5

Orbit of a Planet in Schwarzschild Spacetime
Show that the general solution of the first-order approximation differential equation (4.34) is

$$ {}^{(1)}u(\varphi) = \frac{1}{p_1}\left[1 + e\cos\varphi\right] + \frac{3m^3c^4}{\ell^4} e\varphi\sin\varphi + \frac{m^3c^4}{2\ell^4} e^2\left(3 - \cos 2\varphi\right), \tag{4.35} $$

where the new semi-latus rectum is

$$ p_1 = \frac{\ell^2}{mc^2}\left[1 + \frac{3m^2c^2}{\ell^2}\right]^{-1}. \tag{4.36} $$

The general first-order approximation solution (4.35) shows the correction terms to the Newtonian conic equation. In particular, the correction term in the new semi-latus rectum (4.36) is estimated to be small. Using the values in Tables A.1 and A.2, the correction term for the Solar System is

$$ \frac{3m^2c^2}{\ell^2} \sim 10^{-6}. \tag{4.37} $$

The third term in the right hand side of (4.35) is a periodic function of the angle φ and therefore, its effect is averaged to zero after some revolutions. On the other hand, the second term in the right hand side of (4.35) is a secular contribution, producing an accumulative effect due to its linear dependence on φ.

Advance of the Perihelion

The secular contribution to the orbit of the planets in the Solar System will produce an advance in the perihelion. Considering only the first two terms in the right hand side of equation (4.35) gives the function

$$ u(\varphi) = \frac{1}{p_1}\left[1 + e\left(\cos\varphi + \frac{3m^3c^4p_1}{\ell^4}\varphi\sin\varphi\right)\right]. \tag{4.38} $$

The coefficient of the last term is small,

$$ \frac{3m^3c^4p_1}{\ell^4} = \frac{3m^3c^4}{\ell^4}\frac{\ell^2}{mc^2}\left[1 + \frac{3m^2c^2}{\ell^2}\right]^{-1} \sim \frac{3m^2c^2}{\ell^2} \sim 10^{-6}, \tag{4.39} $$

and therefore, it is possible to approximate the function $u(\varphi)$ as

$$ u(\varphi) = \frac{1}{p_1}\left[1 + e\cos\left(\varphi - \frac{3m^2c^2}{\ell^2}\varphi\right)\right] \tag{4.40} $$

or equivalently

$$ r(\varphi) = \frac{p_1}{1 + e\cos\left(\varphi - \frac{3m^2c^2}{\ell^2}\varphi\right)}. \tag{4.41} $$

The extra term in the cosine function introduces an aperiodicity which produces a shift in the location of the perihelion. To calculate this effect note that the perihelia occur when $\varphi - \frac{3m^2c^2}{\ell^2}\varphi = 2\pi n$ with $n = 0, 1, 2, 3, \ldots$. Hence, two successive perihelia occur with the difference $\Delta\varphi = 2\pi\left(1 + \frac{3m^2c^2}{\ell^2}\right)$ which clearly are not separated by 2π as in the case of the two-body problem. The second term in the parenthesis corresponds to the perihelion shift per orbit, which can be written as

$$\Delta\omega = \frac{6\pi m^2 c^2}{\ell^2} = \frac{6\pi GM}{c^2 a(1-e^2)} \tag{4.42}$$

after introducing the Newtonian angular momentum as the first approximation of the angular momentum, $\ell \sim \ell_N = GMa(1-e^2)$ and using the notation of Sections 1.2 and 2.2. Using the astronomical data in Table A.1, the calculated value for the perihelion advance for Mercury due to the relativistic effect is the well known value

$$\Delta\omega_{\text{Mercury}} = 5 \times 10^{-7} \text{ rad/orbit} = 42.9 \text{ arcsec/century}. \tag{4.43}$$

In Table 4.1 we include the relativistic perihelion shift for all the planets in the Solar System. As expected, the greatest value corresponds to Mercury.

Planet	a (AU)	e	T (yr)	Perihelion shift (arcsec/orbit)	Perihelion shift (arcsec/century)
Mercury	0.387098	0.205630	0.240846	0.1033	42.91
Venus	0.723332	0.006772	0.615198	5.297×10^{-2}	8.610
Earth	1.	0.0167086	1.00001	3.832×10^{-2}	3.832
Mars	1.523679	0.0934	1.88082	2.536×10^{-2}	1.349
Jupiter	5.2044	0.0489	11.862	7.379×10^{-3}	6.220×10^{-2}
Saturn	9.5826	0.0565	29.4571	4.011×10^{-3}	1.362×10^{-2}
Uranus	19.2184	0.046381	84.0205	1.998×10^{-3}	2.378×10^{-3}
Neptune	30.11	0.009456	164.8	1.272×10^{-3}	7.721×10^{-4}

Table 4.1: Relativistic Perihelion Advance for the planets in the Solar System.

4.3.2 The Motion of Light

For massless particles, $\delta = 0$, the differential equation of the orbit becomes

$$u'' + u = 3mu^2. \tag{4.44}$$

Following the same procedure as described above, the zeroth-order approximation solution (i.e. without the relativistic term $3mu^2$) is

$$^{(0)}u(\varphi) = \frac{1}{r_0}\cos\varphi \tag{4.45}$$

which corresponds to a straight line†. The constant r_0 corresponds to the distance of maximum approach of the massless particle to the center of force, occurring at $\varphi = 0$.

Plugging the zeroth-order approximation solution in the right hand side of equation (4.44), we obtain the differential equation for the first-order approximation,

$$^{(1)}u'' + {}^{(1)}u = \frac{3m}{r_0^2}\cos^2\varphi. \tag{4.46}$$

The complete solution, up to first-order, is

$$u(\varphi) = \frac{2m}{r_0^2} + \frac{1}{r_0}\cos\varphi - \frac{m}{r_0^2}\cos^2\varphi. \tag{4.47}$$

†As expected from Newtonian mechanics, light should not be affected by gravity and this equation corresponds to the straight line $x = r\cos\varphi = r_0$ in cartesian coordinates.

The new terms in this solution have a relativistic origin and produce a change in the straight line path of a massless particle. The first effect corresponds to a change in the distance of maximum approach of the massless particle to the center of force. Evaluating at $\varphi = 0$, we obtain

$$r_{min} = r_0 \left(1 + \frac{m}{r_0}\right)^{-1}. \tag{4.48}$$

Note that the correction term is proportional to the mass of the Schwarzschild source. For example, in the case of the Solar System, the mass of the Sun gives a value of $m = \frac{GM_\odot}{c^2} \sim 10^3$ m. Therefore, for a light ray passing tangential to the surface of the Sun $r_0 \sim R_\odot \sim 7 \times 10^8$ m., the correction term is very small, $\frac{m}{r_0} \sim 10^{-5}$. This estimate let us approximate the minimum distance of approach as

$$r_{min} \approx r_0 - m. \tag{4.49}$$

Deflection of a Light Ray by the Sun

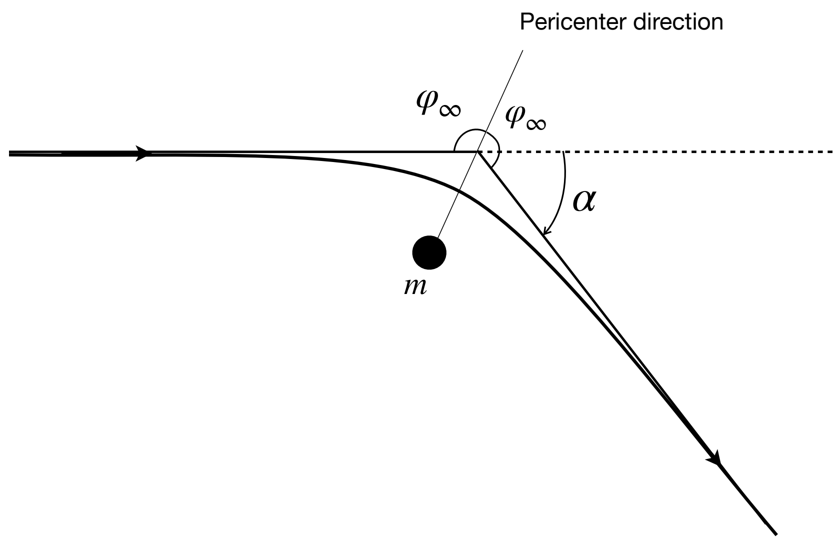

Figure 4.3: The deflection angle α fro a light-ray moving in Schwarzschild's spacetime. The incident and scattered rays have directions given by the angle φ_∞ with respect to the direction of the perihelion.

The second and most important effect of gravity on the motion of massless particles corresponds to a deviation of the trajectory. For example, we define the *deflection angle* for a light ray passing near the surface of the Sun as the angle between the incident and the scattered directions. In order to calculate this effect, we consider the asymptotic rays defined by the condition $r \to \infty$ or equivalently $u \to 0$. Imposing this condition in (4.47) gives an equation for the asymptotic direction, φ_∞,

$$\cos^2 \varphi_\infty - \frac{r_0}{m} \cos \varphi_\infty - 2 = 0. \tag{4.50}$$

Exercise 4.6

Incidence Angle of a Light Ray

Show that the physically acceptable solution of equation (4.50) is

$$\cos \varphi_\infty = -\frac{2m}{r_0}. \tag{4.51}$$

According to the geometry presented in Figure 4.3 and due to the symmetry of the trajectory of the light ray with respect to the direction of the perihelion, the deflection angle α satisfies the condition $2\varphi_\infty - \alpha = \pi$. Using this relation in (4.51) we obtain the equation defining the deflection angle,

$$\sin\left(\frac{\alpha}{2}\right) = \frac{2m}{r_0}. \tag{4.52}$$

Since this angle is expected to be very small, we approximate to obtain

$$\alpha \approx \frac{4m}{r_0}. \tag{4.53}$$

For a light ray passing near to the surface of the Sun, the deflection angle gives the well-known value obtained by Einstein and observed during a Solar eclipse in 1919 by S. A. Eddington and F. W.Dyson,

$$\alpha = 8.47 \times 10^{-6} \text{ rad}\frac{R_\odot}{r_0} = 1.75 \text{ arcsec}\frac{R_\odot}{r_0}, \tag{4.54}$$

where $R_\odot$ is the radius of the Sun.

4.4 Shapiro's Time Delay

Another effect of general relativity is the time delay in electromagnetic signals due to gravity. An interesting setup for testing this effect was proposed by I. Shapiro in 1964 and consist in a radio signal transmitted from Earth to another planet but passing near the Sun. The radio signal is reflected from the planet back to Earth in a similar trajectory and it is calculated the time delay in the signal due to the effect of Sun's gravity.

Our simplified model is shown in Figure 4.4 and will suppose that the electromagnetic signal is emitted at the point $\mathbf{x}_1 = (r_1, \frac{\pi}{2}, \varphi_1)$ on Earth and it is received at the point $\mathbf{x}_2 = (r_2, \frac{\pi}{2}, \varphi_2)$ on the surface of the other planet.

The radial equation of motion for the electromagnetic signal is given by equation (4.21) with the effective potential (4.2),

$$\left(\frac{dr}{d\lambda}\right)^2 = \frac{\mathcal{E}^2}{c^2} - \left(1 - \frac{2m}{r}\right)\frac{\ell^2}{r^2}, \tag{4.55}$$

where λ is an affine parameter. We introduce the time coordinate using equation (4.18) to obtain

$$\frac{r^2}{\left(1 - \frac{2m}{r}\right)^3}\frac{\dot{r}^2}{c^4} = \frac{r^2}{c^2\left(1 - \frac{2m}{r}\right)} - \left(\frac{\ell}{\mathcal{E}}\right)^2. \tag{4.56}$$

At the point of closest approach, located at a distance $r = r_0$, the radial velocity vanishes, $\dot{r} = 0$, and therefore the conserved quantities can be evaluated as

$$\left(\frac{\ell}{\mathcal{E}}\right)^2 = \frac{r_0^2}{c^2\left(1 - \frac{2m}{r_0}\right)}. \tag{4.57}$$

Using this value in the equation of motion gives the relation

$$\frac{\dot{r}^2}{c^2\left(1 - \frac{2m}{r}\right)^2} = 1 - \frac{r_0^2}{r^2}\frac{\left(1 - \frac{2m}{r}\right)}{\left(1 - \frac{2m}{r_0}\right)}, \tag{4.58}$$

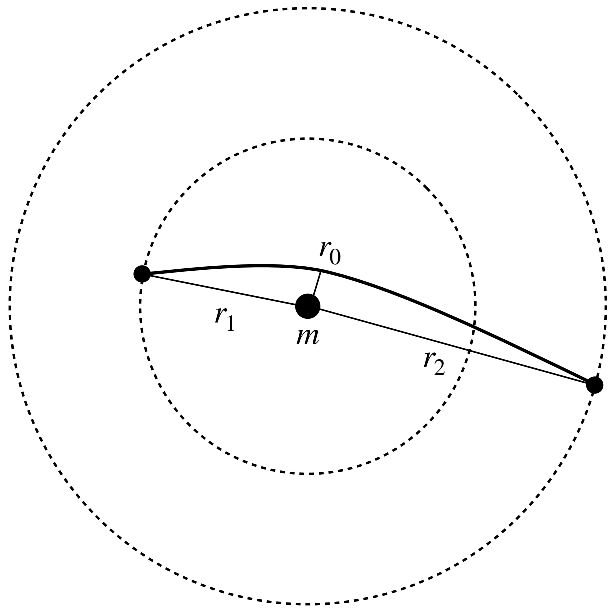

Figure 4.4: An electromagnetic radio signal is send from point $\mathbf{x}_1$ to point $\mathbf{x}_2$ in the spacetime background generated by a particle with mass m.

which is integrated to obtain the time interval that the radio signal requires to travel between two points. Evaluating the travel time from the point r_0 to an arbitrary point with coordinate r gives

$$c\left(t(r)-t(r_0)\right)=c\Delta t_{r,r_0}=\int_{r_0}^{r}\frac{dr}{\left(1-\frac{2m}{r}\right)}\left[1-\frac{r_0^2}{r^2}\frac{\left(1-\frac{2m}{r}\right)}{\left(1-\frac{2m}{r_0}\right)}\right]^{-1/2} \tag{4.59}$$

Assuming that $r \gg 2M$ and $r_0 \gg 2M$, it is possible to approximate the integrand to obtain

$$c\Delta t_{r,r_0}=\int_{r_0}^{r}\frac{r}{\sqrt{r^2-r_0^2}}\left[1+\frac{2m}{r}+\frac{mr_0}{r(r+r_0)}\right]dr, \tag{4.60}$$

and the integration gives the time interval

$$c\Delta t_{r,r_0}=\sqrt{r^2-r_0^2}+2m\ln\left[\frac{r+\sqrt{r^2-r_0^2}}{r_0}\right]+m\sqrt{\frac{r-r_0}{r+r_0}}. \tag{4.61}$$

The above results let us write the total time that the radio signal requires to go from point $\mathbf{r}_1$ to point $\mathbf{r}_2$, when $|\varphi_1-\varphi_2|>\frac{\pi}{2}$, as

$$\Delta t_{r_1,r_2}=\Delta t_{r_1,r_0}+\Delta t_{r_2,r_0}. \tag{4.62}$$

Now, if the spacetime were flat, the total time that the radio signal requires to go from point $\mathbf{r}_1$ to point $\mathbf{r}_2$ would be given by

$$c\Delta t_{r_1,r_2}^{(flat)}=\sqrt{r_1^2-r_0^2}+\sqrt{r_2^2-r_0^2}. \tag{4.63}$$

We can evaluate the difference between the time interval that the radio signal requires to travel in a curved space and the time interval required in a flat space,

$$\Delta t_{1,2}=\Delta t_{r_1,r_2}-\Delta t_{r_1,r_2}^{(flat)}. \tag{4.64}$$

The Shapiro's delay time is defined as twice this result, i.e. the time difference when considering the travel from point 1 to point 2 and back,

$$\Delta t_S = \frac{4m}{c}\ln\left[\frac{\left(r_1+\sqrt{r_1^2-r_0^2}\right)\left(r_2+\sqrt{r_2^2-r_0^2}\right)}{r_0^2}\right] + \frac{2m}{c}\left(\sqrt{\frac{r_1-r_0}{r_1+r_0}}+\sqrt{\frac{r_2-r_0}{r_2+r_0}}\right). \tag{4.65}$$

When the geometry of the system is such that $r_1 \gg r_0$ and $r_2 \gg r_0$, the above expression for the time delay may be approximated to

$$\Delta t_S \approx \frac{4m}{c}\left[1+\ln\frac{4r_1r_2}{r_0^2}\right]. \tag{4.66}$$

As an example, consider a radio signal emitted from Earth to Mars and back in a trajectory passing near the surface of the Sun. Using the values in the Appendix A, equation (4.66) gives a Shapiro's time delay of 0.26694185 ms.

4.5 Gravitational Redshift

Another consequence of the interaction of electromagnetic radiation with gravity, is a shift of its frequency. In order to calculate this effect in a Schwarzschild's background, consider an observer located at the position $\mathbf{x}_1 = (r_1, \theta, \varphi)$ that emits an electromagnetic signal radially to be received by another observer located at the point $\mathbf{x}_2 = (r_2, \theta, \varphi)$, with $r_2 > r_1$. Suppose that the electromagnetic emission is a monochromatic ray with frequency ν_1, as measured by the observer at $\mathbf{x}_1$. If the signal is produced during a time interval Δt_1, the number of wavefronts emitted is $n = \nu_1 \Delta t_1$. The receiver, located at $\mathbf{x}_2$ will report a radiation signal with frequency ν_2 during a time interval Δt_2, corresponding to a number of wavefronts $n = \nu_2 \Delta t_2$. Since this number is an invariant, the relation between frequencies and time intervals is

$$\frac{\nu_1}{\nu_2} = \frac{\Delta t_2}{\Delta t_1}. \tag{4.67}$$

Using the condition $ds^2 = 0$ in the line element (4.2) the ratio of time intervals for the observers, giving the condition

$$\frac{\nu_1}{\nu_2} = \sqrt{\frac{\left(1-\frac{2m}{r_2}\right)}{\left(1-\frac{2m}{r_1}\right)}}. \tag{4.68}$$

Hence, the relative change in frequency, using $\Delta\nu = \nu_2 - \nu_1$ is given by

$$\frac{\Delta\nu}{\nu_2} = \frac{\sqrt{\left(1-\frac{2m}{r_1}\right)} - \sqrt{\left(1-\frac{2m}{r_2}\right)}}{\sqrt{\left(1-\frac{2m}{r_1}\right)}} \tag{4.69}$$

Assuming that $r_1 \gg 2m$ and $r_2 \gg 2m$, the above expression can be approximated as

$$\frac{\Delta\nu}{\nu_2} \approx -\frac{2m}{r_1} + \frac{2m}{r_2}. \tag{4.70}$$

Since $r_2 > r_1$, this expression implies that $\Delta\nu < 0$, i.e. there is a reduction in the frequency, indicating a redshift due to gravity.

Exercise 4.7

Gravitational Redshift 1.
Estimate the relative gravitational redshift for a electromagnetic signal emitted from the surface of Earth and received by a satellite located on a geostationary orbit, at a constant height of 35786 *km above Earth's surface.*

Exercise 4.8

Gravitational Redshift 2.
The gravitational redshift was first measured in 1960 by R. Pound and G. Rebka at a building at the University of Harvard [8, 11, 12]. Estimate the relative gravitational redshift for a electromagnetic signal emitted at the first floor of the building and received 22.5 *m. above.*

Part III

Weak Gravitational Field

Chapter 5

Linearized Gravity

Almost all of the gravitational systems in nature are weak, including the Solar system, binaries and other planetary systems including stars, galaxies and star clusters, etc*. In this chapter, we will study the first and simplest approximation of the equations of General Relativity theory, which corresponds to a linearized theory describing the behavior of a weak gravitational field that can be directly compared with the Newtonian gravity.

For physical systems bound by gravitational forces, the adimensional Newtonian gravitational potential ($\frac{\phi}{c^2} = \frac{GM}{c^2 R}$ with R a typical separation distance between the particles in the system) is related with the typical internal speeds of the particles v via the Virial theorem,

$$\frac{v^2}{c^2} \sim \frac{\phi}{c^2}. \tag{5.1}$$

We will said that a system is *non-relativistic* when its components have slow velocities, $v \ll c$, and weak gravitational fields $\frac{\phi}{c^2} \ll 1$. For example, in scenarios such as the Solar system, we have that the order of the adimensional potential is

$$\frac{|\phi|}{c^2} \lesssim \frac{GM_\odot}{c^2 R_\odot} \sim 10^{-6}. \tag{5.2}$$

Table 5.1 contains some typical values of the adimensional potential for astrophysical objects. Note that this parameter can be considered as small in all scenarios except for the neutron star and black hole cases.

Astrophysical object	$\frac{\phi}{c^2}$
Earth	$\sim 10^{-9}$
Sun	$\sim 10^{-6}$
White dwarf	$\sim 10^{-4}$
Neutron star	$\sim 10^{-1}$
Black Hole	~ 1

Table 5.1: Typical values of the adimensional Newtonian gravitational potential on the surface of some astrophysical objects. Note the very small values, except for the neutron star and black hole cases.

Following these considerations and due to equation (5.1), we will introduce a small parameter ϵ which will

*The relativistic description of strong gravitational fields is only important when modeling systems including neutron stars and/or black holes help us to keep track of the small quantities in the approximation,

$$\epsilon \sim \frac{v}{c} \sim \left(\frac{\phi}{c^2}\right)^{1/2}. \tag{5.3}$$

It is also true that in many weakly interacting gravitational systems, the source is weakly stressed. This condition is written in terms of the components of the energy-momentum tensor as

$$\frac{\left|T^{ij}\right|}{T^{00}} \sim \mathcal{O}(\epsilon^2). \tag{5.4}$$

For example, for a perfect fluid characterized by a density ρ and a pressure p this condition states that $\frac{p}{\rho} \sim \mathcal{O}(\epsilon^2)$. More about these conditions will be presented below.

5.1 Nearly Flat Spacetimes and the Newtonian Limit

In a region where the gravitational field is sufficiently weak, the spacetime will be nearly flat. Hence, there will exist a coordinate system[†] for which the metric tensor can be decomposed as

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \tag{5.5}$$

where the symmetric tensor $h_{\mu\nu} = h_{\nu\mu}$ represents a collection of ten small independent fields satisfying $\left|h_{\mu\nu}\right| \sim \mathrm{h} \ll 1$. Along this chapter, we will present a linearized theory of gravity, in which we will expand all the equations up to first order in h.

Our first calculation will be the inverse metric, defined by the relation $g^{\mu\sigma}g_{\sigma\nu} = \delta^\mu_\nu$. Hence, neglecting terms of order $\mathcal{O}(\mathrm{h}^2)$ and higher, this is

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + \mathcal{O}(\mathrm{h}^2) \tag{5.6}$$

where $h^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}h_{\alpha\beta}$. With the same approximation, the connections are expanded as

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2}\eta^{\alpha\sigma}\left[\partial_\mu h_{\sigma\nu} + \partial_\nu h_{\mu\sigma} - \partial_\sigma h_{\mu\nu}\right] + \mathcal{O}(\mathrm{h}^2). \tag{5.7}$$

5.1.1 Equations of Motion for a Slowly Moving Particle

Consider a slowly moving particle, i.e. with a velocity $|\dot{\mathbf{x}}| \ll c$. From the geodesic equations (3.44) describing its motion, it is possible to conclude that its proper time τ satisfies

$$\frac{dt}{d\tau} \approx 1, \tag{5.8}$$

and that the equations of motion are approximated to

$$\frac{d^2x^j}{dt^2} \approx \frac{d^2x^j}{d\tau^2} \approx -c^2\Gamma^j_{00}. \tag{5.9}$$

Using the linear approximation of the connections (5.7), this gives

$$\frac{d^2x^j}{dt^2} = \frac{1}{2}c^2\left(\partial_j h_{00} - 2\partial_0 h_{0j}\right) + \mathcal{O}(\mathrm{h}^2). \tag{5.10}$$

[†] From now on, we will assume that we are using quasi-galilean or cartesian coordinates, such that the flat spacetime metric will be given by the Minkowski tensor $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$

If we additionally consider that the gravitational field is quasi-stationary, the second term in the connection can be neglected, $\partial_0 h_{0j}$, and therefore

$$\frac{d^2x^j}{dt^2} = \frac{1}{2}c^2\partial_j h_{00} + \mathcal{O}(\mathrm{h}^2). \tag{5.11}$$

Comparison with the Newtonian equation of motion of a particle under the influence of a gravitational potential ϕ as in equation (1.12),

$$\ddot{\mathbf{r}} = -\boldsymbol{\nabla}\phi(r), \tag{5.12}$$

implies the identification of the metric component h_{00} with the potential,

$$h_{00} = -\frac{2\phi}{c^2}. \tag{5.13}$$

Although this result is very interesting, it is not complete. In fact, the linear approximation of general relativity implies the existence of other metric contributions that will be obtained by writing the linearized field equations.

5.1.2 Linearized Curvature Tensors

Using the linearized connections (5.7), we calculate the Riemann tensor, defined in (3.6), as

$$R_{\alpha\mu\beta\nu} = \frac{1}{2}\left[\partial_\mu\partial_\beta h_{\alpha\nu} + \partial_\alpha\partial_\nu h_{\mu\beta} - \partial_\mu\partial_\nu h_{\alpha\beta} - \partial_\alpha\partial_\beta h_{\mu\nu}\right] + \mathcal{O}(\mathrm{h}^2). \tag{5.14}$$

The Ricci tensor of curvature is obtained contracting Riemann,

$$R_{\mu\nu} = \frac{1}{2}\left[\partial_\sigma\partial_\mu h^\sigma_\nu + \partial_\sigma\partial_\nu h^\sigma_\mu - \partial_\mu\partial_\nu h - \Box h_{\mu\nu}\right] + \mathcal{O}(\mathrm{h}^2), \tag{5.15}$$

where $h = h^\sigma_\sigma = \eta^{\sigma\rho}h_{\sigma\rho}$ and the D'Alambert operator is defined as $\Box = \eta^{\alpha\beta}\partial_\alpha\partial_\beta$. Contraction of this tensor gives the curvature scalar

$$R = \partial_\mu\partial_\nu h^{\mu\nu} - \Box h + \mathcal{O}(\mathrm{h}^2). \tag{5.16}$$

Finally, using the above expressions we write the linearized Einstein tensor as

$$G_{\mu\nu} = \frac{1}{2}\left[\partial_\sigma\partial_\mu h^\sigma_\nu + \partial_\sigma\partial_\nu h^\sigma_\mu - \partial_\mu\partial_\nu h - \Box h_{\mu\nu}\right] + \frac{1}{2}\eta_{\mu\nu}\left[\partial_\mu\partial_\nu h^{\mu\nu} - \Box h\right] + \mathcal{O}(\mathrm{h}^2) \tag{5.17}$$

5.2 The Energy Momentum Tensor

In order to have a consistent picture of a nearly flat spacetime, we must consider that the energy-momentum contribution of the source is small and that the characteristic velocities of the matter distribution are small compared with the speed of light, as in equations (5.3) and (5.4). This assumptions let us neglect the pressure contribution of a matter distribution such an ideal gas. Therefore, in situations such as the description of the Solar system, we may consider the energy-momentum tensor of an incoherent matter source (5.18),

$$T^{\mu\nu} = \frac{\varepsilon}{c^2}u^\mu u^\nu. \tag{5.18}$$

The assumption of small velocities, $\frac{v}{c} \sim \mathcal{O}(\varepsilon)$, gives the approximation

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} = 1 + \mathcal{O}(\epsilon^2), \tag{5.19}$$

and therefore, the energy-momentum tensor components for an incoherent matter source satisfy

$$T^{00} = \frac{\varepsilon}{c^2} u^0 u^0 = \varepsilon + \mathcal{O}(\epsilon^2) \tag{5.20a}$$

$$T^{0j} = \frac{\varepsilon}{c^2} u^0 u^j = \varepsilon \frac{v^j}{c} + \mathcal{O}(\epsilon^3) \tag{5.20b}$$

$$T^{jk} = \frac{\varepsilon}{c^2} u^j u^k = \mathcal{O}(\epsilon^2). \tag{5.20c}$$

We must note that for non-relativistic velocities v, space and time derivatives are of order

$$\partial_0 \sim \frac{1}{R}\frac{v}{c} \sim \frac{\epsilon}{R} \tag{5.21}$$

$$\partial_j \sim \frac{1}{R} \tag{5.22}$$

where R is a typical separation distance between the particles in the system. This expression are summarized in the relation

$$\partial_0 = \mathcal{O}(\varepsilon)\, \partial_j. \tag{5.23}$$

Hence, the conservation of energy $\nabla_n T^{\mu\nu} = 0$, will be linearized as

$$\partial_\nu T^{\mu\nu} = 0 + \mathcal{O}(\epsilon^2). \tag{5.24}$$

The presence of a partial derivative, and not a covariant one, in this equation implies that the gravitational field generated by the tensor $T^{\mu\nu}$ doesn't react back on the source, allowing the exchange of energy and momentum between the matter fields but not with the gravitational field. As a result of this fact, for the incoherent matter source described in equation (5.18), we can use the continuity equation $\partial_\nu \left(\rho u^\nu\right) = 0$ and (5.24) to obtain the condition $u^\nu \partial_\nu u^\mu = \frac{du^\mu}{d\tau} = 0$, implying that the integral lines of the velocity are straight lines.

5.3 The Linearized Field Equations

Under the above considerations concerning the small velocities and weak field approximations, Einstein's field equations,

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{5.25}$$

are linearized using equation (5.17) for Einstein's tensor and assuming conditions (5.20) for the energy-momentum tensor,

$$\frac{1}{2}\left[\partial_\sigma \partial_\mu h^\sigma_\nu + \partial_\sigma \partial_\nu h^\sigma_\mu - \partial_\mu \partial_\nu h - \Box h_{\mu\nu}\right] + \frac{1}{2}\eta_{\mu\nu}\left[\partial_\alpha \partial_\beta h^{\alpha\beta} - \Box h\right] + \mathcal{O}(\mathrm{h}^2) = \frac{8\pi G}{c^4} T_{\mu\nu}. \tag{5.26}$$

In the linear approximation, and similarly to what happens with equation (5.24), the Binachi identities (3.14) are written using partial derivatives

$$\partial_\nu G^{\mu\nu} = 0 + \mathcal{O}(h^2). \tag{5.27}$$

5.4 Gauge Transformations

Although General Relativity permits any coordinate transformation, the particular choice of the metric tensor (5.5) restricts the coordinate freedom to those that preserve the desired decomposition. Thus, the allowed transformations in the linearized theory include *Lorentz boosts*, *spatial rotations* (because both of them doesn't change the Minkowskian metric and transform $h_{\mu\nu}$ as a Minkowskian tensor) and *infinitesimal transformations* such as

$$x'^{\mu} = x^{\mu} + \xi^{\mu}(x), \tag{5.28}$$

where $\xi^{\mu} \sim \mathcal{O}(\mathrm{h})$. The inverse transformation, neglecting orders $\mathcal{O}(\mathrm{h}^2)$ and higher, is

$$x^{\mu} = x'^{\mu} - \xi^{\mu}(x). \tag{5.29}$$

The transformation law for the metric tensor gives

$$g'_{\mu\nu} = \eta_{\mu\nu} + h'_{\mu\nu} + \mathcal{O}(\mathrm{h}^2), \tag{5.30}$$

where

$$h'_{\mu\nu} = h_{\mu\nu} - \partial_{\mu}\xi_{\nu} - \partial_{\nu}\xi_{\mu} \tag{5.31}$$

and $\xi_{\mu} = \eta_{\mu\sigma}\xi^{\sigma}$. Because of the resemblance of this equation with the gauge transformations of the electromagnetic potential in electrodynamics, equation (5.31) is called the *gauge transformation* generated by the vector field ξ^{μ} ‡.

Exercise 5.1

Connections and Curvature under Gauge Transformations

Show that, under the gauge transformation (5.31), the linearized connections (5.7) change with terms involving $\partial_{\alpha}\partial_{\beta}\xi^{\mu}$, while the Riemann and Ricci tensors, as well as the curvature scalar in equations (5.14), (5.15) and (5.16) are invariant.
This result implies that the complete information about the gravitational field is contained in the curvature tensors and not in the connections.

The gauge freedom may be used to simplify the form of the field equations. First, we change $h_{\mu\nu}$ in equation (5.26) by the *trace-reversed* function

$$\bar{h}_{\mu\nu} := h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h, \tag{5.32}$$

which satisfies $\bar{h} = -h$. It is easy to show that the inverse relation is

$$h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h}. \tag{5.33}$$

The field equations in terms of the trace-reversed function is

$$-\frac{1}{2}\left[\Box\bar{h}_{\mu\nu} - \partial_{\mu}\partial_{\sigma}\bar{h}^{\sigma}_{\nu} - \partial_{\sigma}\partial_{\nu}\bar{h}^{\sigma}_{\mu} - \eta_{\mu\nu}\partial_{\sigma}\partial_{\rho}\bar{h}^{\sigma\rho}\right] + \mathcal{O}(\mathrm{h}^2) = \frac{8\pi G}{c^4}T_{\mu\nu}. \tag{5.34}$$

The gauge freedom is used now by imposing the *Lorenz-Hilbert gauge* condition,

$$\partial_{\nu}\bar{h}^{\mu\nu} = 0, \tag{5.35}$$

‡It is important to note that this gauge transformation freedom is just the usual general covariance of General Relativity applied to the linearized theory and doesn't represent a new symmetry.

which reduces the field equations to the simple, linearized expression

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}. \tag{5.36}$$

Exercise 5.2

Generating Field for the Lorenz Gauge

In this exercise, it will be shown that the Lorentz-Hilbert gauge may be generated by a infinitesimal coordinate transformation.

Consider a perturbation $h^{\mu\nu}$ and apply to it an infinitesimal transformation with the form (5.28) to obtain the new perturbation $h'^{\mu\nu}$. Now, define the corresponding trace-reversed function to obtain a perturbation $\bar{h}'^{\mu\nu}$ in terms of $h'_{\mu\nu}$ and the derivatives of the field ξ^μ. Show that, although $h'^{\mu\nu}$ doesn't satisfy Lorenz condition, i.e. $\partial_\nu h'^{\mu\nu} \neq 0$, we can make that $\partial_\nu \bar{h}'^{\mu\nu} = 0$ by imposing that the vector field ξ^μ satisfies the condition $\Box \xi^\mu = \partial_\nu \tilde{h}^{\mu\nu}$.

5.4.1 Solution of the Wave Equation

The wave equation (5.36) is solved by using the retarded Green's function $\mathcal{G}(x,x')$, which is a solution of the equation

$$\Box \mathcal{G}(x,x') = -4\pi\delta(x-x') = -4\pi\delta(ct-ct')\delta(\mathbf{x}-\mathbf{x}'). \tag{5.37}$$

As is shown in Appendix C, the retarded Green function is

$$\mathcal{G}_R(x,x') = 2\Theta\left(ct-ct'\right)\delta\left((ct-ct')^2 - \left|\mathbf{x}-\mathbf{x}'\right|^2\right), \tag{5.38}$$

where

$$\Theta(z) = \begin{cases} 1 \text{ if } z>0 \\ 0 \text{ if } z<0 \end{cases} \tag{5.39}$$

is the Heaviside step function. Equivalently, the retarded Green function can be written as

$$\mathcal{G}_R(x,x') = \frac{\delta\left(ct-ct'-|\mathbf{x}-\mathbf{x}'|\right)}{|\mathbf{x}-\mathbf{x}'|}. \tag{5.40}$$

In terms of this function, the general solution of equation (5.36) is

$$\bar{h}_{\mu\nu}(x) = \frac{4G}{c^4}\int \mathcal{G}_R(x,x')T_{\mu\nu}(x')d^4x' = \frac{4G}{c^4}\int \frac{\delta\left(ct-ct'-|\mathbf{x}-\mathbf{x}'|\right)}{|\mathbf{x}-\mathbf{x}'|}T_{\mu\nu}(x')d^4x', \tag{5.41}$$

which, after integration, gives

$$\bar{h}_{\mu\nu}(x) = \frac{4G}{c^4}\int \frac{T_{\mu\nu}\left(ct-|\mathbf{x}-\mathbf{x}'|,\mathbf{x}'\right)}{|\mathbf{x}-\mathbf{x}'|}d^3x'. \tag{5.42}$$

Note that the energy-momentum tensor in the integrand is evaluated at the retarded time $ct_{ret} = ct - |\mathbf{x}-\mathbf{x}'|$.

5.5 The Nearly Newtonian Limit Metric

From the order of the partial derivative operators in the small velocities approximation, given in equation (5.23), we can approximate the D'Alambertian operator as

$$\Box = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2 = \left[1 + \mathcal{O}(\epsilon^2)\right]\nabla^2, \tag{5.43}$$

so that it reduces to the Laplacian operator when neglecting orders $\mathcal{O}(\epsilon^2)$ an higher. Hence, the wave equation (5.36) reduces to Poisson equation,

$$\nabla^2 \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}, \tag{5.44}$$

and the solution of this equation can be expressed in components as

$$\bar{h}^{00}(x) = -\frac{4}{c^2}\phi + \mathcal{O}\left(\epsilon^2\right) \tag{5.45a}$$

$$\bar{h}^{0j}(x) = -\frac{4}{c^3}\zeta^j + \mathcal{O}\left(\epsilon^2\right) \tag{5.45b}$$

$$\bar{h}^{jk}(x) = \mathcal{O}\left(\epsilon^2\right), \tag{5.45c}$$

where

$$\phi(x) = -\frac{G}{c^2}\int \frac{T^{00}(x')}{|\mathbf{x}-\mathbf{x}'|} d^3x' = -G\int \frac{\rho(x')}{|\mathbf{x}-\mathbf{x}'|} d^3x' \tag{5.46}$$

is the Newtonian gravitational potential, satisfying the Poisson equation

$$\nabla^2\phi = 4\pi G\rho, \tag{5.47}$$

while the potential ζ^j is given by

$$\zeta^j(x) = -\frac{G}{c}\int \frac{T^{0j}(x')}{|\mathbf{x}-\mathbf{x}'|} d^3x' = -G\int \frac{\rho(x')v^j}{|\mathbf{x}-\mathbf{x}'|} d^3x' \tag{5.48}$$

and satisfies the differential equation

$$\nabla^2\zeta^j = 4\pi G\rho v^j. \tag{5.49}$$

Given these functions, we obtain the trace of the perturbation as

$$\bar{h} = \frac{4}{c^2}\phi + \mathcal{O}\left(\epsilon^4\right) \tag{5.50}$$

and using the equation (5.33) we get back the coefficients for the original perturbation,

$$h^{00}(x) = -\frac{2}{c^2}\phi + \mathcal{O}\left(\epsilon^4\right) \tag{5.51a}$$

$$h^{0j}(x) = -\frac{4}{c^3}\zeta^j + \mathcal{O}\left(\epsilon^5\right) \tag{5.51b}$$

$$h^{jk}(x) = -\frac{2}{c^2}\phi\delta^{jk} + \mathcal{O}\left(\epsilon^4\right). \tag{5.51c}$$

Lowering the indices with $\eta_{\mu\nu}$, we obtain the components

$$h_{00}(x) = -\frac{2}{c^2}\phi + \mathcal{O}\left(\epsilon^4\right) \tag{5.52a}$$

$$h_{0j}(x) = \frac{4}{c^3}\zeta_j + \mathcal{O}\left(\epsilon^5\right) \tag{5.52b}$$

$$h_{jk}(x) = -\frac{2}{c^2}\phi\delta_{jk} + \mathcal{O}\left(\epsilon^4\right). \tag{5.52c}$$

and therefore the complete metric tensor is recovered as

$$g_{00}(x) = -1 - \frac{2}{c^2}\phi + \mathcal{O}\left(\epsilon^4\right) \tag{5.53a}$$

$$g_{0j}(x) = \frac{4}{c^3}\zeta_j + \mathcal{O}\left(\epsilon^5\right) \tag{5.53b}$$

$$g_{jk}(x) = \left(1 - \frac{2}{c^2}\phi\right)\delta_{jk} + \mathcal{O}\left(\epsilon^4\right). \tag{5.53c}$$

The results in this chapter show that the Newtonian gravitational potential appear in the metric coefficients and that the linear approximation makes sense because the functions $h_{\mu\nu}$ are small. Indeed, the order of the perturbations in the metric can be related with the order ϵ defined at the beginning of this chapter,

$$\left|h_{\mu\nu}\right| \sim \mathcal{O}\left(\mathrm{h}\right) \sim \frac{|\phi|}{c^2} \sim \mathcal{O}(\epsilon^2). \tag{5.54}$$

Therefore, from now on, we will use only the infinitesimal quantity ϵ to keep track of the order of the expansions. Finally, it is important to note that the term appearing in the coefficient g_{0j}, in equation (5.53b), is not exactly a Newtonian contribution. In fact, while the terms proportional to $\frac{|\phi|}{c^2}$ are of order $\mathcal{O}(\mathrm{h}) \sim \mathcal{O}(\epsilon^2)$, we have that $\frac{|\zeta_j|}{c^2} \sim \mathcal{O}(\epsilon^3)$. However, we include it here because it appears in a natural way and because we have decided to neglect orders $\mathcal{O}(\mathrm{h}^2) \sim \mathcal{O}(\epsilon^4)$ in all equations consistently, and this term is not of that order. Thus, we must emphasize that this term corresponds to the first contribution beyond the Newtonian approximation appearing in the expansions. In the next chapter, we will continue the series expansion, including high order terms in the so-called *post-Newtonian approximation*.

Chapter 6

The Post-Newtonian Approximation

In the previous chapter, we use a linearization of the Einstein's field equations to show that Poisson equation is included into the formulation of general relativity and that the Newtonian gravitational potential is included into the components of the metric tensor. An interesting result was obtained in equation (5.53b), which corresponds to a non-Newtonian contribution to the gravitational field. Following this result, in this chapter we will continue the series expansion of the metric tensor, including high order terms and therefore obtaining new relativistic contributions to the gravitational field. This formulation is called *post-Newtonian approximation* and will be constructed by assuming systems with slow motions and weak gravitational fields.

6.1 Required Order of the Metric Tensor in the Post-Newtonian Approximation

The construction of the post-Newtonian approximation depends on the context in which we are interested. Since we want to study the motion of particles, we will consider a Lagrangian with the form of equation (3.43),

$$L = -m_0 c\sqrt{-g_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau}}, \tag{6.1}$$

describing a particle with proper mass m_0 and with proper time τ. We will use the approximation of slow motion (5.19) and the nearly Newtonian metric given in equations (5.53) to obtain

$$L = -m_0c^2 + \frac{1}{2}m_0v^2 + m_0\phi + \mathcal{O}(\epsilon^2), \tag{6.2}$$

which corresponds to the Newtonian lagrangian, except for the first term which is an irrelevant constant. This result let us conclude that the contribution of order $\mathcal{O}(\epsilon^2)$ in g_{00} is enough to reproduce the Newtonian gravity and that the first post-Newtonian (denoted *1PN*) corrections will be given by the neglected contributions, $\mathcal{O}(\epsilon^4)$ in g_{00}, $\mathcal{O}(\epsilon^3)$ in g_{0j} and $\mathcal{O}(\epsilon^2)$ in g_{jk}.

In general, the nPN (n-th Post-Newtonian) contribution requires to calculate the metric to the orders

$$\mathcal{O}\left(\epsilon^{2n+2}\right) \text{ for } g_{00} \tag{6.3a}$$

$$\mathcal{O}\left(\epsilon^{2n+1}\right) \text{ for } g_{0j} \tag{6.3b}$$

$$\mathcal{O}\left(\epsilon^{2n}\right) \text{ for } g_{jk}. \tag{6.3c}$$

For example, in Table 6.1 we show the required values of the parameter ϵ in the metric expansion for a given nPN order.

Order	0PN	0.5PN	1PN	1.5PN	2PN
g_{00}	ϵ^2	ϵ^3	ϵ^4	ϵ^5	ϵ^6
g_{0j}	ϵ	ϵ^2	ϵ^3	ϵ^4	ϵ^5
g_{jk}	ϵ^0	ϵ	ϵ^2	ϵ^3	ϵ^4

Table 6.1: Order of the metric terms needed to calculate the nPN contribution in terms of the parameter ϵ.

6.2 The 1PN Approximation

According to the discussion in the above section, we will obtain the metric coefficients for a 1PN approximation. This require an expansion of the metric tensor and the energy-momentum tensor in powers of the parameter ϵ. However, before we begin it is important to note that, neglecting gravitational radiation, a gravitational system subject to conservative forces must be invariant under time inversion, $t \to -t$. In order to preserve this invariance, and remembering that the expansions are given in powers of $\epsilon \sim \frac{v}{c}$, the metric coefficients g_{00} and g_{jk} must contain only even powers of ϵ, while the coefficients g_{0j} must contain only odd powers of ϵ. According to equation (5.53) and the above argument, we expand the metric as

$$g_{00} = -1 + {}^{(2)}h_{00} + {}^{(4)}h_{00} + \mathcal{O}\left(\epsilon^6\right) \tag{6.4a}$$
$$g_{0j} = {}^{(3)}h_{0j} + \mathcal{O}\left(\epsilon^5\right) \tag{6.4b}$$
$$g_{jk} = \delta_{jk} + {}^{(2)}h_{jk} + \mathcal{O}\left(\epsilon^4\right). \tag{6.4c}$$

Exercise 6.1

Inverse of the Expanded Metric

Using the relation $g^{\mu\sigma}g_{\sigma\nu} = \delta^\mu_\nu$ show that the inverse of the power expanded metric (6.4) is

$$g^{00} = -1 + {}^{(2)}h^{00} + {}^{(4)}h^{00} + \mathcal{O}\left(\epsilon^6\right) \tag{6.5a}$$
$$g^{0j} = {}^{(3)}h^{0j} + \mathcal{O}\left(\epsilon^5\right) \tag{6.5b}$$
$$g^{jk} = \delta^{jk} + {}^{(2)}h^{jk} + \mathcal{O}\left(\epsilon^4\right), \tag{6.5c}$$

where

$${}^{(2)}h^{00} = -{}^{(2)}h_{00} \tag{6.6a}$$
$${}^{(4)}h^{00} = -{}^{(4)}h_{00} - {}^{(2)}h^2_{00} \tag{6.6b}$$
$${}^{(3)}h^{0j} = \delta^{jk\,(3)}h_{0k} \tag{6.6c}$$
$${}^{(2)}h^{jk} = -\delta^{ij}\delta^{lk\,(2)}h_{il}. \tag{6.6d}$$

6.2.1 Connections in the 1PN Approximation

In order to calculate the connections, we must differentiate the metric components. Remembering that space and time derivatives increase the order of the terms according to equation (5.23), i.e. $\partial_0 \sim \frac{1}{R}\frac{v}{c} \sim \frac{\epsilon}{R}$

and $\partial_j \sim \frac{1}{R}$, the resulting expansion up to 1PN order gives the terms

$$\Gamma^0_{00} = {}^{(3)}\Gamma^0_{00} + \mathcal{O}\left(\frac{\epsilon^5}{R}\right) \tag{6.7a}$$

$$\Gamma^0_{0j} = {}^{(2)}\Gamma^0_{0j} + \mathcal{O}\left(\frac{\epsilon^4}{R}\right) \tag{6.7b}$$

$$\Gamma^0_{jk} = {}^{(3)}\Gamma^0_{jk} + \mathcal{O}\left(\frac{\epsilon^5}{R}\right) \tag{6.7c}$$

$$\Gamma^i_{00} = {}^{(2)}\Gamma^i_{00} + {}^{(4)}\Gamma^i_{00} + \mathcal{O}\left(\frac{\epsilon^6}{R}\right) \tag{6.7d}$$

$$\Gamma^i_{0j} = {}^{(3)}\Gamma^i_{0j} + \mathcal{O}\left(\frac{\epsilon^5}{R}\right) \tag{6.7e}$$

$$\Gamma^i_{jk} = {}^{(2)}\Gamma^i_{jk} + \mathcal{O}\left(\frac{\epsilon^4}{R}\right) \tag{6.7f}$$

with the contributions *

$${}^{(3)}\Gamma^0_{00} = -\frac{1}{2}\partial_0{}^{(2)}h_{00} \tag{6.8a}$$

$${}^{(2)}\Gamma^0_{0j} = -\frac{1}{2}\partial_j{}^{(2)}h_{00} \tag{6.8b}$$

$${}^{(3)}\Gamma^0_{jk} = -\frac{1}{2}\left[\partial_j{}^{(3)}h_{0k} + \partial_k{}^{(3)}h_{0j} - \partial_0{}^{(2)}h_{jk}\right] \tag{6.8c}$$

$${}^{(2)}\Gamma^i_{00} = -\frac{1}{2}\partial_i{}^{(2)}h_{00} \tag{6.8d}$$

$${}^{(4)}\Gamma^i_{00} = -\frac{1}{2}\partial_i{}^{(4)}h_{00} + \partial_0{}^{(3)}h_{0i} + \frac{1}{2}{}^{(2)}h_{ij}\partial_j{}^{(2)}h_{00} \tag{6.8e}$$

$${}^{(3)}\Gamma^i_{0j} = \frac{1}{2}\left[\partial_0{}^{(2)}h_{ij} + \partial_j{}^{(3)}h_{0i} - \partial_i{}^{(3)}h_{0j}\right] \tag{6.8f}$$

$${}^{(2)}\Gamma^i_{jk} = \frac{1}{2}\left[\partial_k{}^{(2)}h_{ij} + \partial_j{}^{(2)}h_{ik} - \partial_i{}^{(2)}h_{jk}\right]. \tag{6.8g}$$

6.2.2 Ricci Tensor in the 1PN Approximation

The Ricci tensor is calculated in terms of the expansion of the connections (6.7), giving

$$R_{00} = {}^{(2)}R_{00} + {}^{(4)}R_{00} + \mathcal{O}\left(\epsilon^6\right) \tag{6.9a}$$

$$R_{0j} = {}^{(3)}R_{00} + \mathcal{O}\left(\epsilon^5\right) \tag{6.9b}$$

$$R_{jk} = {}^{(2)}R_{jk} + \mathcal{O}\left(\epsilon^4\right) \tag{6.9c}$$

where

$${}^{(2)}R_{00} = \partial_i{}^{(2)}\Gamma^i_{00} \tag{6.10a}$$

$${}^{(4)}R_{00} = \partial_i{}^{(4)}\Gamma^i_{00} - \partial_0{}^{(3)}\Gamma^i_{0i} + {}^{(2)}\Gamma^i_{00}{}^{(2)}\Gamma^j_{ij} - {}^{(2)}\Gamma^0_{0i}{}^{(2)}\Gamma^i_{00} \tag{6.10b}$$

$${}^{(3)}R_{0i} = \partial_0{}^{(2)}\Gamma^0_{0i} + \partial_j{}^{(3)}\Gamma^j_{0i} - \partial_i{}^{(3)}\Gamma^0_{00} - \partial_i{}^{(3)}\Gamma^j_{0j} \tag{6.10c}$$

$${}^{(2)}R_{ij} = \partial_k{}^{(2)}\Gamma^k_{ij} - \partial_j{}^{(2)}\Gamma^0_{0i} - \partial_j{}^{(2)}\Gamma^k_{ik}. \tag{6.10d}$$

*In the following expressions we use that $\delta^{ij}\partial_j = \partial_i$.

Using the equations (6.8), these components are written in terms of the metric tensor components as

$$^{(2)}R_{00} = -\frac{1}{2}\nabla^2\left[{}^{(2)}h_{00}\right] \tag{6.11a}$$

$$\begin{aligned} ^{(4)}R_{00} = &-\frac{1}{2}\nabla^2\left[{}^{(4)}h_{00}\right] + \partial_0\partial_i{}^{(3)}h_{0i} - \frac{1}{2}\partial_0\partial_0{}^{(2)}h_{ii} + \frac{1}{2}{}^{(2)}h_{ij}\partial_i\partial j^{(2)}h_{00} \\ &+\frac{1}{2}\partial_j{}^{(2)}h_{ij}\partial_i{}^{(2)}h_{00} - \frac{1}{4}\partial_i{}^{(2)}h_{00}\partial_i{}^{(2)}h_{jj} - \frac{1}{4}\partial_i{}^{(2)}h_{00}\partial_i{}^{(2)}h_{00} \end{aligned} \tag{6.11b}$$

$$^{(3)}R_{0i} = -\frac{1}{2}\partial_0\partial_i{}^{(2)}h_{jj} + \frac{1}{2}\partial_i\partial_j{}^{(3)}h_{0j} + \frac{1}{2}\partial_0\partial_j{}^{(2)}h_{ij} - \frac{1}{2}\nabla^2\left[{}^{(3)}h_{0i}\right] \tag{6.11c}$$

$$^{(2)}R_{ij} = \frac{1}{2}\partial_i\partial_j{}^{(2)}h_{00} - \frac{1}{2}\partial_i\partial_j{}^{(2)}h_{kk} + \frac{1}{2}\partial_k\partial_j{}^{(2)}h_{ik} + \frac{1}{2}\partial_k\partial_i{}^{(2)}h_{kj} - \frac{1}{2}\nabla^2\left[{}^{(2)}h_{ij}\right]. \tag{6.11d}$$

6.2.3 Fixing the Gauge

Imposing a gauge may simplify considerably the curvature tensors, the field equations and the whole mathematical description. Here, we will describe two of the possible gauges due to their applicability to gravitational radiation and to the celestial mechanics theory.

The Harmonic Gauge

The components of the Ricci tensor are conveniently simplified by choosing the so-called *harmonic gauge*. This was introduced by Einstein in 1912 [15] and states that the coordinate functions satisfies the harmonic condition

$$\Box x^\mu = 0. \tag{6.12}$$

However, this condition can be put in terms of the metric tensor components as

$$\partial_\nu\left(\sqrt{-g}g^{\mu\nu}\right) = 0, \tag{6.13}$$

or in terms of the connection as the relation

$$g^{\mu\nu}\Gamma^\alpha_{\mu\nu} = 0. \tag{6.14}$$

Using the post-Newtonian approximation of the metric, the harmonic gauge impose a restriction to the perturbation components $h_{\mu\nu}$. This gauge is appropriate for describing gravitational radiation, just as the Lorentz gauge is appropriate to describe electromagnetic radiation. Since celestial mechanics will be described in the near field approximation, we will not use the harmonic gauge here. Instead, we will introduce a more appropriate gauge.

Exercise 6.2

Harmonic gauge conditions
Consider the $\alpha = 0$ case of equation (6.14). Show that the vanishing of the order $\mathcal{O}(\epsilon^3)$ terms in

this equation corresponds to the condition

$$\frac{1}{2}\partial_0{}^{(2)}h_{00} - \partial_j{}^{(3)}h_{0j} + \frac{1}{2}\partial_0{}^{(2)}h_{jj} = 0. \tag{6.15}$$

Similarly, from the $\alpha = i$ case of equation (6.14), show that the vanishing of the order $\mathcal{O}(\epsilon^2)$ terms is given the condition

$$\frac{1}{2}\partial_i{}^{(2)}h_{00} + \partial_j{}^{(2)}h_{ij} - \frac{1}{2}\partial_i{}^{(2)}h_{jj} = 0. \tag{6.16}$$

The Standard Post-Newtonian Gauge

Relativistic celestial mechanics deals with slowly moving systems with weak gravitational fields in the near region. Therefore, the harmonic gauge is not the adequate for this description. In order to study the effects of general relativity in the celestial mechanics formulation we will follow the works of Will[24] and impose the so-called the *Standard Post-Newtonian Gauge*, given by the conditions

$$\partial_j{}^{(3)}h_{0j} - \frac{1}{2}\partial_0{}^{(2)}h_{jj} = 0 \tag{6.17a}$$

$$\frac{1}{2}\partial_i{}^{(2)}h_{00} + \partial_j{}^{(2)}h_{ij} - \frac{1}{2}\partial_i{}^{(2)}h_{jj} = 0. \tag{6.17b}$$

Comparing with equations (6.15) and (6.16), we note that these conditions are similar to those in the harmonic gauge, except for one term. Indeed, the nature of the standard PN gauge is analog to the conditions in the Coulomb gauge in electrodynamics.

Exercise 6.3

Ricci tensor under the standard PN gauge conditions
Show that under the standard Post-Newtonian gauge imposed by conditions (6.17a) and (6.17b), the Ricci tensor (6.11) simplifies to

$${}^{(2)}R_{00} = -\frac{1}{2}\nabla^2\left[{}^{(2)}h_{00}\right] \tag{6.18a}$$

$$\begin{aligned}{}^{(4)}R_{00} = &-\frac{1}{2}\nabla^2\left[{}^{(4)}h_{00}\right] + \frac{1}{2}\partial_j{}^{(2)}h_{ij}\partial_i{}^{(2)}h_{00} + \frac{1}{2}{}^{(2)}h_{ij}\partial_i\partial_j{}^{(2)}h_{00} \\ &- \frac{1}{4}\partial_i{}^{(2)}h_{00}\partial_i{}^{(2)}h_{jj} - \frac{1}{4}\partial_i{}^{(2)}h_{00}\partial_i{}^{(2)}h_{00}\end{aligned} \tag{6.18b}$$

$${}^{(3)}R_{0i} = -\frac{1}{2}\nabla^2\left[{}^{(3)}h_{0i}\right] + \frac{1}{2}\partial_0\partial_j{}^{(2)}h_{ij} - \frac{1}{4}\partial_0\partial_i{}^{(2)}h_{jj} \tag{6.18c}$$

$${}^{(2)}R_{ij} = -\frac{1}{2}\nabla^2\left[{}^{(2)}h_{ij}\right]. \tag{6.18d}$$

6.3 The Field Equations in the Post-Newtonian Approximation

Now that we have the components of the metric tensor and the Ricci tensor in the 1PN approximation, we are ready to set the field equations. We will write the expansion of the energy-momentum tensor, following the results of equation (5.20), as

$$T^{00} = {}^{(0)}T^{00} + {}^{(2)}T^{00} + \mathcal{O}\left(\epsilon^4\right) \tag{6.19a}$$
$$T^{0j} = {}^{(1)}T^{0j} + {}^{(3)}T^{0j} + \mathcal{O}\left(\epsilon^5\right) \tag{6.19b}$$
$$T^{ij} = {}^{(2)}T^{ij} + {}^{(4)}T^{ij} + \mathcal{O}\left(\epsilon^6\right). \tag{6.19c}$$

We will use the second form of the equations (3.37), for which we need to calculate the trace-reversed tensor $S_{\mu\nu} = T_{\mu\nu} - \frac{1}{2}g_{\mu\nu}T$. According to the expansion of the Ricci tensor in (6.18), the trace-reversed energy-momentum tensor shall be calculated up to the following orders†

$$S_{00} = {}^{(0)}S_{00} + {}^{(2)}S_{00} + \mathcal{O}\left(\epsilon^4\right) \tag{6.20a}$$
$$S_{0j} = {}^{(1)}S_{0j} + \mathcal{O}\left(\epsilon^3\right) \tag{6.20b}$$
$$S_{ij} = {}^{(0)}S_{ij} + \mathcal{O}\left(\epsilon^2\right). \tag{6.20c}$$

Using the trace

$$T = -{}^{(0)}T^{00} - {}^{(2)}T^{00} + {}^{(2)}h_{00}{}^{(0)}T^{00} + \delta_{ij}{}^{(2)}T^{ij} + \mathcal{O}\left(\epsilon^4\right) \tag{6.21}$$

we obtain the relevant components of $S_{\mu\nu}$ in the 1PN approximation as

$${}^{(0)}S_{00} = \frac{1}{2}{}^{(0)}T^{00} \tag{6.22a}$$
$${}^{(2)}S_{00} = \frac{1}{2}\left[{}^{(2)}T^{00} - 2{}^{(2)}h_{00}{}^{(0)}T^{00} + \delta_{ij}{}^{(2)}T^{ij}\right] \tag{6.22b}$$
$${}^{(1)}S_{0j} = -{}^{(1)}T^{0j} \tag{6.22c}$$
$${}^{(0)}S_{ij} = \frac{1}{2}\delta_{ij}{}^{(0)}T^{00}. \tag{6.22d}$$

All the previous results let us write the 1PN approximation of the field equations as

$${}^{(2)}R_{00} = \frac{8\pi G}{c^4}{}^{(0)}S_{00} \tag{6.23a}$$
$${}^{(4)}R_{00} = \frac{8\pi G}{c^4}{}^{(2)}S_{00} \tag{6.23b}$$
$${}^{(3)}R_{0i} = \frac{8\pi G}{c^4}{}^{(1)}S_{0i} \tag{6.23c}$$
$${}^{(2)}R_{ij} = \frac{8\pi G}{c^4}{}^{(0)}S_{ij}. \tag{6.23d}$$

These field equations can be re-written defining some potential functions. Using the Ricci tensor components (6.18) and equation (6.22) in the second order equation (6.23a) gives

$$\nabla^2\left[{}^{(2)}h_{00}\right] = -\frac{8\pi G}{c^4}{}^{(0)}T^{00}. \tag{6.24}$$

Writing the corresponding metric component as

$${}^{(2)}h_{00} = -\frac{2}{c^2}\phi, \tag{6.25}$$

†Remember that the trace reversed tensor $S_{\mu\nu}$ is accompanied by the constant $\frac{8\pi G}{c^4}$, which affects the order of these terms in the field equations.

we obtain the potential

$$\phi(x) = -\frac{G}{c^2}\int \frac{^{(0)}T^{00}(t,\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}d^3x'. \tag{6.26}$$

Note that this component is exactly the same as the one obtained in equation (5.52a), corresponding to the Newtonian potential contribution. Similarly, the second order equation (6.23d) gives

$$\nabla^2\left[^{(2)}h_{ij}\right] = -\frac{8\pi G}{c^4}\delta_{ij}{}^{(0)}T^{00}, \tag{6.27}$$

from which we obtain the metric component

$$^{(2)}h_{ij} = -\frac{2}{c^2}\delta_{ij}\phi \tag{6.28}$$

with the same potential ϕ as in equation (6.26). This result is the same contribution obtained in equation (5.52c).

Now, we lead our attention to the fourth order equation (6.23b). Note that the relevant component of the Ricci tensor, using the above results, is

$$^{(4)}R_{00} = -\frac{1}{2}\nabla^2\left[^{(4)}h_{00}\right] + \frac{2}{c^4}\phi\nabla^2\phi - \frac{2}{c^4}\boldsymbol{\nabla}\phi\cdot\boldsymbol{\nabla}\phi. \tag{6.29}$$

Similarly, the corresponding component of the trace-reversed energy-momentum tensor is

$$^{(2)}S_{00} = \frac{1}{2}\left[^{(2)}T^{00} - \frac{4\phi}{c^2}{}^{(0)}T^{00} + {}^{(2)}T^{jj}\right]. \tag{6.30}$$

Using these relations and the vector identity $\boldsymbol{\nabla}\phi\cdot\boldsymbol{\nabla}\phi = \frac{1}{2}\nabla^2\left(\phi^2\right) - \phi\nabla^2\phi$, the field equation (6.23b) is written as

$$\nabla^2\left[^{(4)}h_{00} + \frac{2}{c^4}\phi^2\right] = -\frac{8\pi G}{c^4}\left(^{(2)}T^{00} + {}^{(2)}T^{jj}\right). \tag{6.31}$$

We introduce a gravitational potential ψ to write the metric component $^{(4)}h_{00}$ as

$$^{(4)}h_{00} = -\frac{2}{c^2}\left(\frac{\phi^2}{c^2} + \psi\right) \tag{6.32}$$

where ϕ is given by equation (6.26). The resulting equation gives

$$\psi(x) = -\frac{G}{c^2}\int\left[^{(2)}T^{00}(t,\mathbf{x}') + {}^{(2)}T^{jj}(t,\mathbf{x}')\right]\frac{d^3x'}{|\mathbf{x}-\mathbf{x}'|}. \tag{6.33}$$

Finally, considering the third order equation (6.23c), we use the above results to obtain the equation

$$\nabla^2\left[^{(3)}h_{0i}\right] = \frac{16\pi G}{c^4}{}^{(1)}T^{0i} + \frac{1}{c^2}\partial_0\partial_i\phi. \tag{6.34}$$

Here we will introduce two gravitational potentials, ζ_i and χ, such that

$$\nabla^2\zeta_i = \frac{4\pi G}{c}{}^{(1)}T^{0i} \tag{6.35}$$

and

$$\nabla^2\chi = \frac{\phi}{c^2}, \tag{6.36}$$

and hence, the component $^{(3)}h_{0i}$ is given by

$$^{(3)}h_{0i} = \frac{4}{c^3}\zeta_i + \partial_0\partial_i\chi. \tag{6.37}$$

Exercise 6.4

Potential ζ_i
Show that integration of the equation (6.35) gives the potential

$$\zeta_i(x) = -\frac{G}{c}\int \frac{{}^{(1)}T^{0i}(t,\boldsymbol{x}')}{|\boldsymbol{x}-\boldsymbol{x}'|}d^3x', \tag{6.38}$$

which is the same term obtained in equation (5.52b) as a first post-Newtonian contribution.

Exercise 6.5

Superpotential χ
The function χ is called a superpotential because it is defined in terms of the potential ϕ through the relation (6.36). Show that integration of this differential equation gives

$$\chi(x) = -\frac{G}{2c^4}\int |\boldsymbol{x}-\boldsymbol{x}'|\,{}^{(0)}T^{00}(t,\boldsymbol{x}')d^3x'. \tag{6.39}$$

6.3.1 Summary of Results in terms of the Post-Newtonian Potentials

The main results of this chapter are the post-Newtonian components of the metric, given by

$$g_{00} = -1 - \frac{2}{c^2}\phi - \frac{2}{c^2}\left(\frac{\phi^2}{c^2}+\psi\right) + \mathcal{O}\left(\epsilon^6\right) \tag{6.40a}$$

$$g_{0j} = \frac{4}{c^3}\zeta_j + \partial_0\partial_i\chi + \mathcal{O}\left(\epsilon^5\right) \tag{6.40b}$$

$$g_{ij} = \delta_{ij}\left(1-\frac{2}{c^2}\phi\right) + \mathcal{O}\left(\epsilon^4\right) \tag{6.40c}$$

where the gravitational post-Newtonian potentials satisfy the equations

$$\nabla^2\phi = \frac{4\pi G}{c^2}\,{}^{(0)}T^{00} \tag{6.41a}$$

$$\nabla^2\psi = \frac{4\pi G}{c^2}\left[{}^{(2)}T^{00} + {}^{(2)}T^{jj}\right] \tag{6.41b}$$

$$\nabla^2\zeta_i = \frac{4\pi G}{c}\,{}^{(1)}T^{0i} \tag{6.41c}$$

$$\nabla^2\chi = \frac{\phi}{c^2} \tag{6.41d}$$

and therefore

$$\phi(x) = -\frac{G}{c^2}\int \frac{{}^{(0)}T^{00}(t,\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}d^3x' \tag{6.42a}$$

$$\psi(x) = -\frac{G}{c^2}\int \left[{}^{(2)}T^{00}(t,\mathbf{x}') + {}^{(2)}T^{jj}(t,\mathbf{x}')\right]\frac{d^3x'}{|\mathbf{x}-\mathbf{x}'|} \tag{6.42b}$$

$$\zeta_i(x) = -\frac{G}{c}\int \frac{{}^{(1)}T^{0i}(t,\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}d^3x' \tag{6.42c}$$

$$\chi(x) = -\frac{G}{2c^4}\int |\mathbf{x}-\mathbf{x}'|\,{}^{(0)}T^{00}(t,\mathbf{x}')d^3x'. \tag{6.42d}$$

For future convenience we also write here the connection components (6.8) in terms of the post-Newtonian potentials,

$$^{(3)}\Gamma^0_{00} = \frac{1}{c^2}\partial_0\phi \tag{6.43a}$$

$$^{(2)}\Gamma^0_{0j} = \frac{1}{c^2}\partial_j\phi \tag{6.43b}$$

$$^{(3)}\Gamma^0_{jk} = -\frac{1}{c^2}\left[\frac{2}{c}\left(\partial_j\zeta_k + \partial_k\zeta_j\right) + c^2\partial_0\partial_j\partial_k\chi + \delta_{jk}\partial_0\phi\right] \tag{6.43c}$$

$$^{(2)}\Gamma^i_{00} = \frac{1}{c^2}\partial_i\phi \tag{6.43d}$$

$$^{(4)}\Gamma^i_{00} = \frac{4}{c^4}\phi\partial_i\phi + \frac{4}{c^3}\partial_0\zeta_i + \frac{1}{c^2}\partial_i\psi + \partial_0\partial_0\partial_i\chi \tag{6.43e}$$

$$^{(3)}\Gamma^i_{0j} = -\frac{1}{c^2}\left[\frac{2}{c}\left(\partial_i\zeta_j - \partial_j\zeta_i\right) + \delta_{ij}\partial_0\phi\right] \tag{6.43f}$$

$$^{(2)}\Gamma^i_{jk} = -\frac{1}{c^2}\left[\delta_{ij}\partial_k\phi + \delta_{ik}\partial_j\phi - \delta_{jk}\partial_i\phi\right]. \tag{6.43g}$$

6.4 The Gauge Condition and the Conservation of the Energy-Momentum Tensor

Now, we will find a relation between the standard post-Newtonian gauge conditions (6.17) and the lower order approximation of the equation of conservation for the energy-momentum tensor. Replacing the metric terms with the post-Newtonian potentials in equation (6.17a), gives

$$c\partial_0\phi + \partial_j\zeta_j = 0 \tag{6.44}$$

while equation (6.17b) becomes an identity. On the other hand, the conservation law for the energy-momentum tensor,

$$\nabla_\nu T^{\mu\nu} = \partial_\nu T^{\mu\nu} + \Gamma^\mu_{\nu\sigma}T^{\sigma\nu} + \Gamma^\nu_{\nu\sigma}T^{\mu\sigma} = 0, \tag{6.45}$$

is not immediately assured in this construction because the metric tensor given in equations (6.40) was obtained using a particular gauge. Considering the $\mu = 0$ component of this equation, the lower order of approximation gives the condition

$$\partial_0 {}^{(0)}T^{00} + \partial_j {}^{(1)}T^{0j} = 0 + \mathcal{O}(\epsilon^2) \tag{6.46}$$

and, using the equations (6.41), this relation becomes

$$\nabla^2\left[c\partial_0\phi + \partial_j\zeta_j\right] = 0. \tag{6.47}$$

Since both potentials, ϕ and ζ_i, vanish at infinity; the solution of this equation corresponds to the gauge condition (6.44).

Considering now the $\mu = i$ component of the conservation equation (6.45), the lowest order of approximation gives

$$\partial_0 {}^{(1)}T^{0i} + \partial_j {}^{(2)}T^{ij} = -\frac{1}{c^2}{}^{(0)}T^{00}\partial_i\phi, \tag{6.48}$$

which corresponds to momentum conservation, as will be shown in a following chapter.

Chapter 7

Equations of Motion in the Post-Newtonian Approximation

The equations of motion for massive and massless particles in a relativistic spacetime under the post-Newtonian approximation will be found using a variational approach. In this chapter, we define the Lagrangian describing a test particle to obtain the general expression for its equations of motion in terms of the PN potentials.

7.1 Equations of Motion for a Massive Test Particle

The action describing a test particle of proper mass m_0 is

$$S = -m_0 c^2 \int_{\tau_1}^{\tau_2} d\tau = -m_0 c^2 \int_{t_1}^{t_2} \frac{d\tau}{dt} dt, \tag{7.1}$$

where τ represents the proper time of the test particle and t is the coordinate time, which will be used as the parameter to describe the motion. From the line element $ds^2 = -c^2 d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu$ and denoting $\dot{x}^\mu = \frac{dx^\mu}{dt}$, we have

$$\left(\frac{d\tau}{dt}\right)^2 = -g_{\mu\nu} \frac{\dot{x}^\mu \dot{x}^\nu}{c^2}. \tag{7.2}$$

Using the post-Newtonian metric (6.4), this relation leads to

$$\left(\frac{d\tau}{dt}\right)^2 = 1 - \frac{\dot{\mathbf{x}}^2}{c^2} - {}^{(2)}h_{00} - {}^{(4)}h_{00} - 2\,{}^{(3)}h_{0j}\frac{\dot{x}^j}{c} - {}^{(2)}h_{ij}\frac{\dot{x}^i \dot{x}^j}{c^2} + \mathcal{O}(\epsilon^6). \tag{7.3}$$

We will use the In order to describe the motion of a test particle with proper mass m_0, we will use the lagrangian

$$L = -m_0 c^2 \left(\frac{d\tau}{dt} - 1\right), \tag{7.4}$$

which, in the post-Newtonian approximation, gives

$$\begin{aligned} L =& \frac{1}{2} m_0 \dot{\mathbf{x}}^2 + \frac{1}{2} m_0 c^{2\,(2)}h_{00} + \frac{1}{8} m_0 \frac{\left(\dot{\mathbf{x}}^2\right)^2}{c^2} + \frac{1}{2} m_0 c^{2\,(4)}h_{00} + m_0 c^{(3)}h_{0j}\dot{x}^j \\ &+ \frac{1}{2} m_0 {}^{(2)}h_{ij}\dot{x}^i \dot{x}^j + \frac{1}{8} m_0 c^2 \left({}^{(2)}h_{00}\right)^2 + \frac{1}{4} m_0 {}^{(2)}h_{00}\dot{\mathbf{x}}^2 + \mathcal{O}(\epsilon^6). \end{aligned} \tag{7.5}$$

Using equation (6.40) to write the metric components in terms of the post-Newtonian potentials, the Lagrangian becomes

$$L = \frac{1}{2}m_0\dot{\mathbf{x}}^2 - m_0\phi + \frac{1}{8}m_0\frac{\left(\dot{\mathbf{x}}^2\right)^2}{c^2} - \frac{1}{2}m_0\frac{\phi^2}{c^2} - m_0\psi - \frac{3}{2}m_0\phi\frac{\dot{\mathbf{x}}^2}{c^2}$$
$$+ 4m_0\frac{\zeta_j\dot{x}^j}{c^2} + m_0 c\dot{x}^j\partial_0\partial_j\chi + \mathcal{O}(\epsilon^6). \tag{7.6}$$

From this expression, it is clear that the first two terms correspond to the Newtonian lagrangian, while the rest terms are all 1PN approximation corrections.

Exercise 7.1

Equation of Motion for a Massive Particle

Show that Euler-Lagrange equations,

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}^j}\right) = \frac{\partial L}{\partial x^j} \tag{7.7}$$

applied to lagrangian (7.5) yields the equations of motion

$$\ddot{x}^j = \frac{1}{2}c^2\partial_j{}^{(2)}h_{00} + \frac{1}{2}c^2\partial_j{}^{(4)}h_{00} - \frac{1}{2}c^{2\,(2)}h_{jk}\partial_k{}^{(2)}h_{00} - c^2\partial_0{}^{(3)}h_{0j}$$
$$- \frac{1}{2}c\dot{x}^j\partial_0{}^{(2)}h_{00} - c\dot{x}^k\partial_0{}^{(2)}h_{jk} - c\left(\partial_k{}^{(3)}h_{0j} - \partial_j{}^{(3)}h_{0k}\right)\dot{x}^k$$
$$- \dot{x}^j\dot{x}^k\partial_k{}^{(2)}h_{00} - \left(\partial_k{}^{(2)}h_{jl} - \frac{1}{2}\partial_j{}^{(2)}h_{kl}\right)\dot{x}^k\dot{x}^l + \mathcal{O}(\epsilon^6) \tag{7.8}$$

.

Using the post-Newtonian potentials, the equations of motion for a massive test particle become

$$\ddot{x}^j = -\partial_j\phi - \frac{4}{c^2}\phi\partial_j\phi - \partial_j\psi - \frac{4}{c}\partial_0\zeta_j - c^2\partial_0\partial_0\partial_j\chi + 3\frac{\dot{x}^j}{c}\partial_0\phi - 4\frac{\dot{x}^k}{c^2}\left(\partial_k\zeta_j - \partial_j\zeta_k\right)$$
$$+ 4\frac{\dot{x}^j\dot{x}^k}{c^2}\partial_k\phi - \frac{\dot{\mathbf{x}}^2}{c^2}\partial_j\phi + \mathcal{O}(\epsilon^6), \tag{7.9}$$

or in vector notation,

$$\ddot{\mathbf{x}} = -\boldsymbol{\nabla}\phi - \boldsymbol{\nabla}\psi - \frac{2}{c^2}\boldsymbol{\nabla}\phi^2 - \frac{4}{c}\partial_0\boldsymbol{\zeta} - c^2\partial_0\partial_0\boldsymbol{\nabla}\chi + \frac{3}{c}\dot{\mathbf{x}}\partial_0\phi$$
$$- \frac{4}{c^2}\dot{\mathbf{x}}\times\left(\boldsymbol{\nabla}\times\boldsymbol{\zeta}\right) + \frac{4}{c^2}\dot{\mathbf{x}}\left(\dot{\mathbf{x}}\cdot\boldsymbol{\nabla}\phi\right) - \frac{\dot{\mathbf{x}}^2}{c^2}\boldsymbol{\nabla}\phi + \mathcal{O}(\epsilon^6). \tag{7.10}$$

Note that the leading term in the right hand side of this equation corresponds to the Newtonian contribution $\ddot{\mathbf{x}} = -\boldsymbol{\nabla}\phi$.

7.2 Equations of motion for a Massless Particle

The equations of motion describing a massless particles (e.g. a photon), will be obtained following a different scheme. First, we use the null condition for a massless particle $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$ and the post-Newtonian metric (6.4) to obtain, after differentiation with respect to time*,

$$2\ddot{x}^k\dot{x}^k = -c^3\partial_0{}^{(2)}h_{00} - c^2\dot{x}^k\partial_k{}^{(2)}h_{00} - c\left(2\partial_j{}^{(3)}h_{0k} + \partial_0{}^{(2)}h_{jk}\right)\dot{x}^j\dot{x}^k - \dot{x}^j\dot{x}^k\dot{x}^l\partial_j{}^{(2)}h_{kl} + \mathcal{O}(\epsilon). \tag{7.11}$$

From the general geodesic equation for a massless particle,

$$\frac{d^2x^\alpha}{d\lambda^2} + \Gamma^\alpha_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0 \tag{7.12}$$

with λ an adequate affine parameter, we obtain the 3-dimensional equations of motion in terms of the coordinate time t,

$$\ddot{x}^i + \Gamma^i_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -\frac{d^2t}{d\lambda^2}\left(\frac{dt}{d\lambda}\right)^{-2}\dot{x}^i. \tag{7.13}$$

Using the post-Newtonian metric (6.4) and the connections (6.8), this equation becomes

$$\begin{aligned}\ddot{x}^i - \frac{c^2}{2}\partial_i{}^{(2)}h_{00} + c\left(\partial_0{}^{(2)}h_{ij} + \partial_j{}^{(3)}h_{0i} - \partial_i{}^{(3)}h_{0j}\right)\dot{x}^j&\\ + \left(\partial_j{}^{(2)}h_{ik} - \frac{1}{2}\partial_i{}^{(2)}h_{jk}\right)\dot{x}^j\dot{x}^k + \mathcal{O}(\epsilon^2) &= -\frac{d^2t}{d\lambda^2}\left(\frac{dt}{d\lambda}\right)^{-2}\dot{x}^i.\end{aligned} \tag{7.14}$$

Multiplying this equation by $\dot{x}^i$ and using the condition $\mathbf{x}^2 = \dot{x}^k\dot{x}^k = c^2$ (valid for massless particles), we obtain

$$-\frac{d^2t}{d\lambda^2}\left(\frac{dt}{d\lambda}\right)^{-2}c^2 = \ddot{x}^k\dot{x}^k - \frac{c^2}{2}\dot{x}^k\partial_k{}^{(2)}h_{00} + c\dot{x}^j\dot{x}^k\partial_0{}^{(2)}h_{jk} + \frac{1}{2}\dot{x}^j\dot{x}^k\dot{x}^l\partial_j{}^{(2)}h_{kl} + \mathcal{O}(\epsilon^2). \tag{7.15}$$

Replacing equation (7.11) in the first term of this relation, gives

$$-\frac{d^2t}{d\lambda^2}\left(\frac{dt}{d\lambda}\right)^{-2} = -\frac{c}{2}\partial_0{}^{(2)}h_{00} - \dot{x}^k\partial_k{}^{(2)}h_{00} - \frac{1}{c}\left(\partial_j{}^{(3)}h_{0k} - \frac{1}{2}\partial_0{}^{(2)}h_{jk}\right)\dot{x}^j\dot{x}^k + mathcalO(\epsilon^3). \tag{7.16}$$

Finally, using this equation in (7.14) gives the equation of motion for the massless particle in the 1PN approximation,

$$\begin{aligned}\ddot{x}^i =& \frac{c^2}{2}\partial_i{}^{(2)}h_{00} - \dot{x}^i\dot{x}^k\partial_k{}^{(2)}h_{00} - \left(\partial_j{}^{(2)}h_{ik} - \frac{1}{2}\partial_i{}^{(2)}h_{jk}\right)\dot{x}^j\dot{x}^k - \frac{c}{2}\dot{x}^i\partial_0{}^{(2)}h_{00}\\ &- c\left(\partial_0{}^{(2)}h_{ij} + \partial_j{}^{(3)}h_{0i} - \partial_i{}^{(3)}h_{0j}\right)\dot{x}^j - \frac{1}{c}\left(\partial_j{}^{(3)}h_{0k} - \frac{1}{2}\partial_0{}^{(2)}h_{jk}\right)\dot{x}^i\dot{x}^j\dot{x}^k + \mathcal{O}(\epsilon^2).\end{aligned} \tag{7.17}$$

In terms of the post-Newtonian potentials (6.42), the motion equation reads

$$\begin{aligned}\ddot{x}^i =& -\partial_i\phi + \frac{4}{c^2}\dot{x}^i\dot{x}^k\partial_k\phi + \frac{3}{c}\dot{x}^i\partial_0\phi + \frac{4}{c^2}\left(\partial_i\zeta_j - \partial_j\zeta_i\right)\dot{x}^j\\ &- \frac{4}{c^4}\dot{x}^i\dot{x}^j\dot{x}^k\partial_j\zeta_k + \frac{1}{c^3}\dot{\mathbf{x}}^2\dot{x}^i\partial_0\phi - \frac{1}{c^2}\dot{\mathbf{x}}^2\partial_i\phi + \mathcal{O}(\epsilon^2).\end{aligned} \tag{7.18}$$

*In order to obtain the order of approximation in this equation, it is important to note that for massless particles, $\dot{\mathbf{x}}^i\frac{dx^i}{dt} \sim \mathcal{O}(c)$ and $\ddot{\mathbf{x}}^i\frac{d^2x^i}{dt^2} \sim \mathcal{O}(1)$. Therefore we consider just the terms with these orders of approximation, neglecting $\mathcal{O}(\epsilon)$ and smaller.

Exercise 7.2

Lagrangian for a Massless Particle in the PN Approximation

Show that the lagrangian function

$$L = -\frac{1}{2}\dot{\boldsymbol{x}}^2 - \frac{1}{2}{}^{(2)}h_{00}\dot{\boldsymbol{x}}^2 - \frac{1}{2}{}^{(2)}h_{jk}\dot{x}^j\dot{x}^k - \frac{1}{2}c^{(3)}h_{0j}\dot{x}^j - \frac{1}{2c}{}^{(3)}h_{0j}\dot{x}^j\dot{\boldsymbol{x}}^2 + \mathcal{O}(\epsilon^2) \tag{7.19}$$

produces the equations of motion (7.17) when the metric tensor does not depends explicitly on time.

Part IV

Relativistic Celestial Mechanics

Chapter 8

(Parameterized) Post-Newtonian description of the Static One Body Problem

Equations (6.40) give a general form of the post-Newtonian metric components in quasi-cartesian coordinates. These components can be used not only in general relativity but also in other theories of gravity. The difference between one and the other lies in the coefficients multiplying each term in the metric components. Hence, we can propose a more generalized metric depending on a set of parameters in such a way that each particular set of values of these parameters distinguishes different gravity theories. The generalized metric is usually called a *Parameterized Post-Newtonian* (PPN) metric and the parameters are called PPN-parameters. In this chapter we will present this kind of methodology to study the spacetime curvature produced by a static, spherically symmetric, non-rotating body.

8.1 A General Static Spherically Symmetric Metric

In order to begin with the description of the static one body problem, we introduce a general static metric with spherical symmetry with the form

$$ds^2 = -\mathcal{A}(r)c^2dt^2 + \mathcal{B}(r)dr^2 + \mathcal{R}^2(r)\left[d\theta^2 + \sin^2\theta d\varphi^2\right]. \tag{8.1}$$

Clearly, Schwarzschild's solution in standard coordinates, (4.2), is recovered by choosing the functions

$$\mathcal{A}(r) = \frac{1}{\mathcal{B}(r)} = 1 - \frac{2m}{r} \tag{8.2}$$

$$\mathcal{R}(r) = r, \tag{8.3}$$

where $m = \frac{GM}{c^2}$ and M is the total mass of the central object. In fact, the general line element (8.1) also represents the Schwazrschild's solution in isotropic (4.5) and harmonic (4.9) coordinates by a suitable choice of the functions $\mathcal{A}(r)$, $\mathcal{B}(r)$ and $\mathcal{R}(r)$. For example, considering that

$$\mathcal{A}(r) = 1 - \frac{2m}{\mathcal{R}(r)} \tag{8.4a}$$

$$\mathcal{B}(r) = \frac{\left(\frac{d\mathcal{R}}{dr}\right)^2}{\mathcal{A}(r)}, \tag{8.4b}$$

it is possible to obtain the Schwarzschild's metric in the three coordinate systems by setting an appropiate function $\mathcal{R}(r)$ (see Table 8.1).

$\mathcal{R}(r)$	Coordinates	Equation
r	Standard	(4.2)
$r\left(1+\frac{m}{2r}\right)^2$	Isotropic	(4.5)
$r+m$	Harmonic	(4.9)

Table 8.1: Schwarzschild's metric in three different coordinates systems.

8.1.1 Motion of Particles

As described in Section 4.2, the motion of a particle in this kind of spaces can be restricted to the $\theta = \frac{\pi}{2}$ and therefore, the lagrangian describing the motion of a particle in the background (8.1) can be written as

$$L = \frac{1}{2}\left[\mathcal{A}(r)c^2\left(\frac{dt}{d\lambda}\right)^2 - \mathcal{B}(r)\left(\frac{dr}{d\lambda}\right)^2 - \mathcal{R}^2(r)\left(\frac{d\varphi}{d\lambda}\right)^2\right], \tag{8.5}$$

where λ is an adequate parameter for the description (for massive particle we chose the proper time $\lambda = \tau$ while for massless particles λ is a so-called *affine parameter*). From this lagrangian it is easy to obtain the conserved quantities

$$\mathcal{E} = \mathcal{A}(r)c^2\left(\frac{dt}{d\lambda}\right) \tag{8.6a}$$

$$\ell = \mathcal{R}^2(r)\left(\frac{d\varphi}{d\lambda}\right) \tag{8.6b}$$

$$H = L = \delta, \tag{8.6c}$$

where $\delta = \frac{c^2}{2}$ for massive particles while $\delta = 0$ for massless particles.

In order to describe the motion of particles in the background given by metric (8.1), we write the radial equation of motion using the conserved quantities as

$$\mathcal{B}(r)\left(\frac{dr}{d\lambda}\right)^2 = \mathcal{V}(r) - 2\delta, \tag{8.7}$$

where we introduce the function

$$\mathcal{V}(r) = \frac{\mathcal{E}^2}{\mathcal{A}(r)c^2} - \frac{\ell}{\mathcal{R}^2(r)}. \tag{8.8}$$

Circular Motion

Circular trajectories are defined by the condition $\frac{dr}{d\lambda} = 0$, or equivalently $\frac{d\mathcal{V}}{dr} = 0$. This condition corresponds to the relation

$$\frac{\mathcal{A}^2(r)}{\mathcal{R}(r)} = \frac{m\mathcal{E}^2}{\ell^2}. \tag{8.9}$$

Following the treatment of Newtonian mechanics, we define the mean motion of the particle as $n = \frac{d\varphi}{dt}$, and therefore, equation (8.9) writes as the generalization of Kepler's third law,

$$n^2\mathcal{R}^3(r) = GM \tag{8.10}$$

Introducing the Proper Time

Since we are working with a general metric, some mathematical quantities manifest differently in different coordinate systems. Hence, we will find some special quantities that are coordinate independent and therefore directly measurable from observation, without depending on the choice of a particular coordinate system. For example, considering the motion of massive particles, the adequate parameter to describe the motion is the proper time of the particle, i.e. $\lambda = \tau$, which is an invariant quantity (independent of the coordinate system). With the definition of the mean motion above and the metric (8.1), we obtain the relation between proper and coordinate times as

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{3m}{\mathcal{R}(r)}}. \tag{8.11}$$

Using this relation, we can write the definition of the motion in terms of the proper time,

$$N = \frac{d\varphi}{d\tau} = \frac{n}{\sqrt{1 - \frac{3m}{\mathcal{R}(r)}}}, \tag{8.12}$$

which gives the interesting relation

$$N^2\mathcal{R}^3(r) = \frac{GM}{1 - \frac{3m}{\mathcal{R}(r)}} \tag{8.13}$$

Since the corresponding sidereal period of revolution, $T_N = \frac{2\pi}{N}$, can be obtained directly by observation, the mean motion N is measurable quantity, independent of the coordinate system.

Differential equation of the orbit

For a general motion case, equation (8.7) may be written as a first order differential equation for the trajectory of the particle,

$$\left(\frac{d\mathcal{U}}{d\varphi}\right)^2 = \frac{\mathcal{E}^2}{\ell^2 c^2} - \frac{2\delta}{\ell^2} + \frac{4m\delta}{\ell^2}\mathcal{U} - \mathcal{U}^2 + 2m\mathcal{U}^3, \tag{8.14}$$

or as a second order differential equation with the form

$$\frac{d^2\mathcal{U}}{d\varphi^2} + \mathcal{U} = \frac{2m}{\ell^2}\delta + 3m\mathcal{U}^2, \tag{8.15}$$

where we have introduced the function $\mathcal{U}(r) = \frac{1}{\mathcal{R}(r)}$*. These equations are valid both for massive and massless particles by choosing the appropriate value for δ and because equation (8.15) is mathematically equal to (4.29), it can be solved as described in section 4.3.

8.1.2 A Parameterized Post-Newtonian Function

Following the proposal of V. Brumberg ([3]) and assuming that $\frac{m}{r} \ll 1$, it is interesting to study the possibility of writing the function $\mathcal{R}(r)$ as a parameterized expansion with the form

$$\mathcal{R}(r) = r\left[1 + (1-\sigma_1)\frac{m}{r} + \sigma_2\frac{m^2}{r^2} + ...\right], \tag{8.16}$$

where we introduced two constant parameters, σ_1 and σ_2, which may be chosen to reproduce Schwarzschild's metric in different coordinates, as shown in Table 8.2.

*Note the similarities with the treatment of equations (1.30) and (4.29)

σ_1	σ_2	$\mathcal{R}(r)$	Coordinates	Equation
1	0	r	Standard	(4.2)
0	$\frac{1}{4}$	$r\left(1+\frac{m}{2r}\right)^2$	Isotropic	(4.5)
0	0	$r+m$	Harmonic	(4.9)

Table 8.2: Values of the parameters in the expansion of the function $\mathcal{R}(r)$ given in equation (8.16) to obtain Schwarzschild's metric in three different coordinates systems.

Using this expansion, it is possible to write Kepler's third law in a parameterized form,

$$N^2 r^3 \left[1 - 3\sigma_1 \frac{m}{r} - 3(1-\sigma_2-\sigma_1^2)\frac{m^2}{r^2} + ...\right] = GM, \tag{8.17}$$

showing the Newtonian term as well as the relativistic corrections which are measured by the parameters σ_1 and σ_2. It is possible to use this expansion into (8.15) to obtain a parameterized orbit equation. However, since we have presented a general post-Newtonian approach which can be applied also in other situations, we will sue a parameterized post-newtonian metric for describing the static one-body problem used by Will[24].

8.2 A PPN-Metric describing a Static Spherically Symmetric Body

All our treatment of the Post-Newtonian approximation in previous chapters was done in cartesian coordinates. Therefore, in order to introduce a suitable Parameterized Post-Newtonian (PPN) metric to describe the geometry around a static spherically symmetric body, we will consider the Schwarzschild line element in harmonic coordinates presented in equation (4.11),

$$ds^2 = -\left(\frac{1-\frac{m}{r_h}}{1+\frac{m}{r_h}}\right)c^2dt^2 + \left(1+\frac{m}{r_h}\right)^2\left[\delta_{ij} + \frac{\left(\frac{m}{r_h}\right)^2}{1-\left(\frac{m}{r_h}\right)^2}\frac{x_i x_j}{r_h^2}\right]dx^i dx^j. \tag{8.18}$$

Considering that $\frac{m}{r_h} \sim \epsilon^2 \ll 1$, this metric can be expanded, up to the orders of the 1PN approximation, as

$$ds^2 = -\left[1 - \frac{2m}{r_h} + \frac{2m^2}{r_h^2} + \mathcal{O}(\epsilon^6)\right]c^2dt^2 + \left[\delta_{ij} + \frac{2m}{r_h}\delta_{ij} + \mathcal{O}(\epsilon^6)\right]dx^i dx^j, \tag{8.19}$$

where $m = \frac{GM}{c^2}$ is the mass of the gravitational source in geometric units.

Exercise 8.1

Expansion of the Schwarzschild metric in Isotropic Coordinates
Show explicitly that the expansion, up to the 1PN order, of the line element of the Schwarzschild metric in isotropic coordinates (4.7), is mathematically equal to equation (8.19).

In order to describe the behavior of the Solar System and other systems in the weak gravitational field regime, some authors such as Eddington [6], Robertson [13], Schiff [14], Brumberg [2, 3] and Will[24] proposed a PPN metric modeling a static, spherically symmetric, non-rotating body based on this approximation. Using quasi-cartesian harmonic coordinates, we introduce the PPN metric in the form

$$g_{00} = -\left(1 - 2\kappa\frac{m}{r} + 2\left(\beta - \alpha\right)\frac{m^2}{r^2} + \mathcal{O}(\epsilon^6)\right) \tag{8.20a}$$

$$g_{0j} = 0 \tag{8.20b}$$

$$g_{ij} = \left[\delta_{ij} + \frac{2m}{r}\left((\gamma - \alpha)\delta_{ij} + \alpha\frac{x^i x^j}{r^2}\right) + \mathcal{O}(\epsilon^4)\right], \tag{8.20c}$$

where α, β, γ and κ will be the PPN parameters which may be constrained by observational data and we have dropped the subindex h in the radial coordinate because up to 1PN orders, harmonic and isotropic coordinates coincide†. Equations (8.20) show that the parameter κ measures the contribution of the Newtonian potential while the parameters α and β measure the amount of the non-linear term $\frac{m^2}{r^2}$ in the components of the metric.

Exercise 8.2

Spatial Components of the Riemann Tensor

The significance of the parameter γ is better understood by calculating the spatial components of the Riemann tensor. Explicitly show that they are

$$\begin{aligned} R_{ijkl} = \gamma\frac{m}{r^3}\Bigg[&-\delta_{il}\left(\delta_{jk} - 3\frac{x^j x^k}{r^2}\right) - \delta_{jk}\left(\delta_{il} - 3\frac{x^i x^l}{r^2}\right) \\ &+\delta_{ik}\left(\delta_{jl} - 3\frac{x^j x^l}{r^2}\right) + \delta_{jl}\left(\delta_{ik} - 3\frac{x^i x^k}{r^2}\right) + \mathcal{O}(\epsilon^4)\Bigg]. \end{aligned} \tag{8.21}$$

From this expression it is possible to see that γ directly measures the spatial curvature contribution due to the mass m.

The general relativity solution (both in harmonic and isotropic coordinates) is recovered using the values $\kappa = \beta = \gamma = 1$ and $\alpha = 0$ for the PPN parameters. Now, we will study the motion of test particles in this parameterized background.

8.3 Motion of Test Particles

In order to describe the motion of a test particle in the PPN formulation, we will consider the Lagrangian describing a massive particle given in equation (7.5). Comparing with the PPN metric in (8.20), we identify the contributions

$$^{(2)}h_{00} = 2\kappa\frac{m}{r} \tag{8.22a}$$

$$^{(4)}h_{00} = 2\left(\alpha - \beta\right)\frac{m^2}{r^2} \tag{8.22b}$$

$$^{(3)}h_{0j} = 0 \tag{8.22c}$$

$$^{(2)}h_{ij} = \frac{2m}{r}\left[(\gamma - \alpha)\delta_{ij} + \alpha\frac{x_i x_j}{r^2}\right] \tag{8.22d}$$

†From now on, we will use the variable r instead of r_h in the PPN metric.

and therefore the Lagrangian for the test particle becomes

$$\begin{aligned} L =& \frac{1}{2}m_0\dot{\mathbf{x}}^2 + \kappa\frac{GMm_0}{r} + \frac{1}{8}m_0\frac{\left(\dot{\mathbf{x}}^2\right)^2}{c^2} \\ &+ \frac{mm_0}{r}\left[\left(\gamma + \frac{1}{2}\kappa - \alpha\right)\dot{\mathbf{x}}^2 + \left(\alpha + \frac{1}{2}\kappa^2 - \beta\right)\frac{GM}{r} + \alpha\frac{(\mathbf{x}\cdot\dot{\mathbf{x}})^2}{r^2}\right] + \mathcal{O}(\epsilon^4). \end{aligned} \tag{8.23}$$

The equations of motion derived form this Lagrangian, equivalent to those in equation (7.1), take the form

$$\begin{aligned} \ddot{\mathbf{x}} + \kappa\frac{GM}{r^3}\mathbf{x} = \frac{m}{r^3}\Bigg[&\left(2(\beta+\kappa\gamma-\alpha)\frac{GM}{r} - (\gamma+\alpha)\dot{\mathbf{x}}^2 + 3\alpha\frac{(\mathbf{x}\cdot\dot{\mathbf{x}})^2}{r^2}\right)\mathbf{x} \\ &+ \left(2(\gamma+\kappa-\alpha)\mathbf{x}\cdot\dot{\mathbf{x}}\right)\dot{\mathbf{x}}\Bigg] + \mathcal{O}(\epsilon^4). \end{aligned} \tag{8.24}$$

8.3.1 Conserved Quantities

Now, we will introduce the polar coordinates r and φ in the plane of the orbit following the definition in equations (1.20),

$$\mathbf{x} = r\mathbf{n} \tag{8.25a}$$

$$\dot{\mathbf{x}} = \dot{r}\mathbf{n} + r\dot{\varphi}\boldsymbol{\lambda} \tag{8.25b}$$

$$\ddot{\mathbf{x}} = \left(\ddot{r} - r\dot{\varphi}^2\right)\mathbf{n} + \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\varphi}\right)\boldsymbol{\lambda}. \tag{8.25c}$$

in which we use the time dependent basis in the orbital plane given by the vectors $\mathbf{n}$ and $\boldsymbol{\lambda}$ (see equation (1.18)).

Exercise 8.3

Equation of Motion in the PPN Approximation.
Show that using the polar coordinates defined above, the equations of motion given in (8.24) become

$$\ddot{r} - r\dot{\varphi}^2 = -\kappa\frac{GM}{r^2} + m\left[(\beta+\kappa\gamma-\alpha)\frac{2GM}{r^3} + (\gamma+2\kappa)\frac{\dot{r}^2}{r^2} - (\alpha+\gamma)\dot{\varphi}^2\right] \tag{8.26a}$$

$$\frac{d}{dt}\left(r^2\dot{\varphi}\right) = 2m(\gamma+\kappa-\alpha)\dot{r}\dot{\varphi}. \tag{8.26b}$$

The Lagrangian describing the motion of a test particle (8.23), in polar coordinates, is

$$\begin{aligned} L =& \frac{1}{2}m_0\left(\dot{r}^2 + r^2\dot{\varphi}^2\right) + \kappa\frac{GMm_0}{r} + \frac{1}{8}m_0\frac{\left(\dot{r}^2 + r^2\dot{\varphi}^2\right)^2}{c^2} \\ &+ \frac{mm_0}{r}\left[\left(\gamma + \frac{1}{2}\right)\dot{r}^2 + \left(\gamma + \frac{1}{2}\kappa - \alpha\right)r^2\dot{\varphi}^2 + \left(\alpha + \frac{1}{2}\kappa^2 - \beta\right)\frac{GM}{r}\right] + \mathcal{O}(\epsilon^4). \end{aligned} \tag{8.27}$$

From the functional dependence of this Lagrangian, it is easy to observe that there are two conserved quantities: the angular momentum of the test particle,

$$\ell = \frac{p_\varphi}{m_0} = \frac{1}{m_0}\frac{\partial L}{\partial\dot{\varphi}} = r^2\dot{\varphi}\left[1 + \frac{1}{2}\frac{\left(\dot{r}^2 + r^2\dot{\varphi}^2\right)}{c^2} + \left(\gamma + \frac{1}{2}\kappa - \alpha\right)\frac{2m}{r}\right] + \mathcal{O}(\epsilon^4) \tag{8.28}$$

and the Hamiltonian,

$$\begin{aligned}\mathcal{E} = \frac{H}{m_0} =& \frac{1}{2}\left(\dot{r}^2 + r^2\dot{\varphi}^2\right) - \kappa\frac{GM}{r} + \frac{3}{8}\frac{\left(\dot{r}^2 + r^2\dot{\varphi}^2\right)^2}{c^2} \\ &+ \frac{m}{r}\left[\left(\gamma + \frac{1}{2}\kappa\right)\dot{r}^2 + \left(\gamma + \frac{1}{2}\kappa - \alpha\right) r^2\dot{\varphi}^2 - \left(\alpha + \frac{1}{2}\kappa^2 - \beta\right)\frac{GM}{r}\right] + \mathcal{O}(\epsilon^4).\end{aligned} \tag{8.29}$$

In order to eliminate some of the time derivatives in the right hand side of equation (8.28) we use the zeroth order of the Hamiltonian (i.e. the Newtonian value) to obtain the relation

$$r^2\dot{\varphi} = \ell\left[1 - \frac{\mathcal{E}}{c^2} - \frac{2m}{r}\left(\gamma - \alpha + \kappa\right)\right]. \tag{8.30}$$

Similarly, using the Newtonian values for the Hamiltonian and for the angular momentum, equation (8.29) can be written as a differential equation for $\dot{r}$,

$$\begin{aligned}\dot{r}^2 =& 2\mathcal{E} - r^2\dot{\varphi}^2 + 2\kappa\frac{GM}{r} \\ &+ 2m\left[\alpha\frac{\ell^2}{r^3} + \left(\alpha - \beta - 2\kappa^2 - 2\kappa\gamma\right)\frac{GM}{r^2} - 2\left(\gamma + 2\kappa\right)\frac{\mathcal{E}}{r} - \frac{3}{2}\frac{\mathcal{E}^2}{GM}\right] + \mathcal{O}(\epsilon^4).\end{aligned} \tag{8.31}$$

8.3.2 Differential Equation of the Orbit

Using the angle φ as the independent variable, instead of time, and defining the new variable $u(\varphi) = \frac{1}{r(\varphi)}$, just as in the Newtonian case, we obtain the differential equation of the orbit as

$$\begin{aligned}\left(\frac{du}{d\varphi}\right)^2 + u^2 =& \frac{2\mathcal{E}}{\ell^2} + 2\kappa\frac{GM}{\ell^2}u \\ &+ 2m\left[\alpha u^3 + \left(\alpha - 4\kappa\alpha - \beta + 2\kappa\gamma + 2\kappa^2\right)\frac{GM}{\ell^2}u^2 + 2\left(\kappa + \gamma - 2\alpha\right)\frac{\mathcal{E}}{\ell^2}u + \frac{1}{2}\frac{\mathcal{E}^2}{GM\ell^2}\right] + \mathcal{O}(\epsilon^4).\end{aligned} \tag{8.32}$$

Differentiation with respect to φ gives the second order differential equation of the orbit,

$$\frac{d^2u}{d\varphi^2} + u = \kappa\frac{GM}{\ell^2} + m\left[3\alpha u^2 + 2\left(\alpha - 4\kappa\alpha - \beta + 2\kappa\gamma + 2\kappa^2\right)\frac{GM}{\ell^2}u + 2\left(\kappa + \gamma - 2\alpha\right)\frac{\mathcal{E}}{\ell^2}\right] + \mathcal{O}(\epsilon^4). \tag{8.33}$$

8.3.3 Circular Motion

A particular solution of the orbital equations obtained above correspond to the circular orbit $u(\varphi) =$ constant or

$$\begin{cases} r = & r_0 \\ \varphi = & nt + \text{const.} \end{cases} \tag{8.34}$$

Replacing these functions into equation ((8.26)) gives the mean motion $n = \frac{d\varphi}{dt}$ as

$$n^2 = \frac{GM}{r_0^3}\left[\kappa + \left((2+\kappa)\alpha - 2\beta - \kappa\gamma\right)\frac{m}{r_0}\right] + \mathcal{O}(\epsilon^4) \tag{8.35}$$

8.3.4 Elliptic Motion

In order to solve the orbit equation, we will factorize the third order polynomial in the right hand side of equation (8.32) as

$$\frac{\ell^2}{GM}\left(\frac{du}{d\varphi}\right)^2 = (u-u_1)(u-u_2)\left(2m\alpha\ell^2 u - u_3\right). \tag{8.36}$$

Two of the roots of this polynomial will correspond to the radii of the pericenter and the apocenter of the orbit, which we will write as $u_1 = \frac{1}{r_p} = \frac{1}{a(1-e)}$ and $u_2 = \frac{1}{r_a} = \frac{1}{a(1+e)}$, respectively, introducing the semi-major axis a and the eccentricity e of the orbit. Comparison of coefficients in equations (8.32) and (8.36) gives the third root as

$$u_3 = \kappa a(1-e^2) + m(\alpha-\gamma-\kappa)(1-e^2) + \mathcal{O}(\epsilon^4), \tag{8.37}$$

together with the values of the angular momentum and the Hamiltonian,

$$\ell^2 = GMa(1-e^2)\left[\kappa + \frac{m}{a}\left(\alpha-\gamma-\kappa+2\frac{(3-4\kappa)\alpha-\beta+2\kappa\gamma+2\kappa^2}{1-e^2}\right)\right] + \mathcal{O}(\epsilon^4) \tag{8.38}$$

$$\mathcal{E} = -\frac{GM}{2a}\left[\kappa + \frac{m}{a}\left(\alpha-\gamma-\kappa+\frac{1}{4}\right)\right] + \mathcal{O}(\epsilon^4). \tag{8.39}$$

Considering a conic function with the form

$$u(\varphi) = \frac{1+e\cos f}{a(1-e^2)} \tag{8.40}$$

as an ansatz for the solution of the differential equation of the orbit, we obtain the following differential equation for the true anomaly, $f = f(\varphi)$,

$$\left(\frac{df}{d\varphi}\right)^2 = 1 - \frac{m}{\kappa a(1-e^2)}\left(4\alpha-4\kappa\alpha-\beta+2\kappa\gamma+2\kappa^2\right) - \frac{\alpha m}{\kappa a(1-e^2)} e\cos f + \mathcal{O}(\epsilon^4). \tag{8.41}$$

In order to integrate this equation note that it can be written, up to second order in ϵ, as

$$\frac{df}{d\varphi} = \nu\left(1 - \frac{\alpha m}{\kappa a(1-e^2)} e\cos f\right) + \mathcal{O}(\epsilon^4) \tag{8.42}$$

where

$$\nu = 1 - \frac{m}{\kappa a(1-e^2)}\left(4\alpha-4\kappa\alpha-\beta+2\kappa\gamma+2\kappa^2\right). \tag{8.43}$$

Hence, the zeroth order contribution is obtained by integrating just the first term,

$$\frac{d^{(0)}f}{d\varphi} = \nu, \tag{8.44}$$

which gives

$${}^{(0)}f = \nu(\varphi-\omega) \tag{8.45}$$

where the constant ω corresponds to the longitude of the pericenter. Replacing this solution into the right hand side of equation (8.42) we get the equation for the second order contribution

$$\frac{d^{(2)}f}{d\varphi} = -\nu\frac{\alpha m}{\kappa a(1-e^2)} e\cos[\nu(\varphi-\omega)], \tag{8.46}$$

which is integrated to obtain

$$^{(2)}f = -\frac{\alpha m}{\kappa a(1-e^2)} e \sin[\nu(\varphi - \omega)]. \tag{8.47}$$

Therefore, the true anomaly up to second order is given by the expression

$$f(\varphi) = \nu(\varphi - \omega) - \frac{\alpha m}{\kappa a(1-e^2)} e \sin[\nu(\varphi - \omega)] + \mathcal{O}(\epsilon^4). \tag{8.48}$$

8.3.5 Advance of the Perihelion

The true anomaly obtained above shows a secular contribution to the orbit of the test particle which reflects as an advance in the location of the perihelion. Just as calculated in Section 4.3.1, two successive perihelia occur with the angular difference $\Delta\varphi = \frac{2\pi}{\nu}$ and therefore the perihelion shift is given by $\Delta\omega = \frac{2\pi}{\nu} - 2\pi$. This expression is approximated as

$$\Delta\omega = \frac{2\pi m}{\kappa a(1-e^2)} \left(4\alpha - 4\kappa\alpha - \beta + 2\kappa\gamma + 2\kappa^2\right) + \mathcal{O}(\epsilon^4). \tag{8.49}$$

It is clear that the Schwarzschild's perihelion advance (4.42) is recovered when using the PPN parameters $\alpha = 0$ and $\kappa = \beta = \gamma = 1$.

8.3.6 Time Dependence and Kepler's Equation

We recover time dependence following a similar treatment as that in Section 1.1.3. Using equation (8.30) and replacing the expressions for the angular momentum and the Hamiltonian in equations (8.28) and (8.39), the equation (8.42) becomes a differential equation for the true anomaly as function of time,

$$\frac{df}{dt} = \frac{na^2\sqrt{1-e^2}}{r^2}\left[1 - \frac{m}{r}\left(2\gamma - 2\alpha + \frac{\alpha}{\kappa} + 2\kappa\right) - \frac{m}{2\kappa a}\left((\kappa+1)\alpha - 2\beta - (\kappa-1)\gamma + \kappa - \kappa^2\right)\right]. \tag{8.50}$$

Now we introduce the eccentric anomaly exactly as in Section 1.1.3, using equations (1.43). The generalization of equation (1.47c) is

$$\dot{E} = \frac{na}{r}\left[1 - \frac{m}{2\kappa a}\left((\kappa+1)\alpha - 2\beta - (\kappa-1)\gamma + \kappa - \kappa^2\right) - \frac{m}{r}\left(2\gamma - 2\alpha + \frac{\alpha}{\kappa} + 2\kappa\right)\right]. \tag{8.51}$$

Replacing the radial coordinate in terms of the eccentric anomaly as in equation (1.47b) an integrating in both sides gives the generalization of Kepler's equation,

$$E - \left[1 + \frac{m}{\kappa a}\left((2\kappa - 1)\alpha - 2\kappa\gamma - 2\kappa^2\right)\right] e \sin E = l, \tag{8.52}$$

where the mean anomaly is defined through the equation

$$\frac{dl}{dt} = n\left[1 + \frac{m}{2\kappa a}\left(3(\kappa-1)\alpha + 2\beta - (3\kappa+1)\gamma - \kappa - 3\kappa^2\right)\right]. \tag{8.53}$$

8.3.7 Osculating Elements

Following the celestial mechanics approach, we have now the required relations to describe the motion of a test particle moving in the spacetime described by the metric (8.20) using the osculating elements method.

Note that, from the equation of motion (8.24) one can identify a disturbing force to the Newtonian problem with the form

$$\delta\mathbf{A} = (1-\kappa)\frac{GM}{r^3}\mathbf{x} + \frac{m}{r^3}\left[\left(2(\beta+\kappa\gamma-\alpha)\frac{GM}{r} - (\alpha+\gamma)\dot{\mathbf{x}}^2 + 3\alpha\frac{(\mathbf{x}\cdot\dot{\mathbf{x}})^2}{r^2}\right)\mathbf{x} \right.$$
$$\left. +2(\gamma+\kappa-\alpha)(\mathbf{x}\cdot\dot{\mathbf{x}})\dot{\mathbf{x}}\right] + \mathcal{O}(\epsilon^4). \tag{8.54}$$

The components of this force in the time dependent basis $(\mathbf{n}, \boldsymbol{\lambda}, \mathbf{k})$ are given by equations (2.5),

$$\mathcal{R} = \frac{n^2a^2}{\kappa r^2}\left\{(1-\kappa)a + m\left[\left(1-\frac{1}{\kappa}\right)((2+\kappa)\alpha - 2\beta - \kappa\gamma) - \kappa(\gamma+2\kappa)\right.\right.$$
$$\left.\left. +2(\beta+2\kappa\gamma-\alpha+2\kappa^2)\frac{a}{r} - \kappa(\alpha+2\gamma+2\kappa)\frac{a^2(1-e^2)}{r^2}\right]\right\} + \mathcal{O}(\epsilon^4) \tag{8.55}$$

$$\mathcal{T} = \frac{2n^2a^3}{r^3}m(\gamma-\alpha+\kappa)e\sin f + \mathcal{O}(\epsilon^4) \tag{8.56}$$

$$\mathcal{W} = 0. \tag{8.57}$$

Replacing these relations into the osculating elements in equations (2.14) gives the evolution equations for the orbital parameters,

$$\frac{da}{dt} = \frac{2}{\sqrt{1-e^2}}\frac{na^2}{\kappa r^2}e\sin f\left\{(1-\kappa)a + m\left[\left(1-\frac{1}{\kappa}\right)((2+\kappa)\alpha - 2\beta - \kappa\gamma) - \kappa(\gamma+2\kappa)\right.\right.$$
$$\left.\left. +2(\beta+2\kappa\gamma-\alpha+2\kappa^2)\frac{a}{r} - 3\kappa\alpha\frac{a^2(1-e^2)}{r^2}\right]\right\} + \mathcal{O}(\epsilon^4) \tag{8.58}$$

$$\frac{de}{dt} = \sqrt{1-e^2}\frac{na}{\kappa r^2}\sin f\left\{(1-\kappa)a + m\left[\left(1-\frac{1}{\kappa}\right)((2+\kappa)\alpha - 2\beta - \kappa\gamma) + \kappa(2\alpha - 3\gamma - 4\kappa)\right.\right.$$
$$\left.\left. +2(\beta+2\kappa\gamma-\alpha+2\kappa^2)\frac{a}{r} - 3\kappa\alpha\frac{a^2(1-e^2)}{r^2}\right]\right\} \tag{8.59}$$

$$\frac{d\omega}{dt} = \frac{\sqrt{1-e^2}}{e^2}\frac{na}{\kappa r^2}\left\{(1-\kappa)a\left(1-\frac{a(1-e^2)}{r}\right) + m\left[\left(1-\frac{1}{\kappa}\right)((2+\kappa)\alpha - 2\beta - \kappa\gamma) + \kappa(2\alpha - 3\gamma - 4\kappa)\right.\right.$$
$$\left[2(\beta+4\kappa\gamma-(1+2\kappa)\alpha+4\kappa^2) - \left((2-\kappa)\alpha - 2\beta - \frac{1}{\kappa}((2+\kappa)\alpha - 2\beta - \kappa\gamma)\right)(1-e^2)\right]\frac{a}{r}$$
$$\left.\left. -\left((3\kappa-1)\alpha + 2\beta + 4\kappa\gamma + 2\kappa^2\right)\frac{a^2(1-e^2)}{r^2} + 3\kappa\alpha\frac{a^3(1-e^2)^2}{r^3}\right]\right\} \tag{8.60}$$

$$\frac{d\iota}{dt} = 0 \tag{8.61}$$

$$\frac{d\Omega}{dt} = 0 \tag{8.62}$$

Chapter 9

Post-Newtonian Description of the N-Body Problem

9.1 Energy-Momentum Tensor for a System of N-Particles

The energy momentum tensor for a system of N-particles labeled with the index $a = 1, 2, 3, ..., N$, with masses m_a and located at the positions $x_a^\mu(\tau_a)$ is given by

$$T^{\mu\nu}(x) = \frac{1}{\sqrt{-g}} \sum_a m_a \int \frac{dx_a^\mu}{d\tau_a} \frac{dx_a^\nu}{d\tau_a} \delta^4 \left(x - x_a(\tau_a)\right) d\tau_a, \tag{9.1}$$

where τ_a is the proper time of particle m_a. Using the coordinate time t as the parameter an integrating, the energy-momentum tensor components become

$$T^{\mu\nu}(x) = \frac{1}{\sqrt{-g}} \sum_a m_a \frac{dx_a^\mu}{dt} \frac{dx_a^\nu}{dt} \left(\frac{d\tau_a}{dt}\right)^{-1} \delta^3 \left(\mathbf{x} - \mathbf{x}_a(t)\right). \tag{9.2}$$

In order to obtain the power expansion of these components, we use the definition of the Lagrangian in equation (7.4), to write

$$\frac{d\tau_a}{dt} = 1 - \frac{L_a}{m_a c^2}. \tag{9.3}$$

Using the post-Newtonian Lagrangian (7.5), this equation gives

$$\left(\frac{d\tau_a}{dt}\right)^{-1} = 1 + \frac{1}{2}\frac{\dot{\mathbf{x}}^2}{c^2} + \frac{1}{2}{}^{(2)}h_{00} + \mathcal{O}\left(\epsilon^4\right). \tag{9.4}$$

The determinant of the post-Newtonian metric (6.4), is approximated to

$$g = -1 + {}^{(2)}g + \mathcal{O}\left(\epsilon^4\right) = -1 + {}^{(2)}h_{00} - \delta^{jk\,(2)}h_{jk} + \mathcal{O}\left(\epsilon^4\right), \tag{9.5}$$

or in terms of the post-Newtonian potentials,

$$g = -1 + \frac{4\phi}{c^2} + \mathcal{O}\left(\epsilon^4\right). \tag{9.6}$$

Hence, we obtain the relation

$$\frac{1}{\sqrt{-g}} = 1 + \frac{2\phi}{c^2} + \mathcal{O}\left(\epsilon^4\right), \tag{9.7}$$

and replacing these relations in equation (9.2) , there are obtained the different orders of approximation of the energy-momentum tensor components,

$$ {}^{(0)}T^{00}(x) = \sum_a m_a c^2 \delta^3(\mathbf{x} - \mathbf{x}_a(t)) \tag{9.8a}$$

$$ {}^{(2)}T^{00}(x) = \sum_a m_a \delta^3(\mathbf{x} - \mathbf{x}_a(t)) \left(\frac{1}{2}\frac{\dot{\mathbf{x}}_a^2}{c^2} + \frac{\phi(x)}{c^2} \right) \tag{9.8b}$$

$$ {}^{(1)}T^{0j}(x) = \sum_a m_a c^2 \frac{\dot{x}_a^j}{c} \delta^3(\mathbf{x} - \mathbf{x}_a(t)) \tag{9.8c}$$

$$ {}^{(3)}T^{0j}(x) = \sum_a m_a c^2 \frac{\dot{x}_a^j}{c} \delta^3(\mathbf{x} - \mathbf{x}_a(t)) \left(\frac{1}{2}\frac{\dot{\mathbf{x}}_a^2}{c^2} + \frac{\phi(x)}{c^2} \right) \tag{9.8d}$$

$$ {}^{(2)}T^{jk}(x) = \sum_a m_a c^2 \frac{\dot{x}_a^j}{c} \frac{\dot{x}_a^k}{c} \delta^3(\mathbf{x} - \mathbf{x}_a(t)). \tag{9.8e}$$

9.2 Post-Newtonian Potentials for a System of N-Particles

Integration of the differential equations (6.41) using the above components for the energy-momentum tensor gives the post-Newtonian potentials for the system of N-particles,

$$\phi(x) = -G \sum_a \frac{m_a}{|\mathbf{x} - \mathbf{x}_a(t)|} \tag{9.9a}$$

$$\zeta_j(x) = -G \sum_a \frac{m_a \dot{x}_a^j}{|\mathbf{x} - \mathbf{x}_a(t)|} \tag{9.9b}$$

$$\psi(x) = \frac{G^2}{c^2} \sum_a \sum_{b\neq a} \frac{m_a}{|\mathbf{x} - \mathbf{x}_a(t)|} \frac{m_b}{|\mathbf{x}_a(t) - \mathbf{x}_b(t)|} - \frac{3}{2}\frac{G}{c^2} \sum_a \frac{m_a \dot{\mathbf{x}}_a^2}{|\mathbf{x} - \mathbf{x}_a(t)|} \tag{9.9c}$$

$$\chi(x) = -\frac{G}{2c^2} \sum_a m_a \left|\mathbf{x} - \mathbf{x}_a(t)\right|. \tag{9.9d}$$

9.3 Motion of a Test Particle in the Field of a System of N-Particles

Using the above relations in equations (7.5) and (7.6), we obtain the Lagrangian describing the motion of a test particle with mass m_a in the gravitational field produced by a system of N-particles,

$$\begin{aligned} L_a = & \frac{1}{2} m_a \dot{\mathbf{x}}_a^2 + G m_a \sum_{b\neq a} \frac{m_b}{r_{ab}} + \frac{1}{8} m_a \frac{\left(\dot{\mathbf{x}}_a^2\right)^2}{c^2} - \frac{G^2}{c^2} m_a \sum_{b\neq a}\sum_{c\neq b} \frac{m_b}{r_{ab}}\frac{m_c}{r_{bc}} \\ & - \frac{1}{2}\frac{G^2}{c^2} m_a \sum_{b\neq a}\sum_{c\neq a} \frac{m_b}{r_{ab}}\frac{m_c}{r_{ac}} + \frac{3}{2}\frac{G}{c^2} m_a \sum_{b\neq a} \frac{m_b}{r_{ab}} \left(\dot{\mathbf{x}}_a^2 + \dot{\mathbf{x}}_b^2\right) \\ & - \frac{1}{2}\frac{G}{c^2} m_a \sum_{b\neq a} \frac{m_b}{r_{ab}} \left[7\dot{\mathbf{x}}_a \cdot \dot{\mathbf{x}}_b + \frac{(\mathbf{x}_{ab}\cdot\dot{\mathbf{x}}_a)(\mathbf{x}_{ab}\cdot\dot{\mathbf{x}}_b)}{r_{ab}^2} \right] + \mathcal{O}\left(\epsilon^6\right), \end{aligned} \tag{9.10}$$

where we introduced the quantities $\mathbf{x}_{ab} = \mathbf{x}_a - \mathbf{x}_b$ and $r_{ab} = |\mathbf{x}_{ab}| = |\mathbf{x}_a - \mathbf{x}_b|$. From this equation it is straightforward to define the complete Lagrangian for the system of N-particles by summing over the index $a = 1, 2, 3, ..., N$ and taking into account that the resulting expression must be symmetric in the variables m_a, $\mathbf{x}_a$ and $\dot{\mathbf{x}}_a$, so it is necessary to introduce some coefficients to avoid double sums. The resulting Lagrangian is

$$
\begin{aligned}
L = & \sum_a \frac{1}{2} m_a \dot{\mathbf{x}}_a^2 + \frac{1}{2} G \sum_a \sum_{b \neq a} \frac{m_a m_b}{r_{ab}} + \frac{1}{8} \sum_a m_a \frac{\left(\dot{\mathbf{x}}_a^2\right)^2}{c^2} - \frac{1}{2} \frac{G^2}{c^2} \sum_a \sum_{b \neq a} \sum_{c \neq a} \frac{m_a m_b m_c}{r_{ab} r_{ac}} \\
& + \frac{3}{2} \frac{G}{c^2} \sum_a m_a \dot{\mathbf{x}}_a^2 \sum_{b \neq a} \frac{m_b}{r_{ab}} - \frac{1}{4} \frac{G}{c^2} \sum_a \sum_{b \neq a} \frac{m_a m_b}{r_{ab}} \left[7 \dot{\mathbf{x}}_a \cdot \dot{\mathbf{x}}_b + (\mathbf{n}_{ab} \cdot \dot{\mathbf{x}}_a)(\mathbf{n}_{ab} \cdot \dot{\mathbf{x}}_b) \right] + \mathcal{O}\left(\epsilon^6\right) .
\end{aligned}
\tag{9.11}
$$

Exercise 9.1

Einstein-Infeld-Hoffmann Equations

Obtain the Euler-Lagrange equations associated with the Lagrangian (9.11), known as the Einstein-Infeld-Hoffmann Equations [7] ,

$$
\begin{aligned}
\ddot{\boldsymbol{x}}_a = & - G \sum_{b \neq a} m_b \frac{\boldsymbol{x}_{ab}}{r_{ab}^3} \left[1 - \frac{4G}{c^2} \sum_{c \neq a} \frac{m_c}{r_{ac}} + \frac{G}{c^2} \sum_{c \neq b} m_c \left(- \frac{1}{r_{bc}} + \frac{\boldsymbol{x}_{ab} \cdot \boldsymbol{x}_{bc}}{2 r_{bc}^3} \right) \right. \\
& \left. + \frac{\dot{\boldsymbol{x}}_a^2 - 4 \dot{\boldsymbol{x}}_a \cdot \dot{\boldsymbol{x}}_b + 2 \dot{\boldsymbol{x}}_b^2}{c^2} - \frac{3}{2c^2} \left(\frac{\dot{\boldsymbol{x}}_b \cdot \boldsymbol{x}_{ab}}{r_{ab}} \right)^2 \right] - \frac{7}{2} \frac{G^2}{c^2} \sum_{b \neq a} \frac{m_b}{r_{ab}} \sum_{c \neq b} \frac{m_c \boldsymbol{x}_{bc}}{r_{bc}^3} \\
& + \frac{G}{c^2} \sum_{b \neq a} m_b \frac{\boldsymbol{x}_{ab} \cdot (4 \dot{\boldsymbol{x}}_a - 3 \dot{\boldsymbol{x}}_b)}{r_{ab}^3} \left(\dot{\boldsymbol{x}}_a - \dot{\boldsymbol{x}}_b \right) .
\end{aligned}
\tag{9.12}
$$

Chapter 10

Post-Newtonian Description of a Binary System

In this chapter, we will simplify the equations describing a system of N particles to the simple case of a binary system. This consists of a body with proper mass m_1 and position $\mathbf{x}_1$ together with another body of proper mass m_2 and position $\mathbf{x}_2$. The Lagrangian describing the binary system is obtained by considering $N = 2$ in the equation (9.11), giving

$$\begin{aligned} L =& \frac{1}{2}m_1\dot{\mathbf{x}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{x}}_2^2 + \frac{Gm_1m_2}{r} + \frac{1}{8}m_1\frac{(\dot{\mathbf{x}}_1^2)^2}{c^2} + \frac{1}{8}m_2\frac{(\dot{\mathbf{x}}_2^2)^2}{c^2} - \frac{1}{2}\frac{G^2}{c^2}\frac{m_1m_2}{r^2}(m_1+m_2) \\ &+ \frac{1}{2}\frac{G}{c^2}\frac{m_1m_2}{r}\left[3(\dot{\mathbf{x}}_1^2+\dot{\mathbf{x}}_2^2) - 7\dot{\mathbf{x}}_1\cdot\dot{\mathbf{x}}_2 - \frac{(\mathbf{x}\cdot\dot{\mathbf{x}}_1)(\mathbf{x}\cdot\dot{\mathbf{x}}_2)}{r^2}\right] + \mathcal{O}\left(\epsilon^6\right), \end{aligned} \tag{10.1}$$

where we introduce the relative separation vector

$$\mathbf{x} := \mathbf{x}_1 - \mathbf{x}_2 \tag{10.2}$$

and its magnitude $r := |\mathbf{x}| = |\mathbf{x}_1 - \mathbf{x}_2|$.

As usual, we will separate this Lagrangian into two parts, one describing the baricenter and other describing the relative motion. First, we introduce the Tolman mass for each particle as

$$\tilde{m}_1 = m_1 + \frac{1}{2}m_1\frac{\dot{\mathbf{x}}_1^2}{c^2} - \frac{G}{2c^2}\frac{m_1m_2}{r} \tag{10.3}$$

$$\tilde{m}_2 = m_2 + \frac{1}{2}m_2\frac{\dot{\mathbf{x}}_2^2}{c^2} - \frac{G}{2c^2}\frac{m_1m_2}{r} \tag{10.4}$$

to define the baricenter position,

$$\mathbf{X} = \frac{\tilde{m}_1\mathbf{x}_1 + \tilde{m}_2\mathbf{x}_2}{\tilde{m}_1 + \tilde{m}_2}. \tag{10.5}$$

From these definitions, the acceleration of the baricenter is shown to satisfy

$$(\tilde{m}_1 + \tilde{m}_2)\ddot{\mathbf{X}} = \tilde{m}_1\ddot{\mathbf{x}}_1 + \tilde{m}_2\ddot{\mathbf{x}}_2 + \mathcal{O}\left(\epsilon^4\right). \tag{10.6}$$

Using the Einstein-Infeld-Hoffmann equations to replace the accelerations $\ddot{\mathbf{x}}_1$ and $\ddot{\mathbf{x}}_2$, it is straightforward to show that the baricenter is not accelerated,

$$\frac{d^2\mathbf{X}}{dt^2} = 0 + \mathcal{O}\left(\epsilon^4\right), \tag{10.7}$$

and therefore it is possible to describe the dynamics of the binary system with an inertial system attached to the barycenter, in which $\mathbf{X} = 0$. This assumption, together with equations 10.2 and 10.5, let us solve for $\mathbf{r}_1$ and $\mathbf{r}_2$ in terms of the relative separation,

$$\mathbf{x}_1 = \left[\frac{m_2}{m} + \frac{\mu\Delta m}{2m^2c^2}\left(\dot{\mathbf{x}}^2 - \frac{Gm}{r}\right)\right]\mathbf{x} + \mathcal{O}\left(\epsilon^4\right) \tag{10.8}$$

$$\mathbf{x}_2 = \left[-\frac{m_1}{m} + \frac{\mu\Delta m}{2m^2c^2}\left(\dot{\mathbf{x}}^2 - \frac{Gm}{r}\right)\right]\mathbf{x} + \mathcal{O}\left(\epsilon^4\right), \tag{10.9}$$

where we introduce $m = m_1 + m_2$, $\mu = m_1 + m_2 m$ and $\Delta m = m_1 - m_2$. The corresponding velocities satisfy the relations

$$\dot{\mathbf{x}}_1 = \frac{m_2}{m}\dot{\mathbf{x}} + \mathcal{O}\left(\epsilon^2\right) \tag{10.10}$$

$$\dot{\mathbf{x}}_2 = -\frac{m_1}{m}\dot{\mathbf{x}} + \mathcal{O}\left(\epsilon^2\right). \tag{10.11}$$

In the baricenter frame, the Lagrangian (10.1) describes the relative motion of the particles in the binary system, and takes the form

$$\begin{aligned} L =& \frac{1}{2}\mu\dot{\mathbf{x}}^2 + \frac{G\mu m}{r} + \frac{1}{8}\mu\left(1 - \frac{3\mu}{m}\right)\frac{(\dot{\mathbf{x}}^2)^2}{c^2} - \frac{1}{2}\frac{G^2}{c^2}\frac{\mu m^2}{r^2} \\ &+ \frac{1}{2}\frac{G}{c^2}\frac{\mu m}{r}\left[\left(3 + \frac{\mu}{m}\right)\dot{\mathbf{x}}^2 + \frac{\mu}{m}\left(\frac{\mathbf{x}\cdot\dot{\mathbf{x}}}{r}\right)^2\right] + \mathcal{O}\left(\epsilon^6\right). \end{aligned} \tag{10.12}$$

Note that the first two terms in this Lagrangian correspond to the Newtonian contribution while the remaining terms are post-Newtonian corrections.

Exercise 10.1

Conserved Quantities

Introduce spherical coordinates in the Lagrangian (10.12) to obtain the two conserved quantities: the specific angular momentum,

$$\boldsymbol{\ell} = \left\{1 + \frac{1}{c^2}\left[\frac{1}{2}\left(1 - \frac{3\mu}{m}\right)\dot{x}^2 + \left(3 + \frac{\mu}{m}\right)\frac{Gm}{r}\right]\right\}(\boldsymbol{x}\times\dot{\boldsymbol{x}}) + \mathcal{O}\left(\epsilon^6\right), \tag{10.13}$$

and the Hamiltonian (or specific energy),

$$\begin{aligned} \varepsilon =& \frac{1}{2}\dot{x} - \frac{Gm}{r} \\ &+ \frac{1}{c^2}\left\{\frac{3}{8}\left(1 - \frac{3\mu}{m}\right)\left(\dot{x}^2\right)^2 + \frac{Gm}{2r}\left[\left(3 + \frac{\mu}{m}\right)\dot{x}^2 + \frac{\mu}{m}\dot{r}^2 + \frac{Gm}{r}\right]\right\} + \mathcal{O}\left(\epsilon^6\right). \end{aligned} \tag{10.14}$$

Equation (10.13) corresponds to the post-Newtonian angular momentum and it is interesting to note that, although the whole quantity $\boldsymbol{\ell}$ is conserved (magnitude and direction), the vector $\mathbf{x}\times\dot{\mathbf{x}}$ has not a constant magnitude.

The Euler-Lagrange equations associated to the Lagrangian (10.12) are

$$\begin{aligned} \ddot{x}^k =& -\frac{Gm}{r^3}x^k + \frac{Gm}{c^2r^3}\left[\frac{2Gm}{r}\left(2 + \frac{\mu}{m}\right) - \left(1 + \frac{3\mu}{m}\right)(\dot{\mathbf{x}}^2)^2 + \frac{3}{2}\frac{\mu}{m}\left(\frac{\mathbf{x}\cdot\dot{\mathbf{x}}}{r}\right)^2\right]x^k \\ &+ \frac{2Gm}{c^2r^3}(\mathbf{x}\cdot\dot{\mathbf{x}})\left(2 - \frac{\mu}{m}\right)\dot{x}^k + \mathcal{O}\left(\epsilon^6\right), \end{aligned} \tag{10.15}$$

or, in vector notation,

$$\ddot{\mathbf{x}} = -\frac{Gm}{r^3}\mathbf{x} + \frac{Gm}{c^2r^3}\left[\frac{2Gm}{r}\left(2+\frac{\mu}{m}\right) - \left(1+\frac{3\mu}{m}\right)(\dot{\mathbf{x}}^2)^2 + \frac{3}{2}\frac{\mu}{m}\left(\frac{\mathbf{x}\cdot\dot{\mathbf{x}}}{r}\right)^2\right]\mathbf{x}$$
$$+\frac{2Gm}{c^2r^3}(\mathbf{x}\cdot\dot{\mathbf{x}})\left(2-\frac{\mu}{m}\right)\dot{\mathbf{x}} + \mathcal{O}\left(\epsilon^6\right). \tag{10.16}$$

10.1 Bound Orbits for the Binary System

Because of the conservation of the angular momentum $\boldsymbol{\ell}$, the orbital motion of the binary system lies in a fixed plane, which can be chosen to be the $x-y$ plane. Hence, it is possible to introduce a time dependent basis in the form of equation (1.18) and therefore, the relative position, velocity and acceleration will be given by equations (**??**). Using this basis, the equations of motion (10.16) are written as

$$\ddot{r} = r\dot{\varphi}^2 - \frac{Gm}{r^2} + \frac{Gm}{c^2r^2}\left[\frac{1}{2}\left(6-\frac{7\mu}{m}\right)\dot{r}^2 - \left(1+\frac{3\mu}{m}\right)r^2\dot{\varphi}^2 + 2\left(2+\frac{\mu}{m}\right)\frac{Gm}{r}\right] + \mathcal{O}\left(\epsilon^6\right) \tag{10.17a}$$

$$\frac{d}{dt}\left(r^2\dot{\varphi}\right) = 2\left(2-\frac{\mu}{m}\right)\frac{Gm}{c^2}\dot{r}\dot{\varphi} + \mathcal{O}\left(\epsilon^6\right). \tag{10.17b}$$

10.1.1 Circular Orbits

The post-Newtonian equations of motion for the binary system admit circular orbits. In fact, setting $r =$ constant (i.e. $\dot{r} = \ddot{r} = 0$) in equations (10.17) shows that the orbital angular velocity is a constant

$$\dot{\varphi}^2 = \dot{\varphi}_c^2 = \frac{Gm}{r^3}\left[1-\left(3-\frac{\mu}{m}\right)\frac{Gm}{c^2r}\right] + \mathcal{O}\left(\epsilon^6\right). \tag{10.18}$$

Note that the first term in the right hand side corresponds to the Keplerian orbital angular velocity. Using this relation in equations (10.13) and (10.14), we obtain the corresponding post-Newtonian values of the angular momentum and energy for the circular orbit,

$$\ell = \sqrt{Gmr}\left[1+2\frac{Gm}{c^2r}\right] + \mathcal{O}\left(\epsilon^6\right) \tag{10.19a}$$

$$\varepsilon = -\frac{Gm}{2r}\left[1-\frac{1}{4}\left(7-\frac{\mu}{m}\right)\frac{Gm}{c^2r}\right] + \mathcal{O}\left(\epsilon^6\right). \tag{10.19b}$$

10.1.2 Elliptic Orbits

As can be seen from the Lagrangian (10.12) or the equations of motion (10.17), the zeroth order motion of the binary system corresponds to Keplerian orbits, which can be written as the conics

$$\mathbf{x} = r\cos f\,\mathbf{e}_x + r\sin f\,\mathbf{e}_y, \tag{10.20}$$

where

$$r = \frac{p}{1+e\cos f}, \tag{10.21}$$

p and e are the Keplerian semi-latus rectum and eccentricity, respectively, and $f = \varphi - \omega$ is the true anomaly. In order to obtain a post-Newtonian corrected solution, we will use the perturbation approach given in Section 2.1. Hence, the correction to the Keplerian orbit will be given by the extra terms in equations (10.17), which will be interpreted as a perturbing force in the time dependent basis $(\mathbf{n}, \boldsymbol{\lambda}, \mathbf{k})$. Its components are

$$\begin{cases} \mathcal{R} &= \frac{Gm}{c^2r^2}\left[\frac{1}{2}\left(6-\frac{7\mu}{m}\right)\dot{r}^2-\left(1+\frac{3\mu}{m}\right)r^2\dot{\varphi}^2+2\left(2+\frac{\mu}{m}\right)\frac{Gm}{r}\right] \\ \mathcal{T} &= 2\left(2-\frac{\mu}{m}\right)\frac{Gm}{c^2}\dot{r}\dot{\varphi} \\ \mathcal{W} &= 0. \end{cases} \tag{10.22}$$

Then, the osculating elements (2.14) become

$$\begin{aligned} \frac{da}{dt} =& \frac{2}{n\sqrt{1-e^2}}\frac{G^2m^2}{c^2p^3}(1+e\cos f)^2 e\sin f\left[\left(7-\frac{3\mu}{m}\right)+\frac{1}{2}\left(6-\frac{7\mu}{m}\right)e^2\right. \\ &\left.+2\left(5-\frac{4\mu}{m}\right)e\cos f-\frac{3\mu}{2m}e^2\cos^2 f\right] \qquad (10.23a) \\ \frac{de}{dt} =& \frac{\sqrt{1-e^2}}{na}\frac{G^2m^2}{c^2p^3}(1+e\cos f)^2\sin f\left[\left(3-\frac{\mu}{m}\right)+\frac{1}{2}\left(14-\frac{11\mu}{m}\right)e^2\right. \\ &\left.+2\left(5-\frac{4\mu}{m}\right)e\cos f-\frac{3\mu}{2m}e^2\cos^2 f\right] \qquad (10.23b) \\ \frac{d\iota}{dt} =& 0 \qquad (10.23c) \\ \frac{d\Omega}{dt} =& 0 \qquad (10.23d) \\ \frac{d\omega}{dt} =& \frac{\sqrt{1-e^2}}{nae}\frac{G^2m^2}{c^2p^3}(1+e\cos f)^2\left[4\left(2-\frac{\mu}{m}\right)e+\left(3-\frac{\mu}{m}\right)\cos f\right. \\ &\left.+\left(1+\frac{3\mu}{2m}\right)e^2\cos f-2\left(5-\frac{4\mu}{m}\right)e\cos^2 f+\frac{3\mu}{2m}e^2\cos^3 f\right] \qquad (10.23e) \\ \frac{dl_0}{dt} =& -\sqrt{1-e^2}\left[\frac{d\omega}{dt}-\frac{2p}{na^2(1+e\cos f)}\mathcal{T}\right], \qquad (10.23f) \end{aligned}$$

where $n=\sqrt{\frac{GM}{a^3}}$ is the mean motion. It can also be included the differential equation for the semi-latus rectum,

$$\frac{dp}{dt} = \frac{4}{n}\frac{G^2m^2}{c^2p^3}(1-e^2)^{3/2}\left(2-\frac{\mu}{m}\right)(1+e\cos f)e\sin f. \tag{10.24}$$

However, in order to integrate these equations it is usual to write them in terms of the true anomaly as independent variable, as in equation (2.21),

$$\begin{aligned}\frac{da}{df} =& \frac{2(1-e^2)}{n}\frac{G^2m^2}{c^2p^3} e\sin f\left[\left(7-\frac{3\mu}{m}\right)+\frac{1}{2}\left(6-\frac{7\mu}{m}\right)e^2\right.\\ &\left.+2\left(5-\frac{4\mu}{m}\right)e\cos f-\frac{3\mu}{2m}e^2\cos^2 f\right] \qquad (10.25a)\\ \frac{de}{df} =& \frac{Gm}{c^2p}\sin f\left[\left(3-\frac{\mu}{m}\right)+\frac{1}{2}\left(14-\frac{11\mu}{m}\right)e^2\right.\\ &\left.+2\left(5-\frac{4\mu}{m}\right)e\cos f-\frac{3\mu}{2m}e^2\cos^2 f\right] \qquad (10.25b)\\ \frac{d\omega}{df} =& \frac{1}{e}\frac{Gm}{c^2p}\left[4\left(2-\frac{\mu}{m}\right)e+\left(3-\frac{\mu}{m}\right)\cos f+\left(1+\frac{3\mu}{2m}\right)e^2\cos f\right.\\ &\left.-2\left(5-\frac{4\mu}{m}\right)e\cos^2 f+\frac{3\mu}{2m}e^2\cos^3 f\right] \qquad (10.25c)\\ \frac{dl_0}{df} =& -\sqrt{1-e^2}\left(\frac{d\omega}{df}+\cos\iota\frac{d\Omega}{df}\right)+\frac{2(1-e^2)^3}{n^2a(1+e\cos f)^3}\mathcal{T}, \qquad (10.25d)\end{aligned}$$

together with the corresponding equation for the semi-latus rectum,

$$\frac{dp}{df} = \frac{4Gm}{c^2}\left(2-\frac{\mu}{m}\right)e\sin f. \qquad (10.26)$$

10.1.3 Secular Changes in the Orbital Parameters. Pericenter Shift.

In order to obtain the secular changes in the orbital parameters of the binary system, we integrate equations (10.23) over one orbital period as in equation (2.19) or integration equations (10.25) from $f = 0$ to $f = 2\pi$ as in equation (2.20). This gives no secular changes in the semi-latus rectum, semi-major axis and eccentricity, i.e. $\Delta p = \Delta a = \Delta e = 0$, but it gives a secular shift in the pericenter of

$$\Delta\omega = \frac{6\pi Gm}{c^2p} \qquad (10.27)$$

per orbit. This result is a generalization of Schwarzschild's shift with just one difference: the mass M, appearing in equation (4.42), corresponds to the central object while the mass m, appearing in (10.52), is the sum of the masses of the two bodies in the binary system.

10.2 Post-Newtonian Motion of the Binary System

The angular momentum and energy equations, (10.13) and (10.14), can be used to write $\dot r$ and $\dot\varphi$ in terms of the conserved quantities as

$$\begin{aligned}\dot r^2 =& 2\varepsilon\left[1-\frac{3}{2}\left(1-\frac{3\mu}{m}\right)\frac{\varepsilon}{c^2}\right]+\frac{2Gm}{r}\left[1-\left(6-\frac{7\mu}{m}\right)\frac{\varepsilon}{c^2}\right]-\frac{\ell^2}{r^2}\left[1-2\left(1-\frac{3\mu}{m}\right)\frac{\varepsilon}{c^2}\right]\\ &-5\left(2-\frac{\mu}{m}\right)\frac{G^2m^2}{c^2r^2}+\left(8-\frac{3\mu}{m}\right)\frac{Gm}{c^2r^3}\ell^2+\mathcal{O}\left(\epsilon^4\right) \qquad (10.28a)\\ \dot\varphi =& \frac{\ell}{r^2}\left[1-\left(1-\frac{3\mu}{m}\right)\frac{\varepsilon}{c^2}\right]-2\left(2-\frac{\mu}{m}\right)\frac{Gm}{c^2r^3}\ell. \qquad (10.28b)\end{aligned}$$

An interesting approach to integrate the post-Newtonian equations of motionthis equations describing the binary system was presented in 1985 by Damour and Deruelle [5].

Exercise 10.2

Damour-Dereulle Equation of Motion I
Show that the change of variable

$$r_D = r + \frac{1}{2}\left(8 - \frac{3\mu}{m}\right)\frac{Gm}{c^2} + \mathcal{O}\left(\epsilon^4\right) \tag{10.29}$$

reduces equation (10.28a) to a Newtonian-like form

$$\dot{r}_D^2 = 2\varepsilon_D + 2\frac{Gm_D}{r_D} - \frac{\ell_D^2}{r_D^2} \tag{10.30}$$

where

$$\varepsilon_D = \varepsilon\left[1 - \frac{3}{2}\left(1 - \frac{3\mu}{m}\right)\frac{\varepsilon}{c^2} + \mathcal{O}\left(\epsilon^4\right)\right] \tag{10.31a}$$

$$m_D = m\left[1 - \left(6 - \frac{7\mu}{m}\right)\frac{\varepsilon}{c^2} + \mathcal{O}\left(\epsilon^4\right)\right] \tag{10.31b}$$

$$\ell_D^2 = \ell^2\left[1 - 2\left(1 - \frac{3\mu}{m}\right)\frac{\varepsilon}{c^2} + 2\left(1 - \frac{\mu}{m}\right)\frac{G^2m^2}{c^2\ell^2} + \mathcal{O}\left(\epsilon^4\right)\right] \tag{10.31c}$$

Since the equation of motion in the Damour radial coordinate is equal to the Newtonian equation (1.25), its solution can be written by introducing the eccentric anomaly E, just as in equations (1.47). They are

$$r_D = a_D(1 - e_D\cos E) \tag{10.32}$$

with the time dependence through the relation

$$l = \sqrt{\frac{Gm_D}{a_D^3}}\left(t - t_p\right) = \left(E - e_D\sin E\right), \tag{10.33}$$

where the semi-major axis and the eccentricity are given by

$$\varepsilon_D = -\frac{Gm_D}{2a_D} \tag{10.34}$$

and

$$\ell_D^2 = Gm_Da_D\left(1 - e_D^2\right). \tag{10.35}$$

Exercise 10.3

Damour-Dereulle Equation of Motion II
Apply the inverse transformation

$$r = r_D - \frac{1}{2}\left(8 - \frac{3\mu}{m}\right)\frac{Gm}{c^2} + \mathcal{O}\left(\epsilon^4\right) \tag{10.36}$$

to obtain the post-Newtonian equation

$$r = a\left(1 - e\cos E\right) \tag{10.37}$$

where the post-Newtonian orbital parameters are

$$a = a_D\left[1 - \frac{1}{2}\left(8 - \frac{8\mu}{m}\right)\frac{Gm}{c^2 a_D}\right] \tag{10.38}$$

$$e = e_D\left[1 + \frac{1}{2}\left(8 - \frac{8\mu}{m}\right)\frac{Gm}{c^2 a_D}\right] \tag{10.39}$$

$$\varepsilon = -\frac{Gm}{2a}\left[1 - \frac{1}{4}\left(7 - \frac{\mu}{m}\right)\frac{Gm}{c^2 a} + \mathcal{O}\left(\epsilon^4\right)\right] \tag{10.40}$$

$$\ell^2 = Gma(1-e^2)\left[1 + \frac{4 + \left(2 - \frac{\mu}{m}\right)e^2}{1-e^2}\frac{Gm}{c^2 a} + \mathcal{O}\left(\epsilon^4\right)\right]. \tag{10.41}$$

Now we set, without losing generality, the initial argument of the pericenter as $\omega_0 = 0$. The zeroth order solution has a conserved angular momentum that gives the relation

$$r^2\frac{d\varphi}{dt} = \sqrt{Gmp}. \tag{10.42}$$

In order to introduce the post-Newtonian contributions to the motion of the binary system, we will introduce a small correction term, $\delta\ell \sim \epsilon \ll 1$, to the angular momentum equation

$$r^2\frac{d\varphi}{dt} = \sqrt{Gmp}(1 + \delta\ell), \tag{10.43}$$

and a small correction, $\frac{\delta\dot{\mathbf{x}}}{c} \sim \epsilon \ll 1$, to the velocity

$$\dot{\mathbf{x}} = \frac{d\mathbf{x}}{dt} = \sqrt{\frac{Gm}{p}}\left[-\sin\varphi\mathbf{e}_x + (e + \cos\varphi)\mathbf{e}_y\right] + \delta\dot{\mathbf{x}}. \tag{10.44}$$

Replacing (10.43) into the angular momentum definition $r^2\frac{d\varphi}{dt} = |\mathbf{x}\times\dot{\mathbf{x}}|$, we obtain a differential equation for the correction $\delta\ell$,

$$\frac{d}{dt}(\delta\ell) = \frac{1}{\sqrt{Gmp}}\left|\mathbf{x}\times\ddot{\mathbf{x}}\right|. \tag{10.45}$$

Using equations (10.16) for the relative acceleration and (10.44) for the velocity, we obtain

$$\frac{d}{dt}(\delta\ell) = -\frac{2Gm}{c^2 p}e\left(2 - \frac{\mu}{m}\right)\frac{d\cos\varphi}{dt}, \tag{10.46}$$

which gives immediately the correction to the angular momentum as

$$\delta\ell = -\frac{2Gm}{c^2 p}e\left(2 - \frac{\mu}{m}\right)\cos\varphi. \tag{10.47}$$

Replacing these results into equation (10.16) and integrating, we find the relative velocity in binary system as

$$\dot{\mathbf{x}} = v_x\mathbf{e}_x + v_y\mathbf{e}_y \tag{10.48}$$

with

$$\begin{aligned}
v_x = &-\sqrt{\frac{Gm}{p}}\sin\varphi - \frac{1}{c^2}\left(\frac{Gm}{p}\right)^{3/2}\left[3e\varphi - \left(3-\frac{\mu}{m}\right)\sin\varphi + \left(1+\frac{21\mu}{8m}\right)e^2\sin\varphi\right.\\
&\left.-\frac{1}{2}\left(1-\frac{2\mu}{m}\right)e\sin 2\varphi + \frac{\mu}{8m}e^2\sin 3\varphi\right] + \mathcal{O}\left(\epsilon^2\right) \qquad (10.49a)\\
v_y = &-\sqrt{\frac{Gm}{p}}\left(e+\cos\varphi\right) - \frac{1}{c^2}\left(\frac{Gm}{p}\right)^{3/2}\left[\left(3-\frac{\mu}{m}\right)\cos\varphi + \left(3-\frac{31\mu}{8m}\right)e^2\cos\varphi\right.\\
&\left.+\frac{1}{2}\left(1-\frac{2\mu}{m}\right)e\cos 2\varphi - \frac{\mu}{8m}e^2\cos 3\varphi\right] + \mathcal{O}\left(\epsilon^2\right) \qquad (10.49b)
\end{aligned}$$

10.2.1 Pericenter Shift

Exercise 10.4

Post-Newtonian Correction to the Orbit of the Binary System

Use equations (10.20),(10.43), (10.47) and (10.49), together with the relation

$$\frac{d}{d\varphi}\left(\frac{1}{r}\right) = -\left(r^2\frac{d\varphi}{dt}\right)^{-1}\frac{\boldsymbol{x}\cdot\dot{\boldsymbol{x}}}{r}, \qquad (10.50)$$

to obtain the post-Newtonian corrected orbit of the binary system,

$$\begin{aligned}
r = p&\left[1 + e\cos\varphi + \frac{Gm}{c^2p}\left[-\frac{1}{2}\left(7-\frac{2\mu}{m}\right)e + \frac{\mu}{4m}e^2 + 3e\varphi\sin\varphi\right.\right.\\
&\left.\left.+\frac{1}{2}\left(7-\frac{2\mu}{m}\right)e\cos\varphi - \frac{\mu}{4m}e^2\cos 2\varphi\right] + \mathcal{O}\left(\epsilon^2\right)\right]^{-1}, \qquad (10.51)
\end{aligned}$$

which replaces the conic equation (10.21).

In order to obtain an estimate for the motion of the pericenter of a binary system, it is necessary to analyze the post-Newtonian contributions in equation (10.51) which are very similar to those obtained for the motion of a particle in Schwarzschild's background in equation (4.35). Just as studied in Section 4.3.1, it is clear that the secular contribution to the pericenter motion is given by the third correction term, $3\frac{Gm}{c^2p}e\varphi\sin\varphi$, and it gives a shift of

$$\Delta\omega = \frac{6\pi Gm}{c^2p} \qquad (10.52)$$

per orbit. This result looks exactly as Schwarzschild's shift, with just one difference: the mass M, appearing in equation (4.42), corresponds to the central object while the mass m, appearing in (10.52), is the sum of the masses of the two bodies in the binary system.

Part V

Appendix

Appendix A

Astronomical Data

Some important data about the Solar System

Sun

- Mass: $M_\odot = 1.99 \times 10^{30}$ kg
- Mean Radius: $R_\odot = 696342$ km

Earth

- Mass: $M_\oplus = 5.972 \times 10^{24}$ kg
- Mean Radius: $R_\oplus = 6371$ km

Distance Sun-Earth: 1 au $= 1.496 \times 10^8$ km

Object	m ($M_\odot$) Mass	a (AU) Semi-major axis	e Eccentricity	ι Inclination (to ecliptic)	Ω Longitude of the Ascending Node	ω Argument of the Perihelion	r_p (AU) Perihelion	r_a (AU) Aphelion	T (yr) Orbital Period
Sun	1	–	–	–	–	–	–	–	–
Mercury [19]	1.652×10^{-7}	0.387098	0.205630	7.005°	48.331°	29.124°	0.307499	0.466697	0.240846
Venus [23]	2.447×10^{-6}	0.723332	0.006772	3.39458°	76.680°	54.884°	0.718440	0.728213	0.615198
Earth [16]	3.0027×10^{-6}	1.0	0.0167086	0.0	- 11.26064°	114.20783°	0.98327	1.017	1.00001
Mars [18]	3.213×10^{-7}	1.523679	0.0934	1.850°	49.558°	286.502°	1.382	1.666	1.88082
Jupiter [17]	0.0009543	5.2044	0.0489	1.303°	100.464°	273.867°	4.9501	5.4588	11.862
Saturn [21]	0.0002857	9.5826	0.0565	2.485°	113.665°	339.392°	9.0412	10.1238	29.4571
Uranus [22]	0.00004365	19.2184	0.046381	0.773°	74.006°	96.998857°	18.33	20.11	84.0205
Neptune [20]	00005149	30.11	0.009456	1.767975°	131.784°	276.336°	29.81	30.33	164.8

Table A.1: Some useful data about the Solar System taken from [16–23] (Epoch J2000)

Planet	Tangential Velocity at the perihelion (m/s)
Mercury	47362
Venus	35020
Earth	29780
Mars	24077

Table A.2: Tangential velocity of the planets in Solar System at the perihelion. Taken from [16–23]

Appendix B

Deduction of the Green Function for Poisson Equation

The solution of the Poisson equation (1.1),

$$\nabla^2\phi(t,\mathbf{x}) = 4\pi G\rho(t,\mathbf{x}), \tag{B.1}$$

can be written formally in terms of a Green function.

The standard mathematical definition of the Green's function for a linear differential operator $\hat{L}$ is the solution to $\hat{L}\mathcal{G} = \delta$, where δ represents the Dirac delta function for all relevant coordinates. For the gravitational system, the coordinates are time t and space $\mathbf{x}$. Therefore, our defining equation is

$$\nabla^2\mathcal{G}(t,\mathbf{x};t',\mathbf{x}') = \delta(t-t')\delta^{(3)}(\mathbf{x}-\mathbf{x}'). \tag{B.2}$$

However, it is important to note that the differential operator ∇^2 contains only spatial derivatives, so it does not act on the time coordinate (remember that in Newtonian mechanics, gravitational influence propagates instantaneously). Because time acts merely as a parameter rather than a dynamic variable in this operator, the Green's function separates completely into a temporal part and a spatial part

$$\mathcal{G}(t,\mathbf{x};t',\mathbf{x}') = \delta(t-t')\mathcal{G}_s(\mathbf{x},\mathbf{x}'). \tag{B.3}$$

Substituting this back into the definition, yields the spatial Green's function equation

$$\nabla^2\mathcal{G}_s(\mathbf{x},\mathbf{x}') = \delta^{(3)}(\mathbf{x}-\mathbf{x}'). \tag{B.4}$$

Due to translation invariance of free space, $\mathcal{G}_s$ depends only on the relative distance $\mathbf{R} = \mathbf{x}-\mathbf{x}'$, so the equation can be written as

$$\nabla^2\mathcal{G}_s(\mathbf{R}) = \delta^{(3)}(\mathbf{R}). \tag{B.5}$$

While this can be solved using Fourier transforms, a highly elegant and rigorous physical method is to use Gauss's Divergence Theorem. Thus, we integrate both sides of the spatial equation over a spherical volume V of radius $R = |\mathbf{R}|$ centered at the origin,

$$\int_V \nabla^2\mathcal{G}_s(\mathbf{R})\, d^3R = \int_V \delta^{(3)}(\mathbf{R})\, d^3R. \tag{B.6}$$

From the definition of the 3D Dirac delta function, the volume integral over a space containing the origin is exactly $\int_V \delta^{(3)}(\mathbf{R})\, d^3R = 1$. Then,

$$\int_V \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \mathcal{G}_s(\mathbf{R}))\, d^3R = 1. \tag{B.7}$$

Applying the Divergence Theorem ($\int_V \boldsymbol{\nabla} \cdot \mathbf{A}\, dV = \oint_{\partial V} \mathbf{A} \cdot \mathbf{da}$), the volume integral becomes a surface integral over the sphere's boundary ∂V,

$$\oint_{\partial V} \boldsymbol{\nabla} \mathcal{G}_s(\mathbf{R}) \cdot \mathbf{da} = 1. \tag{B.8}$$

Because the source $\delta^{(3)}(\mathbf{R})$ is symmetric, the solution $\mathcal{G}_s(\mathbf{R})$ must depend only on the radial distance R and therefore, the gradient $\boldsymbol{\nabla} \mathcal{G}_s(\mathbf{R})$ points purely in the radial direction $\hat{\mathbf{R}}$,

$$\boldsymbol{\nabla} \mathcal{G}_s(\mathbf{R}) = \frac{d\mathcal{G}_s}{dR} \hat{\mathbf{R}}. \tag{B.9}$$

The differential area vector on the sphere's surface is $\mathbf{da} = \hat{\mathbf{R}} R^2 d\Omega$ (where $d\Omega$ is the solid angle). The integral becomes

$$\oint_{\partial V} \left(\frac{d\mathcal{G}_s}{dR} \hat{\mathbf{R}} \right) \cdot \left(\hat{\mathbf{R}} R^2 d\Omega \right) = 1 \tag{B.10}$$

$$\frac{d\mathcal{G}_s}{dR} R^2 \oint_{\partial V} d\Omega = 1. \tag{B.11}$$

Integrating the solid angle gives the total surface area of a sphere, i.e. 4π, and hence we obtain

$$\frac{d\mathcal{G}_s}{dR} (4\pi R^2) = 1 \implies \frac{d\mathcal{G}_s}{dR} = \frac{1}{4\pi R^2}. \tag{B.12}$$

Finally, we integrate with respect to R,

$$\mathcal{G}_s(\mathbf{R}) = \int \frac{1}{4\pi R^2} dR = -\frac{1}{4\pi R} + C. \tag{B.13}$$

For an unbounded domain, we demand that the potential vanishes at infinity, i.e. $\lim_{R\to\infty} \mathcal{G}_s(\mathbf{R}) = 0$, so the constant of integration is $C = 0$. Replacing $R = |\mathbf{x} - \mathbf{x}'|$, we get

$$\mathcal{G}_s(\mathbf{x}, \mathbf{x}') = -\frac{1}{4\pi |\mathbf{x} - \mathbf{x}'|} \tag{B.14}$$

and recombining the spatial and temporal parts, the explicit spatio-temporal Green's function is finally

$$\mathcal{G}(t, \mathbf{x}; t', \mathbf{x}') = -\frac{\delta(t - t')}{4\pi |\mathbf{x} - \mathbf{x}'|}. \tag{B.15}$$

In order to prove this deduction, we will apply it to find the solution $\phi(t, \mathbf{x})$ for the original equation (1.1). The general solution via Green's functions is the integral of the Green's function multiplied by the entire right-hand side source term, i.e.

$$\phi(t, \mathbf{x}) = \int dt' \int d^3x'\, \mathcal{G}(t, \mathbf{x}; t', \mathbf{x}') \Big[4\pi G \rho(t', \mathbf{x}') \Big]. \tag{B.16}$$

Substitution of the Green function (B.15) gives

$$\phi(t, \mathbf{x}) = \int dt' \int d^3x' \left(-\frac{\delta(t - t')}{4\pi |\mathbf{x} - \mathbf{x}'|} \right) 4\pi G \rho(t', \mathbf{x}'). \tag{B.17}$$

It is clear that the 4π terms cancel out and evaluating the time integral over t' (which collapses via the $\delta(t - t')$ function) gives the desired result,

$$\phi(t, \mathbf{x}) = -G \int \frac{\rho(t, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x'. \tag{B.18}$$

This final formula is precisely the standard form of Newton's Law of Universal Gravitation in continuous integral form, successfully verifying the derived Green's function.

Appendix C

Deduction of the Retarded Green Function

The solution of the linearized field equations (5.36),

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu} + \mathcal{O}\left(\epsilon^4\right), \tag{C.1}$$

can be written formally in terms of a Green function. We consider the auxiliary equation

$$\Box \mathcal{G}(x, x') = -4\pi\delta(x - x') = -4\pi\delta(ct - ct')\delta(\mathbf{x} - \mathbf{x}') \tag{C.2}$$

and propose that its solution has the form

$$\mathcal{G}(x, x') = \frac{1}{2\pi} \int \mathcal{K}(\mathbf{x}, \mathbf{x}'; k) e^{-ik(x^0 - x'^0)} dk. \tag{C.3}$$

Replacing this transform into the wave equation (C.2) gives

$$\begin{aligned} \frac{1}{2\pi} \int \Box \left[\mathcal{K}(\mathbf{x}, \mathbf{x}'; k) e^{-ik(x^0 - x'^0)}\right] dk &= -4\pi\delta(ct - ct')\delta(\mathbf{x} - \mathbf{x}') \\ \frac{1}{2\pi} \int \left[k^2 + \nabla^2\right] \mathcal{K}(\mathbf{x}, \mathbf{x}'; k) e^{-ik(x^0 - x'^0)} dk &= -4\pi\delta(ct - ct')\delta(\mathbf{x} - \mathbf{x}'). \end{aligned}$$

Using the integral representation of the Dirac delta function,

$$\delta(x^0 - x'^0) = \frac{1}{2\pi} \int e^{-ik(x^0 - x'^0)} dk, \tag{C.4}$$

the right hand side becomes

$$\frac{1}{2\pi} \int \left[k^2 + \nabla^2\right] \mathcal{K}(\mathbf{x}, \mathbf{x}'; k) e^{-ik(x^0 - x'^0)} dk = -4\pi \frac{1}{2\pi} \int e^{-ik(x^0 - x'^0)} \delta(\mathbf{x} - \mathbf{x}') dk \tag{C.5}$$

and we conclude that

$$\left[k^2 + \nabla^2\right] \mathcal{K}(\mathbf{x}, \mathbf{x}'; k) = -4\pi\delta(\mathbf{x} - \mathbf{x}'). \tag{C.6}$$

The form of this equation let us write that the function $\mathcal{K}$ is

$$\mathcal{K}(\mathbf{x}, \mathbf{x}'; k) = \begin{cases} \frac{1}{|\mathbf{x} - \mathbf{x}'|} \text{ if } k = 0 \\ \frac{\bar{\mathcal{K}}(|\mathbf{x} - \mathbf{x}'|; k)}{|\mathbf{x} - \mathbf{x}'|} \text{ if } k \neq 0 \end{cases} \tag{C.7}$$

where $\mathcal{K}\left(|\mathbf{x}-\mathbf{x}'|\,;k\right)$ must be a function with a regular behavior when $R = |\mathbf{x}-\mathbf{x}'| \to 0$. Replacing the $k \neq 0$ case in equation (C.6) gives

$$\left[k^2 + \nabla^2\right]\left[\frac{\bar{\mathcal{K}}\left(|\mathbf{x}-\mathbf{x}'|\,;k\right)}{|\mathbf{x}-\mathbf{x}'|}\right] = -4\pi\delta(\mathbf{x}-\mathbf{x}'). \tag{C.8}$$

For $R \neq 0$ we have $\delta(\mathbf{x}-\mathbf{x}') = 0$ and the Laplacian becomes

$$\begin{aligned}
\nabla^2\left[\frac{\bar{\mathcal{K}}\left(|\mathbf{x}-\mathbf{x}'|\,;k\right)}{|\mathbf{x}-\mathbf{x}'|}\right] &= \frac{1}{R^2}\frac{d}{dR}\left[R^2\frac{d}{dR}\left(\frac{\bar{\mathcal{K}}\left(R;k\right)}{R}\right)\right] \\
&= \frac{1}{R^2}\frac{d}{dR}\left[R^2\left(\frac{1}{R}\frac{d\bar{\mathcal{K}}\left(R;k\right)}{dR} - \frac{\bar{\mathcal{K}}\left(R;k\right)}{R^2}\right)\right] \\
&= \frac{1}{R^2}\frac{d}{dR}\left[R\frac{d\bar{\mathcal{K}}\left(R;k\right)}{dR} - \bar{\mathcal{K}}\left(R;k\right)\right] \\
&= \frac{1}{R^2}\left[\frac{d\bar{\mathcal{K}}\left(R;k\right)}{dR} + R\frac{d^2\bar{\mathcal{K}}\left(R;k\right)}{dR^2} - \frac{d\bar{\mathcal{K}}\left(R;k\right)}{dR}\right] \\
&= \frac{1}{R}\frac{d^2\bar{\mathcal{K}}\left(R;k\right)}{dR^2}.
\end{aligned} \tag{C.9}$$

Then, equation (C.8) becomes

$$\frac{d^2\bar{\mathcal{K}}\left(R;k\right)}{dR^2} + k^2\bar{\mathcal{K}}\left(R;k\right) = 0 \tag{C.10}$$

and its solution is

$$\bar{\mathcal{K}}_{\pm}\left(R;k\right) = e^{\pm ikR}. \tag{C.11}$$

The Green functions are obtained by replacing this results back in (C.2),

$$\mathcal{G}_{\pm}(x,x') = \frac{1}{2\pi}\int\frac{e^{\pm ik|\mathbf{x}-\mathbf{x}'|}}{|\mathbf{x}-\mathbf{x}'|}e^{-ik(x^0-x'^0)}dk = \frac{1}{2\pi\,|\mathbf{x}-\mathbf{x}'|}\int e^{-ik(x^0-x'^0\mp|\mathbf{x}-\mathbf{x}'|)}dk. \tag{C.12}$$

Using the definition of the Dirac delta function let us write

$$\mathcal{G}_{\pm}(x,x') = \frac{\delta\left(x^0 - x'^0 \mp |\mathbf{x}-\mathbf{x}'|\right)}{|\mathbf{x}-\mathbf{x}'|} \tag{C.13}$$

or using $x^0 = ct$,

$$\mathcal{G}_{\pm}(x,x') = \frac{\delta\left(ct - ct' \mp |\mathbf{x}-\mathbf{x}'|\right)}{|\mathbf{x}-\mathbf{x}'|}. \tag{C.14}$$

Due to causality relations, we are interested in the solution which is non-zero when $ct - ct' = |\mathbf{x}-\mathbf{x}'|$, corresponding to $\mathcal{G}_{\pm}(x,x') = \mathcal{G}_R(x,x')$ and which is known as the *retarded Green function*,

$$\mathcal{G}_R(x,x') = \frac{\delta\left(ct - ct' - |\mathbf{x}-\mathbf{x}'|\right)}{|\mathbf{x}-\mathbf{x}'|}. \tag{C.15}$$

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